Main Category	Category	Terms
Disease	OMIM_DISEASE	Breast cancer, somatic
Disease	OMIM_DISEASE	Colorectal cancer, somatic
Disease	OMIM_DISEASE	Hepatocellular carcinoma, somatic
Disease	OMIM_DISEASE	Macrodactyly, somatic
Disease	OMIM_DISEASE	Nevus, epidermal, somatic mosaic
Disease	OMIM_DISEASE	Ovarian cancer, somatic
Disease	OMIM_DISEASE	Keratosis, seborrheic, somatic
Disease	OMIM_DISEASE	Nonsmall cell lung cancer, somatic
Disease	OMIM_DISEASE	Megalencephaly-capillary malformation-polymicrogyria syndrome, somatic
Disease	OMIM_DISEASE	Hemifacial myohyperplasia, somatic
Disease	OMIM_DISEASE	CLOVE syndrome, somatic
Disease	OMIM_DISEASE	CLAPO syndrome, somatic
Disease	OMIM_DISEASE	Gastric cancer, somatic
Disease	OMIM_DISEASE	Cowden syndrome 5
Disease	OMIM_DISEASE	Cerebral cavernous malformations 4, somatic
Disease	UP_KW_DISEASE	Disease variant
Disease	UP_KW_DISEASE	Proto-oncogene
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Angiogenesis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Lipid metabolism
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Phagocytosis
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Kinase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Serine/threonine-protein kinase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Transferase
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:C2 PI3K-type
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:PI3K-ABD
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:PI3K-RBD

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:PI3K/PI4K catalytic
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:PIK helical
Functional Annotations	UP_SEQ_FEATURE	REGION:Activation loop
Functional Annotations	UP_SEQ_FEATURE	REGION:Catalytic loop
Functional Annotations	UP_SEQ_FEATURE	REGION:G-loop
Functional Annotations	UP_SEQ_FEATURE	SITE:Implicated in the recognition of ATP as well as PIP2. Also crucial for autophosphorylation of the p85alpha subunit
Gene Ontology	GOTERM_BP_DIRECT	angiogenesis
Gene Ontology	GOTERM_BP_DIRECT	liver development
Gene Ontology	GOTERM_BP_DIRECT	vasculature development
Gene Ontology	GOTERM_BP_DIRECT	glucose metabolic process
Gene Ontology	GOTERM_BP_DIRECT	lipid metabolic process
Gene Ontology	GOTERM_BP_DIRECT	phagocytosis
Gene Ontology	GOTERM_BP_DIRECT	epidermal growth factor receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	insulin receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of lamellipodium assembly
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of gene expression
Gene Ontology	GOTERM_BP_DIRECT	response to activity
Gene Ontology	GOTERM_BP_DIRECT	response to muscle inactivity
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of macroautophagy
Gene	GOTERM_BP_DIRECT	cell migration

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_BP_DIRECT	actin cytoskeleton organization
Gene Ontology	GOTERM_BP_DIRECT	platelet activation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of actin filament depolymerization
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of TOR signaling
Gene Ontology	GOTERM_BP_DIRECT	cellular response to insulin stimulus
Gene Ontology	GOTERM_BP_DIRECT	response to muscle stretch
Gene Ontology	GOTERM_BP_DIRECT	phosphatidylinositol-3-phosphate biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	vascular endothelial growth factor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	regulation of multicellular organism growth
Gene Ontology	GOTERM_BP_DIRECT	response to L-leucine
Gene Ontology	GOTERM_BP_DIRECT	anoikis
Gene Ontology	GOTERM_BP_DIRECT	regulation of cellular respiration
Gene Ontology	GOTERM_BP_DIRECT	phosphatidylinositol 3-kinase/protein kinase B signal transduction
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of neuron apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	endothelial cell migration
Gene Ontology	GOTERM_BP_DIRECT	phosphatidylinositol phosphate biosynthetic process
Gene Ontology	GOTERM_BP_DIRECT	insulin-like growth factor receptor signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	phosphatidylinositol-mediated signaling
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of smooth muscle cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	T cell receptor signaling pathway
Gene	GOTERM_BP_DIRECT	positive regulation of phosphatidylinositol 3-

Main Category	Category	Terms
Ontology		kinase/protein kinase B signal transduction
Gene Ontology	GOTERM_BP_DIRECT	relaxation of cardiac muscle
Gene Ontology	GOTERM_BP_DIRECT	cardiac muscle contraction
Gene Ontology	GOTERM_BP_DIRECT	adipose tissue development
Gene Ontology	GOTERM_BP_DIRECT	cellular response to glucose stimulus
Gene Ontology	GOTERM_BP_DIRECT	cellular response to hydrostatic pressure
Gene Ontology	GOTERM_BP_DIRECT	response to dexamethasone
Gene Ontology	GOTERM_BP_DIRECT	cardiac muscle cell contraction
Gene Ontology	GOTERM_BP_DIRECT	energy homeostasis
Gene Ontology	GOTERM_BP_DIRECT	regulation of actin filament organization
Gene Ontology	GOTERM_BP_DIRECT	autosome genomic imprinting
Gene Ontology	GOTERM_BP_DIRECT	response to butyrate
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of protein localization to membrane
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of fibroblast apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of anoikis
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	plasma membrane
Gene Ontology	GOTERM_CC_DIRECT	phosphatidylinositol 3-kinase complex
Gene Ontology	GOTERM_CC_DIRECT	phosphatidylinositol 3-kinase complex, class IA
Gene Ontology	GOTERM_CC_DIRECT	phosphatidylinositol 3-kinase complex, class IB
Gene	GOTERM_CC_DIRECT	intercalated disc

Main Category	Category	Terms
Ontology		
Gene Ontology	GOTERM_CC_DIRECT	lamellipodium
Gene Ontology	GOTERM_CC_DIRECT	perinuclear region of cytoplasm
Gene Ontology	GOTERM_MF_DIRECT	nucleotide binding
Gene Ontology	GOTERM_MF_DIRECT	protein serine/threonine kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	ATP binding
Gene Ontology	GOTERM_MF_DIRECT	kinase activity
Gene Ontology	GOTERM_MF_DIRECT	1-phosphatidylinositol-3-kinase activity
Gene Ontology	GOTERM_MF_DIRECT	transferase activity
Gene Ontology	GOTERM_MF_DIRECT	protein kinase activator activity
Gene Ontology	GOTERM_MF_DIRECT	1-phosphatidylinositol-4-phosphate 3-kinase activity
Gene Ontology	GOTERM_MF_DIRECT	insulin receptor substrate binding
Gene Ontology	GOTERM_MF_DIRECT	1-phosphatidylinositol-4,5-bisphosphate 3-kinase activity
Gene Ontology	GOTERM_MF_DIRECT	protein serine kinase activity
General Annotations	CHROMOSOME	3
General Annotations	CYTOGENETIC_LOCATION	3q26.32
General Annotations	ENTREZ_GENE_SUMMARY	Phosphatidylinositol 3-kinase is composed of an 85 kDa regulatory subunit and a 110 kDa catalytic subunit. The protein encoded by this gene represents the catalytic subunit, which uses ATP to phosphorylate PtdIns, PtdIns4P and PtdIns(4,5)P2. This gene has been found to be oncogenic and has been implicated in cervical cancers. A pseudogene of this gene has been defined on chromosome 22. [provided by RefSeq, Apr 2016]

Main Category	Category	Terms
General Annotations	UP_COMMENT_BIOPHYSICOCHEMICAL_PROPERTIES	Kinetic parameters: KM=1.77 uM for PIP2 (1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo- inositol-4,5-bisphosphate)) {ECO:0000269|PubMed:28676499}; KM=2.00 uM for ATP {ECO:0000269|PubMed:28676499}; Vmax=1.70 pmol/min/ng enzyme for the phosphorylation of PIP2 (1,2- dioctanoyl-sn-glycero-3-phospho-(1D-myo-inositol-4,5-bisphosphate)) {ECO:0000269|PubMed:28676499};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo-inositol-4,5-bisphosphate) + ATP = 1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo- inositol-3,4,5-trisphosphate) + ADP + H(+); Xref=Rhea:RHEA:55632, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:83416, ChEBI:CHEBI:83419, ChEBI:CHEBI:456216; Evidence={ECO:0000256|ARBA:ARBA00050641}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:55633; Evidence={ECO:0000256|ARBA:ARBA00050641};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo-inositol-4,5-bisphosphate) + ATP = 1,2-dioctanoyl-sn-glycero-3-phospho-(1D-myo- inositol-3,4,5-trisphosphate) + ADP + H(+); Xref=Rhea:RHEA:55632, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:83416, ChEBI:CHEBI:83419, ChEBI:CHEBI:456216; Evidence={ECO:0000269|PubMed:28676499}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:55633; Evidence={ECO:0000305|PubMed:28676499};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=1-octadecanoyl-2-(5Z,8Z,11Z,14Z)-eicosatetraenoyl-sn-glycero- 3-phospho-1D-myo-inositol 4,5-bisphosphate + ATP = 1-octadecanoyl-2- (5Z,8Z,11Z,14Z-eicosatetraenoyl)-sn-glycero-3-phospho-(1D-myo- inositol 3,4,5-triphosphate) + ADP + H(+); Xref=Rhea:RHEA:43396, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:77137, ChEBI:CHEBI:83243, ChEBI:CHEBI:456216; Evidence={ECO:0000256|ARBA:ARBA00051347}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:43397; Evidence={ECO:0000256|ARBA:ARBA00051347};

Main Category	Category	Terms
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=1-octadecanoyl-2-(5Z,8Z,11Z,14Z)-eicosatetraenoyl-sn-glycero- 3-phospho-1D-myo-inositol 4,5-bisphosphate + ATP = 1-octadecanoyl-2- (5Z,8Z,11Z,14Z-eicosatetraenoyl)-sn-glycero-3-phospho-(1D-myo- inositol 3,4,5-triphosphate) + ADP + H(+); Xref=Rhea:RHEA:43396, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:77137, ChEBI:CHEBI:83243, ChEBI:CHEBI:456216; Evidence={ECO:0000305| PubMed:15135396}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:43397; Evidence={ECO:0000305| PubMed:15135396};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-seryl-[protein] + ATP = O-phospho-L-seryl-[protein] + ADP + H(+); Xref=Rhea:RHEA:17989, Rhea:RHEA-COMP:9863, Rhea:RHEA- COMP:11604, ChEBI:CHEBI:15378, ChEBI:CHEBI:29999, ChEBI:CHEBI:30616, ChEBI:CHEBI:83421, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000256| ARBA:ARBA00048977}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:17990; Evidence={ECO:0000256| ARBA:ARBA00048977};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=L-seryl-[protein] + ATP = O-phospho-L-seryl-[protein] + ADP + H(+); Xref=Rhea:RHEA:17989, Rhea:RHEA-COMP:9863, Rhea:RHEA- COMP:11604, ChEBI:CHEBI:15378, ChEBI:CHEBI:29999, ChEBI:CHEBI:30616, ChEBI:CHEBI:83421, ChEBI:CHEBI:456216; EC=2.7.11.1; Evidence={ECO:0000269| PubMed:23936502, ECO:0000269| PubMed:28676499}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:17990; Evidence={ECO:0000305| PubMed:23936502, ECO:0000305| PubMed:28676499};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol) + ATP = a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol-3-phosphate) + ADP + H(+); Xref=Rhea:RHEA:12709, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:57880, ChEBI:CHEBI:58088, ChEBI:CHEBI:456216; EC=2.7.1.137; Evidence={ECO:0000256| ARBA:ARBA00023985}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:12710; Evidence={ECO:0000256| ARBA:ARBA00023985};

Main Category	Category	Terms
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol) + ATP = a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol-3-phosphate) + ADP + H(+); Xref=Rhea:RHEA:12709, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:57880, ChEBI:CHEBI:58088, ChEBI:CHEBI:456216; EC=2.7.1.137; Evidence={ECO:0000269|PubMed:23936502}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:12710; Evidence={ECO:0000305|PubMed:23936502};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol-4,5- bisphosphate) + ATP = a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo- inositol-3,4,5-trisphosphate) + ADP + H(+); Xref=Rhea:RHEA:21292, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:57836, ChEBI:CHEBI:58456, ChEBI:CHEBI:456216; EC=2.7.1.153; Evidence={ECO:0000256|ARBA:ARBA00023981}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:21293; Evidence={ECO:0000256|ARBA:ARBA00023981};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo-inositol-4,5- bisphosphate) + ATP = a 1,2-diacyl-sn-glycero-3-phospho-(1D-myo- inositol-3,4,5-trisphosphate) + ADP + H(+); Xref=Rhea:RHEA:21292, ChEBI:CHEBI:15378, ChEBI:CHEBI:30616, ChEBI:CHEBI:57836, ChEBI:CHEBI:58456, ChEBI:CHEBI:456216; EC=2.7.1.153; Evidence={ECO:0000269|PubMed:15135396, ECO:0000269|PubMed:28676499}; PhysiologicalDirection=left-to-right; Xref=Rhea:RHEA:21293; Evidence={ECO:0000305|PubMed:15135396, ECO:0000305|PubMed:28676499};
General Annotations	UP_COMMENT_DISEASE	Breast cancer (BC) [MIM:114480]: A common malignancy originating from breast epithelial tissue. Breast neoplasms can be distinguished by their histologic pattern. Invasive ductal carcinoma is by far the most common type. Breast cancer is etiologically and genetically heterogeneous. Important genetic factors have been indicated by familial occurrence and bilateral involvement. Mutations at more than one locus can be involved in different families or even in the same case. {ECO:0000269|PubMed:16353168}. Note=Disease susceptibility is associated

Main Category	Category	Terms
		with variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	CLAPO syndrome (CLAPO) [MIM:613089]: A syndrome characterized by capillary malformation of the lower lip, lymphatic malformation of the face and neck, asymmetry of face and limbs and partial or generalised overgrowth. {ECO:0000269|PubMed:29446767}. Note=The disease may be caused by variants affecting the gene represented in this entry. The tissue distribution of the clinical manifestations in CLAPO seems to follow a pattern of somatic mosaicism. {ECO:0000269|PubMed:29446767}.
General Annotations	UP_COMMENT_DISEASE	Cerebral cavernous malformations 4 (CCM4) [MIM:619538]: A form of cerebral cavernous malformations, a congenital vascular anomaly of the central nervous system that can result in hemorrhagic stroke, seizures, recurrent headaches, and focal neurologic deficits. The lesions are characterized by grossly enlarged blood vessels consisting of a single layer of endothelium and without any intervening neural tissue, ranging in diameter from a few millimeters to several centimeters. CCM4 cases occur sporadically. {ECO:0000269|PubMed:34496175}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Colorectal cancer (CRC) [MIM:114500]: A complex disease characterized by malignant lesions arising from the inner wall of the large intestine (the colon) and the rectum. Genetic alterations are often associated with progression from premalignant lesion (adenoma) to invasive adenocarcinoma. Risk factors for cancer of the colon and rectum include colon polyps, long-standing ulcerative colitis, and genetic family history. {ECO:0000269|PubMed:15930273, ECO:0000269|PubMed:15994075}. Note=The gene represented in this entry may be involved in disease pathogenesis.
General Annotations	UP_COMMENT_DISEASE	Congenital lipomatous overgrowth, vascular malformations, and epidermal nevi (CLOVE) [MIM:612918]: A sporadically occurring, non- hereditary disorder characterized by asymmetric somatic hypertrophy and anomalies in multiple organs. It is defined by four main clinical findings: congenital lipomatous overgrowth, vascular malformations, epidermal nevi, and skeletal/ spinal abnormalities. The presence of truncal overgrowth and characteristic

Main Category	Category	Terms
		patterned macrodactyly at birth differentiates CLOVE from other syndromic forms of overgrowth. {ECO:0000269|PubMed:22658544}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Cowden syndrome 5 (CWS5) [MIM:615108]: A form of Cowden syndrome, a hamartomatous polyposis syndrome with age-related penetrance. Cowden syndrome is characterized by hamartomatous lesions affecting derivatives of ectodermal, mesodermal and endodermal layers, macrocephaly, facial trichilemmomas (benign tumors of the hair follicle infundibulum), acral keratoses, papillomatous papules, and elevated risk for development of several types of malignancy, particularly breast carcinoma in women and thyroid carcinoma in both men and women. Colon cancer and renal cell carcinoma have also been reported. Hamartomas can be found in virtually every organ, but most commonly in the skin, gastrointestinal tract, breast and thyroid. {ECO:0000269|PubMed:23246288}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Hemifacial myohyperplasia (HFMH) [MIM:606773]: A rare disease characterized by facial asymmetry due to unilateral muscular hypertrophy mimicking spasm and orofacial dystonia. {ECO:0000269|PubMed:37712948}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Hepatocellular carcinoma (HCC) [MIM:114550]: A primary malignant neoplasm of epithelial liver cells. The major risk factors for HCC are chronic hepatitis B virus (HBV) infection, chronic hepatitis C virus (HCV) infection, prolonged dietary aflatoxin exposure, alcoholic cirrhosis, and cirrhosis due to other causes. {ECO:0000269|PubMed:15608678}. Note=The gene represented in this entry may be involved in disease pathogenesis.
General Annotations	UP_COMMENT_DISEASE	Keratosis, seborrheic (KERSEB) [MIM:182000]: A common benign skin tumor. Seborrheic keratoses usually begin with the appearance of one or more sharply defined, light brown, flat macules. The lesions may be sparse or numerous. As they initially grow, they develop a velvety to finely verrucous surface, followed by an uneven warty surface with multiple plugged follicles and a dull or lackluster appearance.

Main Category	Category	Terms
		{ECO:0000269|PubMed:17673550}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Macrodactyly (MADAC) [MIM:155500]: A congenital anomaly characterized by fibrofatty tissue enlargement and bony overgrowth affecting the digits or the entire hand or foot. {ECO:0000269|PubMed:23100325}. Note=The disease may be caused by variants affecting the gene represented in this entry. The tissue distribution of the clinical manifestations in MADAC seems to follow a pattern of somatic mosaicism. {ECO:0000269|PubMed:23100325}.
General Annotations	UP_COMMENT_DISEASE	Megalencephaly-capillary malformation-polymicrogyria syndrome (MCAP) [MIM:602501]: A syndrome characterized by a spectrum of anomalies including primary megalencephaly, prenatal overgrowth, brain and body asymmetry, cutaneous vascular malformations, digital anomalies consisting of syndactyly with or without postaxial polydactyly, connective tissue dysplasia involving the skin, subcutaneous tissue, and joints, and cortical brain malformations, most distinctively polymicrogyria. {ECO:0000269|PubMed:22729224, ECO:0000269|PubMed:26593112}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Note=PIK3CA mutations are involved in various type of cancer. Most of the cancer-associated mutations are missense mutations and map to one of the three hotspots: Glu-542; Glu-545 and His-1047. Mutated isoforms participate in cellular transformation and tumorigenesis induced by oncogenic receptor tyrosine kinases (RTKs) and HRAS/KRAS. Interaction with HRAS/KRAS is required for Ras-driven tumor formation. Mutations increasing the lipid kinase activity are required for oncogenic signaling. The protein kinase activity may not be required for tumorigenesis. {ECO:0000269|PubMed:15016963, ECO:0000269|PubMed:15289301, ECO:0000269|PubMed:15520168, ECO:0000269|PubMed:15712344, ECO:0000269|PubMed:15784156, ECO:0000269|PubMed:15924253, ECO:0000269|PubMed:15930273, ECO:0000269|PubMed:15994075, ECO:0000269|PubMed:16114017, ECO:0000269|PubMed:16322209, ECO:0000269|PubMed:16353168, ECO:0000269|PubMed:16432179, ECO:0000269|PubMed:16533766,

Main Category	Category	Terms
		ECO:0000269|PubMed:17673550, ECO:0000269|PubMed:19805105, ECO:0000269|PubMed:21708979, ECO:0000269|PubMed:22658544, ECO:0000269|PubMed:22729224}.
General Annotations	UP_COMMENT_DISEASE	Ovarian cancer (OC) [MIM:167000]: The term ovarian cancer defines malignancies originating from ovarian tissue. Although many histologic types of ovarian tumors have been described, epithelial ovarian carcinoma is the most common form. Ovarian cancers are often asymptomatic and the recognized signs and symptoms, even of late-stage disease, are vague. Consequently, most patients are diagnosed with advanced disease. {ECO:0000269|PubMed:15520168}. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DOMAIN	The PI3K-ABD domain and the PI3K-RBD domain interact with the PI3K/PI4K kinase domain. The C2 PI3K-type domain may facilitate the recruitment to the plasma membrane. The inhibitory interactions with PIK3R1 are mediated by the PI3K-ABD domain and the C2 PI3K-type domain with the iSH2 (inter-SH2) region of PIK3R1, and the C2 PI3K-type domain, the PI3K helical domain, and the PI3K/PI4K kinase domain with the nSH2 (N-terminal SH2) region of PIK3R1. {ECO:0000269|PubMed:18079394, ECO:0000269|PubMed:19805105}.
General Annotations	UP_COMMENT_FUNCTION	Phosphoinositide-3-kinase (PI3K) phosphorylates phosphatidylinositol (PI) and its phosphorylated derivatives at position 3 of the inositol ring to produce 3-phosphoinositides (PubMed:15135396, PubMed:23936502, PubMed:28676499). Uses ATP and PtdIns(4,5)P2 (phosphatidylinositol 4,5-bisphosphate) to generate phosphatidylinositol 3,4,5-trisphosphate (PIP3) (PubMed:15135396, PubMed:28676499). PIP3 plays a key role by recruiting PH domain- containing proteins to the membrane, including AKT1 and PDPK1, activating signaling cascades involved in cell growth, survival, proliferation, motility and morphology. Participates in cellular signaling in response to various growth factors. Involved in the activation of AKT1 upon stimulation by receptor tyrosine kinases ligands such as EGF, insulin, IGF1, VEGFA and PDGF. Involved in signaling via insulin-receptor substrate (IRS) proteins. Essential in endothelial cell migration during vascular development through VEGFA signaling, possibly by regulating RhoA activity. Required for lymphatic vasculature

Main Category	Category	Terms
		development, possibly by binding to RAS and by activation by EGF and FGF2, but not by PDGF. Regulates invadopodia formation through the PDPK1-AKT1 pathway. Participates in cardiomyogenesis in embryonic stem cells through a AKT1 pathway. Participates in vasculogenesis in embryonic stem cells through PDK1 and protein kinase C pathway. In addition to its lipid kinase activity, it displays a serine-protein kinase activity that results in the autophosphorylation of the p85alpha regulatory subunit as well as phosphorylation of other proteins such as 4EBP1, H-Ras, the IL-3 beta c receptor and possibly others (PubMed:23936502, PubMed:28676499). Plays a role in the positive regulation of phagocytosis and pinocytosis (By similarity). {ECO:0000250|UniProtKB:P42337, ECO:0000269|PubMed:15135396, ECO:0000269|PubMed:21708979, ECO:0000269|PubMed:23936502, ECO:0000269|PubMed:26593112, ECO:0000269|PubMed:28676499}.
General Annotations	UP_COMMENT_INTERACTION	P42336; P54253: ATXN1; NbExp=3; IntAct=EBI-2116585, EBI-930964; P42336; P54252: ATXN3; NbExp=3; IntAct=EBI-2116585, EBI-946046; P42336; P21860: ERBB3; NbExp=18; IntAct=EBI-2116585, EBI-720706; P42336; P01100: FOS; NbExp=3; IntAct=EBI-2116585, EBI-852851; P42336; P62993: GRB2; NbExp=6; IntAct=EBI-2116585, EBI-401755; P42336; P42858: HTT; NbExp=3; IntAct=EBI-2116585, EBI-466029; P42336; P35568: IRS1; NbExp=41; IntAct=EBI-2116585, EBI-517592; P42336; P27986: PIK3R1; NbExp=64; IntAct=EBI-2116585, EBI-79464; P42336; P27986-2: PIK3R1; NbExp=3; IntAct=EBI-2116585, EBI-9090282; P42336; O00459: PIK3R2; NbExp=49; IntAct=EBI-2116585, EBI-346930; P42336; Q92569: PIK3R3; NbExp=47; IntAct=EBI-2116585, EBI-79893;
General Annotations	UP_COMMENT_MISCELLANEOUS	The avian sarcoma virus 16 genome encodes an oncogene derived from PIK3CA. {ECO:0000305|PubMed:18418043}.
General Annotations	UP_COMMENT_PATHWAY	Lipid metabolism. {ECO:0000256|ARBA:ARBA00005189}.
General Annotations	UP_COMMENT_PATHWAY	Phospholipid metabolism; phosphatidylinositol phosphate biosynthesis. {ECO:0000256|ARBA:ARBA00004805}.

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General Annotations	UP_COMMENT_PATHWAY	Phospholipid metabolism; phosphatidylinositol phosphate biosynthesis. {ECO:0000305|PubMed:15135396, ECO:0000305|PubMed:28676499}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the PI3/PI4-kinase family. Type III PI4K subfamily. {ECO:0000256|ARBA:ARBA00006209}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the PI3/PI4-kinase family. {ECO:0000255|PROSITE-ProRule:PRU00877, ECO:0000255|PROSITE-ProRule:PRU00879, ECO:0000255|PROSITE-ProRule:PRU00880}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the PI3/PI4-kinase family. {ECO:0000256|PROSITE-ProRule:PRU00877}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the PI3/PI4-kinase family. {ECO:0000256|PROSITE-ProRule:PRU00880}.
General Annotations	UP_COMMENT_SUBUNIT	Heterodimer of a catalytic subunit PIK3CA and a p85 regulatory subunit (PIK3R1, PIK3R2 or PIK3R3) (PubMed:26593112). Interacts with IRS1 in nuclear extracts (By similarity). Interacts with RUFY3 (By similarity). Interacts with RASD2 (By similarity). Interacts with APPL1. Interacts with HRAS and KRAS (By similarity). Interaction with HRAS/KRAS is required for PI3K pathway signaling and cell proliferation stimulated by EGF and FGF2 (By similarity). Interacts with FAM83B; activates the PI3K/AKT signaling cascade (PubMed:23676467). {ECO:0000250|UniProtKB:P42337, ECO:0000269|PubMed:23676467, ECO:0000269|PubMed:26593112}.
General Annotations	UP_COMMENT_SUBUNIT	Heterodimer of a catalytic subunit PIK3CA and a p85 regulatory subunit (PIK3R1, PIK3R2 or PIK3R3). Interacts with IRS1 in nuclear extracts. Interacts with RUFY3. Interacts with RASD2. Interacts with APPL1. Interacts with HRAS and KRAS. Interaction with HRAS/KRAS is required for PI3K pathway signaling and cell proliferation stimulated by EGF and FGF2. Interacts with FAM83B; activates the PI3K/AKT signaling cascade. {ECO:0000256|ARBA:ARBA00065166}.
Identifiers	ENTREZ_GENE_ID	5290
Identifiers	ENSEMBL_GENE_ID	ENSG00000121879
Identifiers	UNIPROT_ID	Q4LE51_HUMAN
Identifiers	UNIPROT_ID	PK3CA_HUMAN

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Identifiers	UNIPROT_ID	E2I6G1_HUMAN
Identifiers	UNIPROT_ID	C9J951_HUMAN
Identifiers	UNIPROT_ID	C9JAM9_HUMAN
Identifiers	UNIPROT_ID	A0A8A2FPJ2_HUMAN
Identifiers	UNIPROT_ID	A0A6Q8PGV1_HUMAN
Identifiers	UNIPROT_ID	A0A6Q8PGN9_HUMAN
Identifiers	UNIPROT_ID	A0A6B9QIS5_HUMAN
Identifiers	UNIPROT_ID	A0A6B9QKK2_HUMAN
Identifiers	UNIPROT_ID	A0A3G2C3N5_HUMAN
Identifiers	UNIPROT_ID	A0A2R8Y2F6_HUMAN
Identifiers	UNIPROT_ID	A0A2P0XI22_HUMAN
Interactions	DRUGBANK	ATP
Interactions	DRUGBANK	Caffeine
Interactions	DRUGBANK	XL765
Interactions	DRUGBANK	Wortmannin
Interactions	DRUGBANK	Pilaralisib
Interactions	DRUGBANK	Alpelisib
Interactions	DRUGBANK	Copanlisib
Interactions	UP_KW_LIGAND	ATP-binding
Interactions	UP_KW_LIGAND	Nucleotide-binding
Literature	GENERIF_SUMMARY	In squamous cell carcinomas the negative effect of p53 induction on cell survival involves transcriptional inhibition of PIK3CA that is independent of PTEN activity, as PTEN is not expressed in the primary tumors
Literature	GENERIF_SUMMARY	agonist binding to Gq-coupled receptors blocks Akt activation via the release of active Galphaq subunits that inhibit PI3K via an inhibitory interaction between Galphaq and p110alpha PI3K.
Literature	GENERIF_SUMMARY	data suggest that mutant PIK3CA is likely to function as an oncogene in human cancers; PIK3CA may prove useful for diagnostic and therapeutic purposes
Literature	GENERIF_SUMMARY	Missense Mutation in PIK3CA is associated with breast cancers
Literature	GENERIF_SUMMARY	Mutations of PIK3CA is associated with anaplastic oligodendrogliomas, high-grade

Main Category	Category	Terms
		astrocytomas, and medulloblastomas
Literature	GENERIF_SUMMARY	oncogenic mutations in human cancers
Literature	GENERIF_SUMMARY	Genetic alterations of class IA PI3K subunit genes can occasionally play a role in human glioblastoma by activating the PI3K-AKT signaling pathway independently of PTEN mutation.
Literature	GENERIF_SUMMARY	Data show that glucose is a crucial regulator of V-ATPase in renal epithelial cells and that the effect of glucose is mediated by phosphatidylinositol 3-kinase (PI3K).
Literature	GENERIF_SUMMARY	determined the growth-regulatory and signaling properties of the three most frequently observed PIK3CA mutations
Literature	GENERIF_SUMMARY	PIK3CA mutations in exons 9 & 20 were directly sequenced. A novel silent mutation was found. PIK3CA mutation may be an early event in ovarian carcinogenesis.
Literature	GENERIF_SUMMARY	Activation of the PI3K signalling pathway in gastric cancer may be achieved through up-regulation or mutation of PIK3CA, in which the latter may be a consequence of mismatch repair deficiency.
Literature	GENERIF_SUMMARY	Mutation of PIK3CA is frequent, occurs early in carcinoma development, and has prognostic and therapeutic implications for breast cancer.
Literature	GENERIF_SUMMARY	Data indicate that mutations of PIK3CA play an oncogenic role in substantial fractions of ovarian and breast carcinomas.
Literature	GENERIF_SUMMARY	mutation of the PIK3CA gene is not common, but its amplification is relatively common and may be a novel mechanism in activating the PI3K/Akt pathway in some thyroid tumors
Literature	GENERIF_SUMMARY	Colon cancer-associated PIK3CA mutations are functionally active so that they are likely to be involved in carcinogenesis.
Literature	GENERIF_SUMMARY	Amino acid substitutions in PIK3CA from human colorectal cancer cell lines constitutively activate the AKT pathway, and consequently, promote tumor cell growth and invasion.
Literature	GENERIF_SUMMARY	PI-3 kinase p110alpha subunit expression and recombinant heparin-binding epidermal growth factor-like growth factor (HB-EGF) synergistically decrease inducible nitric oxide synthase expression and nitric oxide production in intestinal epithelial cells

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Literature	GENERIF_SUMMARY	copy number gains, instead of mutation, may be a common mechanism for activation of PIK3CA in tumorigenesis of primary nasopharyngeal carcinoma
Literature	GENERIF_SUMMARY	an important role of PIK3CA gene aberrations in the molecular pathogenesis of primary glioblastomas
Literature	GENERIF_SUMMARY	this study has shown that the mutations detected in the helical and kinase domains of PIK3CA do not seem to affect the PI3-K pathway characteristics of the eight cell lines studied
Literature	GENERIF_SUMMARY	The frequent and clustered mutations within PIK3CA make it an attractive molecular marker for early detection and a promising therapeutic target in breast cancer.
Literature	GENERIF_SUMMARY	analysis of PIK3CA mutations in ovarian cancer [letter]
Literature	GENERIF_SUMMARY	increased activation state of AKT kinase appears to be present in cervical carcinogenesis, and may be accounted for by PIK3CA amplification, whereas PTEN mutation seems to be of little importance
Literature	GENERIF_SUMMARY	Mutant PIK3CA is likely to function as an oncogene in anaplastic thyroid cancer and less frequently well-differentiated thyroid carcinomas.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with breast cancer
Literature	GENERIF_SUMMARY	Data show that alpha6 and alpha3 integrin subunits interact with laminin 5 to increase expression of E-cadherin, and suggest that phosphoinositide 3-kinase (PI 3-kinase) activation plays a key role in this cross-talk.
Literature	GENERIF_SUMMARY	although introduction of activating mutations from p110alpha at the corresponding sites in p110beta failed to render the enzyme oncogenic in human cells, the possibility remains that other mutations might activate the beta isoform
Literature	GENERIF_SUMMARY	cancer-specific mutations in PIK3CA have made p110alpha an ideal drug target [Review]
Literature	GENERIF_SUMMARY	A mutation status profile of the PIK3Ca gene in the National Cancer Institute (NCI)-60 panel of human cancer cell lines provides insights into the role of mutant PIK3Ca in oncogenic signaling
Literature	GENERIF_SUMMARY	study extends previous observations in other tumor types by demonstrating

Main Category	Category	Terms
		presence of somatic PIK3CA mutations in both SCC and adenocarcinoma of the esophagus, thus implicating the PI3K pathway in the initiation and/or progression of esophageal cancers
Literature	GENERIF_SUMMARY	These results indicate that PI3K plays different roles in the activation of Ras/ERK1/2 signaling by insulin and EGF, and that insulin-stimulated, but not EGF-stimulated, ERK1/2 and Akt signalings diverge at PI3K.
Literature	GENERIF_SUMMARY	AT(1) receptor-mediated activation of PI 3-K/Akt cascades occurs at least partially via the transactivation of EGF receptor, which is under a negative control by AT(2) receptor in hypertrophic scar fibroblasts.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with head and neck squamous cell carcinoma
Literature	GENERIF_SUMMARY	Although a key factor in TRAF6-dependent activation of PI 3-kinase, ectopic expression of Src was insufficient for NF-kappaB activation and, in contrast to NF-kappaB, was not inhibited by IRAK2.
Literature	GENERIF_SUMMARY	Mutation of PIK3CA is associated with breast cancer
Literature	GENERIF_SUMMARY	Somatic mutations of PIK3CA, encoding p110alpha catalytic subunit of Class IA PI3Ks, have been found in various cancers.
Literature	GENERIF_SUMMARY	inhibition of PI 3-kinase activity reduced hypoxic HIF-1alpha protein levels to a similar extent as serum deprivation
Literature	GENERIF_SUMMARY	PI3K-IA activity is necessary for both high NaCl- and ionizing radiation-induced activation of ATM and (ii) high NaCl activates PI3K-IA, which, in turn, contributes to full activation of TonEBP/OREBP via ATM.
Literature	GENERIF_SUMMARY	Various mutations in a variety of cancers, including a pseudogene and polymorphisms.
Literature	GENERIF_SUMMARY	p110alpha play an important role in tumor growth by inducing angiogenesis and by increasing HIF-1alpha and VEGF expression. This work provides a better understanding of the molecular mechanism of human cancer induced by the activation of PI3K signaling.
Literature	GENERIF_SUMMARY	First report of PIK3CA mutation in pancreatic cancer, providing evidence of its oncogenic properties.
Literature	GENERIF_SUMMARY	Increased expression for PIK3CA mrna is

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		associated with regional lymph node metastasis in esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	Acidic pH exposure protects HEMEC through induction of Hsps and activation of MAPK and PI3 kinase pathway.
Literature	GENERIF_SUMMARY	PIK3CA amplification is uncommon and appears to be late event in the development of ethmoid sinus adenocarcinoma.
Literature	GENERIF_SUMMARY	TGF-beta1 regulates COX-2 expression in human mesangial cells through the activation of ERK1/2, p38 MAPK, and PI3K.
Literature	GENERIF_SUMMARY	Pik3ca, Ras family, and Braf activating mutations occur rarely in human neuroblastomas or in a murine model driven by neural-restricted MYCN overexpression.
Literature	GENERIF_SUMMARY	These studies demonstrate that induction of fibulin 5 gene expression in lung fibroblasts is mediated via canonical TGF-beta/Smad signaling and requires the PI3-kinase/Akt pathway.
Literature	GENERIF_SUMMARY	The expression of p110alpha siRNA significantly decreased cell migration, invasion and proliferation.
Literature	GENERIF_SUMMARY	Outside-in signaling via P2Y12 and both PI3Kbeta and PI3Kgamma isoforms is required for maintenance of a platelet aggregate.
Literature	GENERIF_SUMMARY	PIK3CA mutations were strongly associated with FGFR3 mutations in superficial papillary bladder tumors.
Literature	GENERIF_SUMMARY	PIK3CA mutation incidence was significantly lower in lung adenocarcinoma than in lung squamous cell carcinoma; among eight patients with a PIK3CA mutation, three patients also harbored an EGFR somatic mutation.
Literature	GENERIF_SUMMARY	results confirm that PIK3CA mutations are frequent in endometrial carcinoma and support the hypothesis that PIK3CA mutations may have an additive effect to PTEN monoallelic inactivation in endometrial carcinoma
Literature	GENERIF_SUMMARY	PI3K is an important intracellular regulator of epidermal homeostasis and repair.
Literature	GENERIF_SUMMARY	It has recently been shown that Akt activation is associated with a worse outcome among endocrine treated breast cancer patients and that it also inhibits the progesterone receptor (PR) expression via

Main Category	Category	Terms
		the PI3K/Akt pathway in breast cancer cells.
Literature	GENERIF_SUMMARY	This is the first study showing PIK3CA mutations in pediatric glioblastomas, thus providing a molecular target in this important pediatric malignancy.
Literature	GENERIF_SUMMARY	Mutations of the PIK3CA gene is associated with the development of oral squamous cell carcinoma
Literature	GENERIF_SUMMARY	Activation of PI3K signaling by mutations in PTEN or PIK3CA can lead to activation of p53-mediated growth suppression.
Literature	GENERIF_SUMMARY	PIK3CA may serve as a marker of invasion in endometrial cancer.
Literature	GENERIF_SUMMARY	Results show that coculture with ECs enhances VSMC adhesion and spreading by up-regulating beta(1)-integrin expression and activating the PI3K/Akt pathway. Interaction between ECs and VSMCs serves a role in vascular homeostasis and remodeling.
Literature	GENERIF_SUMMARY	PIK3CA mutations might exert their effects through activation of the phosphatidylinositol 3-kinase/AKT/ER-alpha pathway
Literature	GENERIF_SUMMARY	Results indicate that stimulation of heme oxygenase-1 expression by hypericin-PDT is a cytoprotective mechanism governed by the p38(MAPK) and PI3K pathways.
Literature	GENERIF_SUMMARY	PIK3CA exon 9 and exon 20 rarely mutate in nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA alterations in primary (de novo) and secondary glioblastomas.
Literature	GENERIF_SUMMARY	Thus EPO promotes survival of endothelial cells through PI3K-dependent Bcl-x(L)-induction and BIM regulation, as well as through a separate mechanism involving the ERK pathway.
Literature	GENERIF_SUMMARY	These results demonstrate that hypoxic condition-and high cell density-induced expression of Redd1 is mediated by coactivation of Sp1 and HIF-1alpha downstream of the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	These results suggest that there is a fine distinction between the signaling activators and downstream effectors of the oncogenic PI3K and RAS pathways in breast epithelium and those in other tissues.
Literature	GENERIF_SUMMARY	The data suggest that aberrant activation of

Main Category	Category	Terms
		PI3K/Akt pathway plays an extensive role in thyroid tumorigenesis, particularly in FTC and ATC, and promotes progression of BTA to FTC and to ATC as the genetic alterations of this pathway accumulate.
Literature	GENERIF_SUMMARY	These data demonstrate that IGF-I induces COX-2 expression in human ovarian cancer cells, which is mediated by three parallel signaling cascades--PI3K, MAPK, and PKC pathways that differentially regulate COX-2 expression.
Literature	GENERIF_SUMMARY	studies show that mutant PIK3CA increases the capacity for proliferation and survival under environmental stresses, such as growth factor deprivation stress while also imparting greater dependency on the PI3K pathway for proliferation and survival
Literature	GENERIF_SUMMARY	p110alpha PI-3K appears to uniquely regulate important monocyte functions, where other PI-3K isoforms are uninvolved or unable to fully compensate.
Literature	GENERIF_SUMMARY	PIK3CA mutations may have a role in cancer
Literature	GENERIF_SUMMARY	PIK3CA amplification and Ki-67 index were strong predictors for an early tumor-associated death in ovarian cancer patients.
Literature	GENERIF_SUMMARY	data from a Chinese cohort provide further genetic evidence suggesting that dysregulated PI3K/Akt pathway plays a significant role in the pathogenesis of thyroid tumors
Literature	GENERIF_SUMMARY	TLR3, PI3K, and IRF3 are involved in the poly IC-induced galectin-9 expression in HUVECs
Literature	GENERIF_SUMMARY	PI3K/ILK/Akt/NF-kappaB axis is a promising target for therapeutic intervention in renal cell carcinoma.
Literature	GENERIF_SUMMARY	Data suggest that lipid rafts play critical roles in KSHV infection and gene expression, probably due to their roles in modulating KSHV-induced PI3-K, RhoA-GTPase, and Dia-2 molecules essential for postbinding and entry stages of infection.
Literature	GENERIF_SUMMARY	Data suggest that PIK3CA mutations contribute to the invasion step from intramucosal carcinoma to invasive carcinoma in colorectal carcinogenesis.
Literature	GENERIF_SUMMARY	PIK3CA copy number gain and amplification is associated with head and neck squamous cell carcinoma
Literature	GENERIF_SUMMARY	PIK3CA mutations and PTEN loss were not mutually exclusive events and associated

Main Category	Category	Terms
		with similar prognostic factors
Literature	GENERIF_SUMMARY	Whereas PI3K is downstream of MUC1 activation and negatively regulates TLR5 signaling, it is not responsible for MUC1-induced suppression of TLR5 signaling.
Literature	GENERIF_SUMMARY	PIK3CA mutation was significantly associated with shorter relapse-free survival in stage II/III patients, and shorter disease-specific survival in all patients.
Literature	GENERIF_SUMMARY	Phosphoinositide 3-kinase-independent non-genomic signals transit from the androgen receptor to Akt1 in membrane raft microdomains
Literature	GENERIF_SUMMARY	PIK3CA mutations are mutually exclusive with PTEN loss in diffuse large B-cell lymphoma
Literature	GENERIF_SUMMARY	Coexistent mutations/epigenetic inactivations in PI3K/AKT pathway may be responsible for the unusually aggressive course of ACC.
Literature	GENERIF_SUMMARY	oncogenic PIK3CA mutations contribute to the pathogenesis of skin tumors lacking malignant potential
Literature	GENERIF_SUMMARY	There was significant difference in survival between the lung cancer patients with PIK3CA normal copy number and the patients with PIK3CA amplification
Literature	GENERIF_SUMMARY	PIK3CA has an important role in human lung carcinogenesis, possibly by promoting malignant transformation.
Literature	GENERIF_SUMMARY	IGF-1 plays a role in potentiating platelet aggregation by complementing G(i)- but not G(q)-signaling pathways via PI3-K p110alpha.
Literature	GENERIF_SUMMARY	plexin-B1 promotes endothelial cell motility through RhoA and ROK by regulating the integrin-dependent signaling networks that result in the activation of PI3K and Akt
Literature	GENERIF_SUMMARY	Data suggest that PI3K-C2beta and AKT are epistatic to intersectin, and show that ITSN, independent of its role in endocytosis, regulates a critical cellular signaling pathway necessary for cell survival.
Literature	GENERIF_SUMMARY	The three most common tumour-derived alleles of p110alpha were shown to potently activate phosphoinositide 3-kinase signalling in human epithelial cells.
Literature	GENERIF_SUMMARY	Engagement of NKG2D by itself is sufficient to stimulate the formation of the natural

Main Category	Category	Terms
		killer cells immunological synapse (NKIS), with recruitment of NKG2D to the center synapse.
Literature	GENERIF_SUMMARY	High levels of p-AKT expression occurred independently of the presence of PTEN or PIK3CA mutations in endometrial cancer.
Literature	GENERIF_SUMMARY	In trastuzumab-treated breast cancer patients combined analysis of PTEN and PIK3CA identified twice as many patients at increased risk for progression compared to PTEN alone.
Literature	GENERIF_SUMMARY	rotavirus-induced PI3K activation causes regulation of integrin expression in intestinal cells, leading to prolonged adherence of infected cells to collagen and increased virus production.
Literature	GENERIF_SUMMARY	exon 9 PIK3CA mutations are typical of infiltrating lobular carcinomas
Literature	GENERIF_SUMMARY	The C allele of rs361072 attenuates insulin resistance in superobese European children.
Literature	GENERIF_SUMMARY	The results presented here indicate that actin reorganization through FAK/PI3-K/Rac-1 activation operates in various human cancer cell systems supporting a functional role for FAK/PI-3K/Rac1/actin signaling in controlling cell motility.
Literature	GENERIF_SUMMARY	An impaired control of glycogen synthase kinase-3beta activity by insulin receptor-mediated signalling plays a role in the pathogenesis of AD, facilitating tau protein phosphorylation and neurofibrillary tangle formation.
Literature	GENERIF_SUMMARY	Using both direct genomic DNA sequencing and novel mutant-enriched sequencing methods developed specifically for the 3 hot-spot mutations (H1047R, E545K and E452K) of PIK3CA, we detected 5 mutations of PIK3CA in the 24 pharyngeal cancers (20.8%).
Literature	GENERIF_SUMMARY	A higher incidence of PIK3CA alterations and the possible synergistic effect of these alterations and BRAF mutations suggest their major role in Middle Eastern papillary thyroid neoplasms.
Literature	GENERIF_SUMMARY	Sexual dimorphisms and tissue specific factors might directly or indirectly influence the occurrence of PI3KCA cancer alleles.
Literature	GENERIF_SUMMARY	These data provide the first evidence that IL-18 and FN stimulate each other's expression

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Literature	GENERIF_SUMMARY	Phosphatidylinositol 3-kinase-dependent membrane recruitment of Rac-1 and p47phox is critical for alpha-platelet-derived growth factor receptor-induced production of reactive oxygen species.
Literature	GENERIF_SUMMARY	different PIK3CA mutations differentially contribute to breast cancer transformation
Literature	GENERIF_SUMMARY	study reports a 3.0 angstrom resolution structure of a complex between p110alpha and a polypeptide containing the p110alpha-binding domains of p85alpha; results suggest specific mechanisms for the effect of oncogenic mutations in p110alpha and p85alpha
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with myometrial invasion and tumors with PIK3CA mutations in exon 20 are frequently high-grade, invasive endometrial carcinomas.
Literature	GENERIF_SUMMARY	PIK3CA mutations in HNSCC are likely to be oncogenic and may significantly contribute to HNSCC carcinogenesis and pave attractive target for therapeutic prevention.
Literature	GENERIF_SUMMARY	In this study, we were unable to find a clear association between PIK3CA mutations and gene amplifications, nor with tumor histological subtypes or staging
Literature	GENERIF_SUMMARY	somatic PIK3CA mutations contribute to the frequent activation of the PI3K/AKT pathway in carcinomas of the biliary tract and liver
Literature	GENERIF_SUMMARY	PIK3CA mutations are common in invasive ductal carcinomas of the breast and PIK3CA exon 20 mutation is an independent risk factor for poor prognosis in breast cancer patients.
Literature	GENERIF_SUMMARY	analysis of PIK3CA mutations in Saudi Arabian colorectal cancer patience
Literature	GENERIF_SUMMARY	Poliovirus simultaneously activates the phosphatidylinositol 3-kinase (PI3K)/Akt survival signaling pathway in these cells, limiting the extent of JNK activation and thereby cell death.
Literature	GENERIF_SUMMARY	genetic alterations of KRAS and PIK3CA may play equivalent roles in endometrial carcinogenesis
Literature	GENERIF_SUMMARY	Mutations in the PIK3CA is associated with colon cancers
Literature	GENERIF_SUMMARY	In exon 9, a somatic mutation was found in two patients (2.2%) and in two cell lines;

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		mutations included three E545K (G1633A) mutations and one E545Q (G1633C) mutation, in exon 20, no mutation was observed in our esophageal cancer patients
Literature	GENERIF_SUMMARY	The demonstration that p53 binds directly to the PIK3CA promoter and inhibits its activity identifies a novel mechanism whereby these two mediators regulate cellular functions.
Literature	GENERIF_SUMMARY	PI3Kalpha and -beta present distinct activation requirements and kinetics in G(1) phase, with a selective action of PI3Kalpha at the G(0)/G(1) phase transition
Literature	GENERIF_SUMMARY	point mutations in the p110alpha subunit of class IA PI3K confer factor independence to hematopoietic cells in vitro and leukemogenic potential in vivo, but have lower transforming activity than a deregulated class III receptor tyrosine kinase.
Literature	GENERIF_SUMMARY	identified and characterized the PIK3CA 5' upstream transcriptional regulatory region and confirmed that PIK3CA is transcriptionally regulated through NF-kappaB pathway
Literature	GENERIF_SUMMARY	Our data provide evidence that PIK3CA and BRAF contribute to the tumorigenesis of IPMN/IPMC, but at a lower frequency than KRAS.
Literature	GENERIF_SUMMARY	high frequency of mutations in the PIK3CA, HRAS and KRAS genes leads us to believe that dysregulation of the phosphatidylinositol 3-kinase or Ras pathway is significant for the development and progression of penile carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA mutation was associated with a decreased rate of node-positive disease, particularly among ER-positive tumors.
Literature	GENERIF_SUMMARY	PIK3CA is rarely mutated in diffuse large B-cell lymphoma
Literature	GENERIF_SUMMARY	structure of the complex between the catalytic subunit of PI3Kalpha, p110alpha, & a portion of its regulatory subunit, p85alpha reveals the majority of the oncogenic mutations occur at the interfaces between p110 domains & between p110 & p85 domains
Literature	GENERIF_SUMMARY	results provide the first in vivo evidence for p110-isoform selectivity in endothelial PI3K signalling during angiogenesis
Literature	GENERIF_SUMMARY	FGFR3 and PIK3CA germline mutations can be excluded as an underlying genetic basis,

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		therefore alternative mechanisms have to contribute to familial Seborrheic Keratoses.
Literature	GENERIF_SUMMARY	PIK3CA mutation is significantly associated with key molecular events in colorectal cancer, and MGMT loss likely contributes to the development of PIK3CA G>A mutation
Literature	GENERIF_SUMMARY	Overexpression of PI3K and Akt is associated with malignant rhabdoid tumor
Literature	GENERIF_SUMMARY	KRAS mutations and PI3KCA/PTEN deregulation significantly correlate with resistance to cetuximab.
Literature	GENERIF_SUMMARY	PIK3CA mutations, in comparison with PTEN loss and AKT1 mutations, were associated with significantly less and inconsistent activation of AKT and of downstream PI3K/AKT signaling in tumors and cell lines
Literature	GENERIF_SUMMARY	Exon 20 PIK3CA mutations are relatively frequent in Her-2+ breast tumors and shorten survival, whereas neither exons 9 and 20 mutations seem related with ""triple negative"" breast carcinomas"
Literature	GENERIF_SUMMARY	study identified a potential hotspot for resistance mutations (I800), a drug-sensitizing mutation (L814C), and a surprising lack of resistance mutations at the ""gatekeeper"" residue"
Literature	GENERIF_SUMMARY	RNA interference-mediated knockdown of PIK3CA inhibited colony formation of cell lines with PIK3CA mutations or gains but was not effective in PIK3CA wild-type cells. PIK3CA mutations or gains are present in a subset of lung cancers
Literature	GENERIF_SUMMARY	TGFbeta activates PI3K to downregulate PTEN for enhancement of cell proliferation that is independent of SMAD proteins
Literature	GENERIF_SUMMARY	BRAF, KRAS and PIK3CA mutations occur prior to malignant transformation demonstrating that these oncogenic alterations are primary genetic events in colorectal carcinogenesis
Literature	GENERIF_SUMMARY	Inhibitors that target exclusively the cancer-specific mutants of p110 alpha constitute an important goal and challenge for current drug development.
Literature	GENERIF_SUMMARY	Higher BRCA1 mRNA expression is significantly correlated with advanced disease and ERBB2 overexpression in breast cancers.
Literature	GENERIF_SUMMARY	In tumors with co-occurring mutations in

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		multiple components of the PI3K pathway, selective inhibition of the alpha isoform of p110 is an attractive therapeutic strategy, especially for late-stage tumors.
Literature	GENERIF_SUMMARY	These data establish a new function for the NBS1 protein as a regulator of PI3K activity via SFK members.
Literature	GENERIF_SUMMARY	strong genetic evidence supporting a role of the PI3K/AKT signaling pathway in the tumorigenesis of pituitary tumors, particularly the invasive types
Literature	GENERIF_SUMMARY	Mutant-specific differences in the catalytic cleft and activation loop behavior suggest that structure-based mutant-specific inhibitors can be designed for PIK3CA-positive breast cancers.
Literature	GENERIF_SUMMARY	our data reveal a novel function of p110alpha in medulloblastoma growth and survival.
Literature	GENERIF_SUMMARY	ARF1 regulates epidermal growth factor-dependent breast cancer cell growth and invasion during cancer progression by controlling the activation of the phosphatidylinositol 3-kinase pathway
Literature	GENERIF_SUMMARY	JAK2, EGFR and PI3KCA hot spot mutations are uncommon in endocrine tumors
Literature	GENERIF_SUMMARY	The result showed that the mutation rate of PIK3CA in nasopharyngeal carcinomas (n = 73) was 9.6%, whereas both BRAF (n = 65) and RAS (n = 45) were wild type in every specimen with adequate DNA for analysis.
Literature	GENERIF_SUMMARY	data showed a higher incidence of PIK3CA mutations in Malaysian breast cancers compared to colorectal & nasopharyngeal tumor tissue; findings also indicate PIK3CA mutations play a pivotal role in activation of the PI3 K signaling pathway in breast cancer
Literature	GENERIF_SUMMARY	PIK3CA amplification and single nucleotide polymorphism may be a marker predicting ovarian cancer response and resistance to chemotherapy.
Literature	GENERIF_SUMMARY	inhibitors of the PI3K-mTOR pathway may be active in cancers with PIK3CA mutations and, when combined with MEK inhibitors, may effectively treat KRAS mutated lung cancers
Literature	GENERIF_SUMMARY	Zinc sulfate stimulation could activate the EGFR and the PI3K/Akt signaling pathway, further leading to upregulation of ICAM-1 expression in human airway epithelial cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	results suggest that FGFR3 and PIK3CA mutations are involved in the pathogenesis of solar lentigo and occurrence of these mutations in both SL and SK suggests a common genetic basis.
Literature	GENERIF_SUMMARY	a selective downregulation of the p110alpha catalytic isoform in pregnancy
Literature	GENERIF_SUMMARY	Missense mutations of PIK3CA are identified in 81% of cervical cancer cell lines, with mutation rate higher in older women.
Literature	GENERIF_SUMMARY	genetic variations in PIK3CA may be associated with survival in esophageal cancer
Literature	GENERIF_SUMMARY	activation of PIK3CA may be an important molecular event contributing to the maintenance of OSE transformation initiated by oncogenes such as K-ras
Literature	GENERIF_SUMMARY	findings suggest GSK3beta is an important effector of mutant PIK3CA, and that lithium, an FDA-approved therapy for bipolar disorders, has selective antineoplastic properties against cancers that harbor these mutation
Literature	GENERIF_SUMMARY	PIK3CA mutations in colorectal cancer are associated with clinical resistance to EGFR-targeted monoclonal antibodies.
Literature	GENERIF_SUMMARY	In this study they investigated the effect that the two major classes of PIK3CA mutation have on the biochemical properties of p110alpha.
Literature	GENERIF_SUMMARY	PIK3CA mutations are frequent events in endometrial carcinomas of any histological type
Literature	GENERIF_SUMMARY	Compared with patients with PIK3CA wild-type tumors, those with PIK3CA-mutated tumors experienced an increase in colon cancer-specific mortality according to univariate analysis.
Literature	GENERIF_SUMMARY	Amplifications of 3q26.32 harboring the oncogene PIK3CA were associated with poor prognosis and segregated with the aggressive transcriptional cluster.
Literature	GENERIF_SUMMARY	In the current study of primary uterine serous carcinoma (USC) tumors, oncogenic mutations of the PIK3CA gene were seen in 15% of USC cases
Literature	GENERIF_SUMMARY	There was an association of PIK3CA mutation (kinase domain) with increased androgen receptor levels in breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	low frequency of PIK3CA mutations in CLL suggests that such mutations are unlikely to cause any major transforming activity in CLL in general.
Literature	GENERIF_SUMMARY	Frequent activating mutations of PIK3CA is associated with ovarian clear cell carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA mutations are not a major determinant of resistance to the epidermal growth factor receptor inhibitor cetuximab in metastatic colorectal cancer.
Literature	GENERIF_SUMMARY	GILZ1 and SGK1 provide a physical and functional link between the PI3K- and Raf-1-dependent signaling modules
Literature	GENERIF_SUMMARY	PIK3 kinase expression is higher in metastatic melanoma.
Literature	GENERIF_SUMMARY	This study for the first time uncovered PIK3CA alterations in small cell lung cancer
Literature	GENERIF_SUMMARY	complete concordance of PIK3CA and AKT1 mutations in matched samples of invasive and in situ tumor indicates that these mutations occur early in breast cancer development
Literature	GENERIF_SUMMARY	Mutation E17K/AKT1 was not detected in the 24 samples of retinoblastoma analyzed. K-RAS mutations were identified in two samples. There were no mutations in any of the other genes analyzed.
Literature	GENERIF_SUMMARY	RNA interference-mediated silencing of the phosphatidylinositol 3-kinase catalytic subunit attenuates growth of ovarian cancer.
Literature	GENERIF_SUMMARY	oncogenic properties of KRAS and BRAF but not NRAS, HRAS, and PIK3CA contribute to the tumorigenesis of periampullary and ampullary tumors
Literature	GENERIF_SUMMARY	results suggest that genetic alterations in the PI3K pathway are common in prostate cancer, and occur mainly through PIK3CA amplification and PTEN hemizygous or homozygous deletion
Literature	GENERIF_SUMMARY	Absence of PIK3CA/AKT1 mutations in one specimen may not reflect the status at other sites because these mutations arise during progression.
Literature	GENERIF_SUMMARY	PI3-K accelerates the short course of GPIb centralisation for two PAR signal pathways during the course of thrombin receptor activation in platelets.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Phosphatidylinositol 3-kinase (PI3K) signaling pathway occurs frequently in human cancer; PTEN tumor suppressor or PIK3CA oncogene mutations both direct PI3K-dependent tumorigenesis in breast cancers
Literature	GENERIF_SUMMARY	Findings show that PI3K inhibition diminished hepatoblastoma cell growth being accompanied by reduced AKT and GSK-3beta phosphorylation.
Literature	GENERIF_SUMMARY	PIK3CA (P=0.01) mutations predicted reduced progress free survival in response to cetuximab salvage therapy in patients with metastatic colorectal cancer .
Literature	GENERIF_SUMMARY	Data show that no mutations were detected in PIK3CA and CTNNB1/beta-catenin hotspot sequences.
Literature	GENERIF_SUMMARY	SHIP gene can inhibit K562 cell proliferation and promote cell apoptosis via inactivating PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with resistance to HER2-targeted agents.
Literature	GENERIF_SUMMARY	Investigated PI3K/AKT genetic alterations such as PIK3CA amplification, PIK3CA mutation, PTEN protein loss. PIK3CA amplification was seen in 54 of 152 epithelial ovarian cancer (EOC) cases analyzed.
Literature	GENERIF_SUMMARY	Nicotine increases PPARbeta/delta gene expression through alpha7 nAChR-mediated activation of PI3K/mTOR signals that inhibit AP-2alpha protein expression and DNA binding activity to the PPARbeta/delta gene promoter.
Literature	GENERIF_SUMMARY	Positive prognostic significance of PIK3CA mutations in breast cancer.
Literature	GENERIF_SUMMARY	a link between PI3K(p110alpha) and atrial fibrillation
Literature	GENERIF_SUMMARY	Associations between PIK3CA expression and clinicopathological variables, molecular classes, and patients' outcome were investigated. PIK3CA is an oncogenic biomarker associated with poor prognosis in breast cancer.
Literature	GENERIF_SUMMARY	Mutation analysis showed activation mutations in the PIK3CA gene in tumors that also had DNA copy number increases in the PIK3CA locus, suggesting an additive effect of coexisting activating amino acid substitution and dosage increase from amplification.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	association of PIK3CA gene amplification with worse prognosis in nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	PI3K pathway alterations are common in bladder cancer.
Literature	GENERIF_SUMMARY	VDR overexpression in colorectal cancer is independently associated with PIK3CA and KRAS mutations.
Literature	GENERIF_SUMMARY	The structural and biochemical data suggest a previously undescribed mechanism for mutational activation of PIK3CA that involves perturbation of its interaction with the cellular membrane.
Literature	GENERIF_SUMMARY	phosphoinositide 3-kinase isoforms alpha and beta have roles in glycoprotein VI-induced platelet signaling and thrombus formation
Literature	GENERIF_SUMMARY	PIK3CA mutations mostly began to develop at the stage from the DIN1B to the carcinoma in situ (DCIS), which is a late event of breast oncogenesis.
Literature	GENERIF_SUMMARY	Age-associated changes in transgenic murine heart lead to suppression of the p110 alpha isoform of phosphoinositide 3-kinase activity that modifies cardiac aging.
Literature	GENERIF_SUMMARY	Rare mutations in the PIK3CA gene contributing to aggressive endometrial cancer - has been shown.
Literature	GENERIF_SUMMARY	PIK3CA KD mutation was a favorable prognostic factor for relapse-free survival in response to neoadjuvant endocrine therapy for estrogen receptor positive breast cancer.
Literature	GENERIF_SUMMARY	Three of five stucco keratosis samples revealed a PIK3CA mutation(E542K, E545K), but no FGFR3 mutation was found. In contrast, both dermatosis papulosa nigra samples harboured an FGFR3 mutation (R248C, S249C) but no PIK3CA mutation.
Literature	GENERIF_SUMMARY	In hereditary breast cancer, PIK3CA mutations were found only in the BRCA2 group.
Literature	GENERIF_SUMMARY	PI3K physically interacts with the glucocorticoid receptor (GR)through two putative PI3K recruitment consensus YxxM binding motifs in the GR suggesting that some functions regulated by this receptor might occur through kinase interaction.
Literature	GENERIF_SUMMARY	Aurora-A, via activating Akt, stimulated nuclear factor-kappaB signaling pathway to promote cancer cell survival.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	approximately two-thirds of papillomas are driven by mutations in the PI3CA/AKT pathway
Literature	GENERIF_SUMMARY	PIK3CA mutations may have a role in recurrence of rectal cancer
Literature	GENERIF_SUMMARY	Tumors expressing the p110alpha helical domain mutant showed a increase in tumor cell intravasation into the blood and tumor cell extravasation into the lung compared with tumors expressing wild-type p110alpha or the kinase domain mutant.
Literature	GENERIF_SUMMARY	Leptin plays an important role in papillary thyroid carcinoma pathogenesis through PI3K/AKT pathway via Ob-R and is a potential prognostic marker associated with an aggressive phenotype and poor disease-free survival.
Literature	GENERIF_SUMMARY	PI3K/AKT mutations may not play a major role in multiple myeloma.
Literature	GENERIF_SUMMARY	These findings demonstrate the importance of cross-talk between the PI3K/Akt and beta-catenin pathways in medulloblastoma
Literature	GENERIF_SUMMARY	PI3-kinase regulates the thrombin-induced actin cytoskeleton reconstitution in platelets by the RhoA-mDia1 pathway.
Literature	GENERIF_SUMMARY	Expression of PIK3CA was higher in the hepatocellular carcinoma tissue than in the corresponding adjacent tissue.
Literature	GENERIF_SUMMARY	mutations in critical genes within the PI3-kinase (PI3K) pathway are not functionally equivalent, and that other cooperative genetic events may be necessary to achieve oncogenic PI3K pathway activation in cancers that contain the AKT1 E17K mutation.
Literature	GENERIF_SUMMARY	Data show that similar responses were seen in cancer cells with wild-type or activated mutant PI3K genes treated with p110alpha/delta or p110alpha/beta/delta inhibitors in cell viability assays.
Literature	GENERIF_SUMMARY	PIK3CA mutations were detected in 13/46 (28%) plasma-derived and 10/46 (21%) serum-derived cfDNA samples from metastatic breast cancer patients.
Literature	GENERIF_SUMMARY	The results showed that inhibition of PI3K and Akt-2 was associated with reduced SREBP-1 activation by genotype 3a hepatitis c virus core protein.
Literature	GENERIF_SUMMARY	TLR2 cooperates with MyD88, PI3K, and Rac1 in lipoteichoic acid-induced cPLA2/

Main Category	Category	Terms
		COX-2-dependent airway inflammatory responses
Literature	GENERIF_SUMMARY	PIK3CA alterations specifically correlate with clinicopathological characteristics of Bulgarian patients with breast cancer.
Literature	GENERIF_SUMMARY	We evaluated the presence of mutations in PIK3CA, AKT1, AKT2, AKT3, PTEN, and PDPK1 genes in 83 papillary thyroid carcinomas
Literature	GENERIF_SUMMARY	Data show that PTPN11 mutations causing Leopard syndrome (LS) facilitate EGF-induced PI3K/AKT/GSK-3beta stimulation through impaired GAB1 dephosphorylation, resulting in deregulation of a novel signaling pathway that could be involved in LS pathology.
Literature	GENERIF_SUMMARY	The p110alpha selectively mediates activation of the Tyr kinase Pyk2 to regulate barrier function.
Literature	GENERIF_SUMMARY	Mutant PIK3CA interferes with apoptosis downstream of death receptor-signal complex, allowing caspase-8 activation without apoptosis. Cleavage of ROCK-1 by caspase-8 triggers amoeboid shape and invasiveness of HCT116 cells in response to TRAIL and CD95L.
Literature	GENERIF_SUMMARY	PIK3CA mutations in gastric carcinoma may be a signature of cancer type
Literature	GENERIF_SUMMARY	No hotspot mutations were detected in any of the samples. PIK3CA hotspot mutations are unlikely to play a major role in the pathogenesis of OSCC in the Greek population.
Literature	GENERIF_SUMMARY	Although the PIK3CA gene signature was not associated with prognosis in ER- and HER2+ breast cancer, it could identify better clinical outcomes in ER+/HER2- disease.
Literature	GENERIF_SUMMARY	Studies indicate that the high-frequency somatic mutations of PIK3CA observed in different human cancers further strengthen the notion that PI3Ks are optimal targets for therapeutic intervention.
Literature	GENERIF_SUMMARY	Direct sequencing analysis of endometrial DNA analysis did not reveal any activating somatic mutations in either PI3KCA or AKT1 gene, indicating that these mutations are rarely associated with endometriosis in South Indian women.
Literature	GENERIF_SUMMARY	Decreased expression of PTEN together with overexpression of PI3K and AKT contribute to gastric cancer progression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the relevance of PIK3CA mutations to a wide range of common human tumor types (Review)
Literature	GENERIF_SUMMARY	Results suggest that mutation of PIK3CA is an early event in breast cancer that is more likely to play a role in breast tumor initiation than in invasive progression. A role for exon 9 mutations in the progression of a subset of DCIS cannot be excluded.
Literature	GENERIF_SUMMARY	Activation of the PI3K/AKT signalling pathway in non-melanoma skin cancer is not mediated by oncogenic PIK3CA and AKT1 hotspot mutations.
Literature	GENERIF_SUMMARY	reports the low frequency of PIK3CA and B-RAF mutations in astrocytomas, despite the presence of activated ERK and AKT proteins.
Literature	GENERIF_SUMMARY	PIK3CA mutations may have a role in KRAS and BRAF wild type colorectal cancer
Literature	GENERIF_SUMMARY	Oncogenic mutations of PI3Kalpha result in a constitutively activated enzyme that triggers downstream pathways that increase tumor aggressiveness and survival (Review)
Literature	GENERIF_SUMMARY	PIK3CA HRMA assay was evaluated by performing repeatability, sensitivity, and comparison with DNA sequencing studies and was further validated in 129 formalin-fixed paraffin-embedded breast tissue samples
Literature	GENERIF_SUMMARY	if KRAS is not mutated, assessing BRAF, NRAS, and PIK3CA exon 20 mutations (in that order) gives additional information about the efficacy of cetuximab plus chemotherapy in metastatic colorectal cancer.
Literature	GENERIF_SUMMARY	PIK3CA rs7651265 GG genotype was associated strongly with colon and rectal cancer.
Literature	GENERIF_SUMMARY	mediates activation of TGF-beta2 promoter response elements leading to high cytokine expression in cancer cells
Literature	GENERIF_SUMMARY	Genetic ablation of the mutant PIK3CA allele in MEK inhibitor-resistant cells restores MEK pathway sensitivity, re-expression of mutant PIK3CA reinstates resistance, highlighting the importance of this mutation in resistance to therapy in cancers.
Literature	GENERIF_SUMMARY	Data show that DLBS1425 significantly reduced phosphoinositide-3 (PI3)-kinase/ protein kinase B (AKT) signalling by

Main Category	Category	Terms
		reducing PI3K transcript level and subsequent reduction in AKT phosphorylation.
Literature	GENERIF_SUMMARY	Data show that the mutations weaken an inhibitory interaction between p85alpha and p110alpha.
Literature	GENERIF_SUMMARY	Sodium butyrate increased PTEN expression, promoted the expression of MUC2 and induced the differentiation of gastric cancer cells through the PTEN/PI3K signalling pathway.
Literature	GENERIF_SUMMARY	study supports that mutations in the PIK3CA gene make at least a minor contribution to OSCC tumorigenesis
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with poor trastuzumab response in HER2-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	The levels of total Akt and phosphorylated Akt (active Akt) are increased in E6-AP overexpressing prostate gland and LNCaP cells suggesting that E6-AP regulates the PI3K-Akt signaling pathway.
Literature	GENERIF_SUMMARY	STC1 induction by thyroid hormone depends on both TRbeta and PI3K activation.
Literature	GENERIF_SUMMARY	Findings suggest that there is p70S6K-mediated, cross-inhibitory regulation between the MEK/ERK and PI3K/mTOR pathways, in which each contribute to the maintenance of the self-renewal and tumorigenic capacity of glioblastoma glioblastoma stem-like cells
Literature	GENERIF_SUMMARY	Mutations in PIK3CA is associated with squamous cell carcinomas of the lung.
Literature	GENERIF_SUMMARY	PIK3CA mutants overall showed a significantly longer time to first recurrence than wild type cases
Literature	GENERIF_SUMMARY	the p110alpha and beta isoforms of class IA PI3K are both required for the proinflammatory response to flagellin via TLR5
Literature	GENERIF_SUMMARY	Up-regulation of PIK3CA may promote the metastasis of gastric cancer through aberrant activation of PI3K/Akt.
Literature	GENERIF_SUMMARY	cIAP2 upregulated by E6 via EGFR/PI3K/AKT cascades may contribute to cisplatin resistance, revealing that the EGFR or PI3K inhibitor combined with cisplatin may improve the chemotherapeutic efficacy in HPV-infected lung cancer.

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Literature	GENERIF_SUMMARY	the PI3K/AKT signaling pathway is aberrantly active and plays a critical role for the cell cycle progression in human pleural mesothelioma cells.
Literature	GENERIF_SUMMARY	findings indicate that somatic mutations in KRAS and PIK3CA but not BRAF oncogenes are closely associated with the development of cholangiocarcinoma in Chinese population
Literature	GENERIF_SUMMARY	Combined inhibition of PI3K and mTOR, which activates autophagy without activating Akt, cooperated with inhibition of autophagy to cause glioma cells to undergo apoptosis.
Literature	GENERIF_SUMMARY	Although the mechanism of load-induced mTORC1 activation remains to be determined, it is clear that it does not require classical growth factor signaling.
Literature	GENERIF_SUMMARY	Data show that of primary tumors 64% were mutant for FGFR3, 11% for RAS, 24% for PIK3CA, and 26% for p53.
Literature	GENERIF_SUMMARY	Expressions of Akt1, Akt2 and PI3K were decreased in ovarian epithelial cancer cells transduced with PTEN.
Literature	GENERIF_SUMMARY	Common genetic variation in PTEN, PIK3CA, AKT1, MLH1, or MSH2 was not statistically significantly associated with endometrial cancer.
Literature	GENERIF_SUMMARY	Data show that among 181 CRC patients, stratified by microsatellite instability status, DNA sequence changes were identified in KRAS (32%), BRAF (16%), PIK3CA (4%), PTEN (14%) and TP53 (51%).
Literature	GENERIF_SUMMARY	PI3K signaling pathway is activated by PIK3CA mRNA overexpression and copy gain in prostate tumors.
Literature	GENERIF_SUMMARY	Activation of the PI3K/AKT in addition to MAPK/ERK signaling pathways may underlie biological aggressiveness in pilocytic astrocytoma (PA). Specifically, it may mediate the increased proliferative activity observed in histologically anaplastic PA.
Literature	GENERIF_SUMMARY	In hyperglycemic leptin-deficient Lep(ob/ob) transgenic mice, leptin acutely and potently improves glucose metabolism, before any change of body fat mass, via a mechanism involving the p110alpha and -beta isoforms of phosphatidylinositol-3-kinase (PI3K).
Literature	GENERIF_SUMMARY	down-regulation of PI3K-C2alpha may be a feature of type 2 diabetes.

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Literature	GENERIF_SUMMARY	Mutations in PIK3CA, KRAS and NRAS were exclusive to tumors of uterine origin and age-adjusted Cox proportional hazards modeling associated advanced age, stage and TP53 mutations with decreased survival in the uterine subset.
Literature	GENERIF_SUMMARY	PIL3Ca mutations may not only be a prognostic factor, but may also be predictive of preoperative radiation benefit.
Literature	GENERIF_SUMMARY	Data support the idea that PI3Kalpha plays an important role in the metastatic process and suggest a more informed strategy for selecting drugs worthy of further development for clinical application.
Literature	GENERIF_SUMMARY	Results suggest that low magnitude stretch can induce cPLA(2) phosphorylation through the MEK/ERK and PI3K-Akt pathways, independently.
Literature	GENERIF_SUMMARY	Expression of matrix metalloproteinase-13 is controlled by IL-13 via PI3K/Akt3 and PKC-delta in normal human dermal fibroblasts.
Literature	GENERIF_SUMMARY	the PI3K/PAK1/ERK signaling pathway has a role in LPA-stimulated breast cancer cell migration
Literature	GENERIF_SUMMARY	PIK3CA mutation and the Wnt pathway have roles in ERalpha-positive breast cancer
Literature	GENERIF_SUMMARY	EGFR, PI3Kp110 and PTEN expression is a common feature of neuroblastoma
Literature	GENERIF_SUMMARY	Here, the authors show that VP11/12 is required for virus-induced activation of PI3K-Akt signaling in herpes simplex virus 1-infected Jurkat T cells and primary fibroblasts.
Literature	GENERIF_SUMMARY	study identified one PIK3CA missense mutation in pancreatic neuroendocrine tumors
Literature	GENERIF_SUMMARY	Data show that the effects of PI3K-Akt signalling on IL-10 responses were mediated at least in part by GSK3.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with drug resistance in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	analysis of somatic PIK3CA (p110alpha) mutations within primary endometrial carcinomas
Literature	GENERIF_SUMMARY	PI3K/AKT/mTOR signalling pathway activation is a common event in high-risk neuroblastoma.

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Literature	GENERIF_SUMMARY	Data suggest that the PI3K/Akt pathway is involved in apoptotic function in PKHD1-silenced cells, and PI3K/Akt inhibition correlates with upregulation of NF-kappaB activity.
Literature	GENERIF_SUMMARY	The presence of activating mutations in PIK3CA specifically in gallbladder cancer has clinical implications in both the diagnosis of this cancer type, as well as the potential utility of targeted therapies such as PI3 kinase inhibitors.
Literature	GENERIF_SUMMARY	The importance of PIK3K/PAK in neoplasm transformation and progression, their disregulation detected in oral squamous cell carcer may provide early diagnosis and aid in the design of new therapies.
Literature	GENERIF_SUMMARY	PIK3CA mutations were rare in colorectal laterally-spreading tumors.
Literature	GENERIF_SUMMARY	Report novel hotspot PIK3CA mutation in Iranian breast cancer patients.
Literature	GENERIF_SUMMARY	PI3K-activating hormones inhibit ROMK by enhancing its endocytosis via a mechanism that involves phosphorylation of WNK1 by Akt1 and SGK1.
Literature	GENERIF_SUMMARY	Phosphatidylinositol 3'-kinase signaling pathway is essential for Rac1-induced hypoxia-inducible factor-1(alpha) and vascular endothelial growth factor expression in hypoxia.
Literature	GENERIF_SUMMARY	demonstrate here that breast cancer cells harbouring PIK3CA mutations are selectively sensitive to mTOR allosteric and kinase inhibitors. However, cells with PTEN loss of function are not sensitive to these drugs
Literature	GENERIF_SUMMARY	Thus, sendai virus infection-mediated apoptosis is temporally regulated by the prevention of XIAP degradation by PI3K.
Literature	GENERIF_SUMMARY	Studies suggest that dual PI3K and mTOR inhibition using NVP-BEZ235 may be an effective novel therapeutic approach in patients with ovarian cancer.
Literature	GENERIF_SUMMARY	Our data demonstrate that PIK3CA is a mediator of collagen I-induced down-regulation of E-cadherin.
Literature	GENERIF_SUMMARY	Results show the mechanism by which EZH2 regulates BRCA1 expression and genomic stability mediated by the PI3K/Akt-1 pathway.
Literature	GENERIF_SUMMARY	Overexpression of PIK3CA was confirmed in

Main Category	Category	Terms
		biopsy washout specimens and in postoperative tissues of papillary thyroid carcinoma, in comparison to macroscopically unchanged thyroid tissue
Literature	GENERIF_SUMMARY	Allele-specific qPCR assays for the most frequent activating mutations in EGFR, KRAS, BRAF and PIK3CA in tumor-positive fine needle cytological aspirates were compared against histological material of primary tumors.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in PI3K gene is not associated with colorectal cancer.
Literature	GENERIF_SUMMARY	Data suggest that glioma cell survival occurs through a PI(3)-kinase/PDK1/PKC-iota/BAD mediated pathway.
Literature	GENERIF_SUMMARY	data indicated that colorectal cancers with KRAS and PIK3CA bi-mutations are more likely to develop into liver metastasis
Literature	GENERIF_SUMMARY	Studies indicate that FoxOs have emerged as an important effector arm of PI3K/Akt signaling by driving multiple apoptotic gene expression.
Literature	GENERIF_SUMMARY	Data suggest that the PI3K/Akt signal transduction pathway is involved in the regulation and release of pro-inflammatory cytokines such as TNF-alpha and plays an important role in the development and progression of ulcerative colitis.
Literature	GENERIF_SUMMARY	CCL5 signaling induces GLI2 through a PI3K-AKT-IkappaBalpha-p65 pathway and requires GLI2 transcriptional activity to modulate IL-6 expression and Ig secretion in vitro and in vivo
Literature	GENERIF_SUMMARY	These results suggest that ERalpha induces the down-regulation of PTEN through PI3K activation in breast cancer cells.
Literature	GENERIF_SUMMARY	controls inflammatory cytokine production in human neutrophils
Literature	GENERIF_SUMMARY	Based on a comparative molecular analysis, we show that activated PIK3CA/AKT is a weaker inducer of cell senescence than is activated RAS.
Literature	GENERIF_SUMMARY	Data show that the Ras/PI3K/Akt pathway can account for Fyn over-expression in cancers.
Literature	GENERIF_SUMMARY	A detachment-induced and PI3K/Akt-mediated resistance to apoptotic effects of TRAIL, is demonstrated.

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Literature	GENERIF_SUMMARY	The PIK3CA H1047R oncogene targets a multipotent progenitor cell and, furthermore, show that this model recapitulates features of human breast tumors with PIK3CA H1047R.
Literature	GENERIF_SUMMARY	PIK3CA mutations were detected in 19 (40%) of primary tumors and 21 (42%) of metastases
Literature	GENERIF_SUMMARY	Data show that hepatocyte growth factor inhibits anoikis of pancreatic carcinoma cells through phosphatidylinositol 3-kinase pathway in which activation of Akt may be involved.
Literature	GENERIF_SUMMARY	Moreover, sustained PI3K/Akt signalling in Aspergillus fumigatus infected cells is an important determinant to resist apoptosis.
Literature	GENERIF_SUMMARY	PIK3CA amplification was frequently found in NSCLC
Literature	GENERIF_SUMMARY	PIK3CA somatic mutations occurred in 39 (19.9%) of 196 endometrial carcinomas. No PIK3CA mutations were found in endometrial hyperplasias. PIK3CA amplification was observed in 24 (12.2%) carcinomas and in 2 (10%) hyperplasias.
Literature	GENERIF_SUMMARY	Inhibition of PI3K reduced rhTRAIL sensitivity independently of the cell line preference for either DR4- or DR5-mediated apoptosis signaling.
Literature	GENERIF_SUMMARY	Findings show that the oncogenic PI3K/Akt signalling pathway participates in maintaining cells in a type II phenotype by excluding CD95 from lipid rafts.
Literature	GENERIF_SUMMARY	Prediction of CH5132799 response on the basis of PIK3CA mutations could enable patient stratification in clinical settings.
Literature	GENERIF_SUMMARY	PIK3CA activating mutations and and PTEN loss is associated with less efficacy of trastuzumab therapy in metastatic breast cancer
Literature	GENERIF_SUMMARY	These studies validate the Kit L575P-activated model of melanoma and establish the PI3K pathway as a dominant signaling pathway downstream of Kit in melanoma.
Literature	GENERIF_SUMMARY	Demonstrate the presence of PIK3CA and AKT1(E17K) mutations from pre-invasive to invasive to metastatic breast cancer tissue.
Literature	GENERIF_SUMMARY	Data show that PI3K dependent phosphorylation of Fbw7 stimulates its ability to ubiquitinate and degrade its substrates.

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Literature	GENERIF_SUMMARY	Retinal pigment epithelial disruption induced by high glucose plus IL-1beta is prevented by erythropoietin through the downstream signaling of JAK2 and PI3K/AKT pathways.
Literature	GENERIF_SUMMARY	Tannerella forsythia epithelial cell entry is dependent on host PI3K signalling, purified BspA protein causes activation of this lipid kinase, and also requires the cooperation of host Rac1 GTPase.
Literature	GENERIF_SUMMARY	Results identify unique sites of PTEN and Akt regulation by means of S-nitrosylation, resulting in an ""on-off"" pattern of control of Akt/PI3K signaling."
Literature	GENERIF_SUMMARY	patients with Non-Hodgkin's lymphoma of mucosa-associated lymphoid tissue that are positive for both PIK3CA and FOXP1 had a worse overall and progression-free survival
Literature	GENERIF_SUMMARY	The present data indicate that activation of the PI3-kinase/PKB/AKT pathway through PIK3CA regulates various transformed phenotypes as well as growth and differentiation of HPV-immortalized cells
Literature	GENERIF_SUMMARY	Investigated PIK3CA mutation in Chinese hepatocellular carcinoma patients.Exons 9 and 20 mutations of PIK3CA gene were detected.Two point mutations in exon 9 were found in only one patient (1/90; 1.11%), no mutation was found in exon 20 in any cases.
Literature	GENERIF_SUMMARY	Studies indicate that PI3K can phosphorylate and convert phosphatidylinositol 4,5-bisphosphate (PIP2) to phosphatidylinositol 3,4,5-trisphosphate (PIP3) that leads to activation of AKT.
Literature	GENERIF_SUMMARY	PIK3CA mutations occur more often in elderly HER2-positive breast cancer patients. Most mutations were at H1947R & E542K. PI3K pathway activation resulting from PIK3CA mutations may lead to drug resistance to lapatinib & trastuzumab.
Literature	GENERIF_SUMMARY	Data suggest that PI3K and PKC are along the same pathway that branches into two separate ones involving each either p38MAP kinase or MEK/ERK.
Literature	GENERIF_SUMMARY	PI3K signaling via p110alpha regulates invadopodia-mediated invasion of breast cancer cells.
Literature	GENERIF_SUMMARY	Beclin1/PI3K-mediated autophagy prevents hypoxia-induced apoptosis in EAhy926 cell line

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Literature	GENERIF_SUMMARY	TGF-beta1 undergo a switch into mesenchymal cells and PI3K/Akt and MAPK/Erk1/2 signaling pathways serve to regulate TGF-beta1-induced epithelial-to-mesenchymal transition of A549 cells.
Literature	GENERIF_SUMMARY	Results indicate that stimulation of the early metastatic steps of motility and invasion by ErbB3 requires activation of the PI3-kinase pathway by the ErbB3 receptor.
Literature	GENERIF_SUMMARY	These findings provide evidence that mutations of the PIK3CA gene occur in the putative precursor lesions of ovarian clear cell adenocarcinoma
Literature	GENERIF_SUMMARY	This paper reports the first purification of a phosphoinositide 3-kinase and shows that the protein is a heterodimer of an 85 kd regulatory subunit that mediates binding to phosphorylated proteins and a 110 kd catalytic subunit.
Literature	GENERIF_SUMMARY	Glycosaminoglycans and glucose prevent apoptosis in 4-methylumbelliferone-treated human aortic smooth muscle cells in a process mediated by Toll-like receptor 4, CD44, and PI3K but not by ERK1/2
Literature	GENERIF_SUMMARY	These findings indicate that the expression of IL-15 up-regulated by IL-6 is associated with MAPKs-ERK1/2 and PI3K-AKT signaling pathways in HaCaT cells.
Literature	GENERIF_SUMMARY	Cells transformed by the p110alpha-H1047R mutant of PI3K show increased tyrosine phosphorylation of Stat3.
Literature	GENERIF_SUMMARY	results show that hotspot mutations in the PI3K, JAK2, FLT3 and NPM1 genes are not common in myelodysplastic syndrome patients.
Literature	GENERIF_SUMMARY	Oncogenic PIK3CA-driven mammary tumors frequently recur via PI3K pathway-dependent and PI3K pathway-independent mechanisms.
Literature	GENERIF_SUMMARY	data confirm that PIK3CA is important to oral squamous cell carcinoma (OSCC) tumorigenesis and can contribute to oncogene activation of the PIK3CA/AKT pathway in OSCC
Literature	GENERIF_SUMMARY	PIK3CA mutations frequently coexist with RAS and BRAF mutations in patients with advanced cancers.
Literature	GENERIF_SUMMARY	PIK3CA mutation in exon 20 is a negative prognostic factor in stage III colon cancer patients. Moreover, this negative effect is not present in stage I and II patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Down-regulation of JAK1 by RNA interference inhibits growth of the lung cancer cell line A549 and interferes with the PI3K/mTOR pathway
Literature	GENERIF_SUMMARY	CREG plays a critical role in protecting the vascular endothelium from apoptosis, and the protective effort of CREG against ECs apoptosis is through the activation of the VEGF/PI3K/AKT signaling pathway
Literature	GENERIF_SUMMARY	FGF2 and EGF affect the motor neuron fate decision in human fetal neural stem cells differently through a fine tuning of the PI3K/AKT/GSK3beta pathway.
Literature	GENERIF_SUMMARY	The OHCI induces mitochondrial oxidative stress, leading to DNA double-strand breaks and elicitation of a phosphoinositol 3-kinase-mediated cellular response in peripheral blood lymphocytes.
Literature	GENERIF_SUMMARY	Accumulation of viral RNA in cells infected with influenza A or B viruses results in PI3K activation.
Literature	GENERIF_SUMMARY	Results demonstrate a PIK3CA mutation simultaneously occuring with a 15-up deletion in KIT exon 11 in gastrointestinal stromal tumors.
Literature	GENERIF_SUMMARY	PIK3CA mutations were uncommonly, but exclusively, seen in tubulovillous adenomas (4/124, 3.2%) and 1/4 (25.0%) tubulovillous adenomas with a focus of cancer
Literature	GENERIF_SUMMARY	PI3 kinase-alpha/gamma-mediated signaling downstream of VEGFR2 activation regulated Akt-dependent survival signals, but it was not required to activate endothelial nitric oxide synthase or to elicit NO production in glomerular endothelial cells.
Literature	GENERIF_SUMMARY	Mutations in PIK3CA were found in 3 of 11 Intraductal tubulopapillary pancreatic neoplasms but not in intraductal papillary pancreatic mucinous neoplasms
Literature	GENERIF_SUMMARY	These findings, representing the first genomewide search for variants influencing human protein phosphorylation, provide useful information about the PI3K/AKT/mTOR pathway.
Literature	GENERIF_SUMMARY	FASN inhibition not only abrogates lipogenesis, which is indispensable for cancer growth, but also downregulates oncogenic PI3K signaling.
Literature	GENERIF_SUMMARY	Data show that siRNA mediated targeting of PIK3CA may specifically knockdown the

Main Category	Category	Terms
		expression of PIK3CA in gastric cancer cells, providing a potential implication for therapy of gastric cancer.
Literature	GENERIF_SUMMARY	Survival, proliferation, leptin production, OB-R isoforms (OB-Ra and OB-Rb), phosphoinositide 3-kinase (PI3k) and BCL-2 expression have been tested after 24h, 48h and 72h of treatment with alpha-linolenic acid.
Literature	GENERIF_SUMMARY	Findings suggest that large HBsAg (LHBs) promotes tumorigenesis of hepatoma cells by triggering a PKCalpha/Raf1 to Src/PI3K/Akt signaling pathway, revealing novel insights into the underlying mechanisms of HBV-associated hepatocarcinogenesis.
Literature	GENERIF_SUMMARY	A frequent PIK3CA mutation appears in a series of rosette-forming glioneuronal tumors.
Literature	GENERIF_SUMMARY	multiple mutations were identified in phosphoinositide-3-kinase (PI3K), catalytic, alpha polypeptide (PIK3CA) (H1047R, E-->lysine at codon 545 [E545K], and H-->proline at codon 701 [H701P]) that were not observed previously in osteosarcoma
Literature	GENERIF_SUMMARY	activating mutation of PIK3CA, loss of PTEN, or IGF1R expression have a role in round cell transformation
Literature	GENERIF_SUMMARY	The overall prevalence of PIK3CA mutations was 23% (7/31) for both tumor types (22% in glioblastoma multiform, and 25% in anaplastic astrocytoma).
Literature	GENERIF_SUMMARY	Common germline variation in PIK3CA does not have a strong influence on the risk of breast cancer
Literature	GENERIF_SUMMARY	the activation of PI3K and MEK pathways is not the only mechanism of EGFR-resistance
Literature	GENERIF_SUMMARY	PIK3CA exon 20 mutations may be a potential biomarker for resistance to anti-EGFR monoclonal antibodies in KRAS wild-type metastatic colorectal cancer. [Review and meta-analysis]
Literature	GENERIF_SUMMARY	Addition of short (6 to 16 amino acids) peptide sequences to the N-terminus of p110alpha induces a gain of function includeds signaling, oncogenic transformation and stimulation of cell growth.
Literature	GENERIF_SUMMARY	PIK3R1, like PIK3CA, is a potential therapeutic target in Glioblastoma multiforme and that it also influences tumor cell growth and motility

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	In endometriotic stromal cells, overactivation of the phosphatidylinositol 3-kinase/AKT signaling pathway contributes to the reduced expression of the decidua-specific gene, IGFBP1, potentially through reduced levels of nuclear FOXO1.
Literature	GENERIF_SUMMARY	combination of low CEN7 copy number, PTEN loss, and PI3KCA gain may be useful for identifying NSCLC patients unlikely to benefit from treatment with EGFR (TKIs), specifically in wild-type EGFR cases
Literature	GENERIF_SUMMARY	proliferation of human IVD cells is regulated by exogenous and autocrine growth factors mainly via the MEK/ERK and PI-3K/Akt pathways.
Literature	GENERIF_SUMMARY	The prognostic impact of Akt (Akt1) phosphorylated at threonine308 and serine473, Akt2, Akt3, PI3K and PTEN, alone and in coexpression with ER and PgR in non-gastrointestinal stromal tumor soft tissue sarcomas.
Literature	GENERIF_SUMMARY	PIK3CA mutations and loss of ARID1A protein expression are early events in the development of cystic ovarian clear cell adenocarcinoma.
Literature	GENERIF_SUMMARY	Data indicate that PIK3CA mutations occur in lung adenocarcinomas, usually concurrently with EGFR, KRAS, and anaplastic lymphoma kinase (ALK).
Literature	GENERIF_SUMMARY	BDNF promotes EGF-induced proliferation and migration of neural stem/progenitor cells through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	PIK3CA rs6443624 and rs9838411 variants either somewhat or significantly decreased risk of endometrial cancer in a dominant model.
Literature	GENERIF_SUMMARY	Loss of ARID1A protein expression occurs as a very early event in ovarian clear-cell carcinoma development and frequently coexists (not mutually exclusive) with PIK3CA mutations.
Literature	GENERIF_SUMMARY	Breast cnacer patients with PIK3CA mutations or increased PI3K pathway activity had a significantly poorer survival despite adequate treatment with adjuvant chemotherapy and trastuzumab.
Literature	GENERIF_SUMMARY	Data suggest that wild-type PTEN gene transfection might increase drug sensitivity or reverse drug resistance via inhibiting the PI3K/Akt pathway and regulating downstream molecules of the cell signaling transduction pathway in K562/ADM cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Mutations in ras PIK3CA are present in benign lichenoid keratosis.
Literature	GENERIF_SUMMARY	Data indicate that NVP-BKM120 inhibits all four class I PI3K isoforms in biochemical assays.
Literature	GENERIF_SUMMARY	Gene mutations of PIK3CA are not associated with the postoperative prognosis of patients with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Results show a novel anti-inflammatory role for the PI3K/Akt signaling pathway in microglia, and show that Ad-IRF3 suppressed proinflammatory genes and enhanced anti-inflammatory genes.
Literature	GENERIF_SUMMARY	PIK3CA mutation status was analyzed by sequencing in 94 tumors. phosphatidylinositol 3'-kinase pathway alteration might have a favorable prognostic impact on endometrial cancers in Chinese women.
Literature	GENERIF_SUMMARY	conclude that exocrine pancreatic expression of phosphatidylinositol 3-kinase p85alpha subunit is attenuated by aging
Literature	GENERIF_SUMMARY	Although PIK3CA mutation has been found in 10-18% of colorectal cancer patients, the results are still inconclusve if it affects the efficacy of cetuximab or panitumumab. [review]
Literature	GENERIF_SUMMARY	Results describe how phosphatidylinositol 3-kinase and protein kinase D1 cooperate to regulate the insulin-like growth factor signaling pathway.
Literature	GENERIF_SUMMARY	Mutational activation of both BRAF and PIK3CA genes does contribute to hepatocellular tumorigenesis at somatic level in Southern Italian population.
Literature	GENERIF_SUMMARY	Patients with PIK3CA mutations treated with PI3K/AKT/mTOR inhibitors demonstrated a higher response rate than patients without mutations.
Literature	GENERIF_SUMMARY	loss of PTEN expression (by immunohistochemistry) and PIK3CA mutation is likely to be predictive of a lack of benefit to anti-EGFR therapy in metastatic colorectal cancer
Literature	GENERIF_SUMMARY	UCA1 regulated cell cycle through CREB via PI3K-AKT dependent pathway in bladder cancer.
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with colorectal serrated adenocarcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	data suggest that lamin A/C proteins are positively involved in malignant behavior of PC cells through the PI3K/AKT/PTEN pathway. Lamin A/C may represent a new oncogenic factor and a novel therapeutic target for PC
Literature	GENERIF_SUMMARY	These findings strengthen the evidence for involvement of the Notch, Hedgehog, NF-KB, and PIK3CA pathways in breast cancer development, and point to novel processes that likely are involved.
Literature	GENERIF_SUMMARY	PIK3CA mutations confer constitutive activation of PI3K, which initiates intracellular kinase signaling cascades that promote cell proliferation and survival.
Literature	GENERIF_SUMMARY	PIK3CA mutations may play an important role in the carcinogenesis and development of breast cancer.
Literature	GENERIF_SUMMARY	PIK3CA mutation was associated with significantly longer metastasis-free survival in breast cancer patients.
Literature	GENERIF_SUMMARY	PIK3CA mutations at exon 9 or 20 may have different impact on the prognosis in endometrial carcinoma.
Literature	GENERIF_SUMMARY	PI3K/mTOR inhibitors can enhance 5-FU cytotoxicity in vitro and in vivo, especially in PIK3CA mutant tumor cells.
Literature	GENERIF_SUMMARY	Studies suggest that the effects of PHLPP1 deletion on prostate carcinogenesis may be explained by the strong dependence of prostate cancer on phosphoinositide 3-kinase (PI3K)/Akt signalling.
Literature	GENERIF_SUMMARY	Data indicate that the mechanisms of action of ON 01910.Na invloved PI3K/AKT inhibition and induction of oxidative stress response.
Literature	GENERIF_SUMMARY	High p110alpha is associated with non-small cell lung cancer.
Literature	GENERIF_SUMMARY	coexistence of PIK3CA (the PI3K p110alpha subunit) exon 9 and 20 mutations, but not PIK3CA mutation in either exon 9 or 20 alone, is associated with poor prognosis of colorectal cancer patients
Literature	GENERIF_SUMMARY	The relationship between expression and genetic alterations of the components of the PI3K/AKT pathway in non-small cell lung cancer, was investigated.
Literature	GENERIF_SUMMARY	Merkel cell carcinoma samples were analyzed for activating PIK3CA and AKT1 mutations.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	analysis of FGFR2 point mutations in 466 endometrioid endometrial tumors and their relationship with MSI, KRAS, PIK3CA, CTNNB1 mutations
Literature	GENERIF_SUMMARY	we have analyzed the prevalence of somatic mutations in the FGFR3, PIK3CA, AKT1, KRAS, HRAS, and BRAF genes in bladder cancers
Literature	GENERIF_SUMMARY	The PIK3CA mutation rate was lower in adenocarcinomas compared to other forms of lung cancers
Literature	GENERIF_SUMMARY	Our findings indicate that the effects of mutant PIK3CA are both cell type- and mutation-specific.
Literature	GENERIF_SUMMARY	PIK3CA mutations were identified in 13/24 columnar cell lesions (54%) and 3/8 invasive carcinomas (37%).
Literature	GENERIF_SUMMARY	Data indicate a functional link between the Notch1 and PI3K/Akt pathways in clear cell renal cell carcinoma (CCRCC).
Literature	GENERIF_SUMMARY	This review outlines the interaction of the PI3K pathway with other signaling cascades, highlights mechanisms involved in hyperactivation, discusses current therapeutics in cancer-related clinical trials that target this pathway.
Literature	GENERIF_SUMMARY	The PI3K/AKT pathway can be an important signaling pathway for the survival of BRCA1-defective breast cancer cells.
Literature	GENERIF_SUMMARY	Both osteopontin-c and osteopontin-b splicing isoforms promote distinct aspects of PCa progression by inducing PI3K signaling.
Literature	GENERIF_SUMMARY	A PIK3CA mutation was observed in 16 (17%) specimens in patients with endometrial cancer.
Literature	GENERIF_SUMMARY	In summary, the PI3K/PKCiota regulates the alternative splicing of Bcl-x pre-mRNA with implications in the cell survival of NSCLC cells.
Literature	GENERIF_SUMMARY	Aberrant DLC1 methylation and PIK3CA mutat-ions may have important roles in extra-mammary Paget's disease pathogenesis.
Literature	GENERIF_SUMMARY	findings describe a PI3K-driven, NF-kappaB-dependent transcriptional profile that may play a critical role in promoting a microenvironment amenable to tumor progression
Literature	GENERIF_SUMMARY	In human airway smooth muscle cells,

Main Category	Category	Terms
		regulation of CD38 expression occurs by specific class I PI3K isoforms.
Literature	GENERIF_SUMMARY	Although there is an association of PI3 kinase signaling and chemoresistance in advanced ovarian cancer, there is no clear evidence that this is driven specifically by PIK3CA mutations.
Literature	GENERIF_SUMMARY	We further identify the activation of the PI3K/AKT signal pathway by T3 as an underlying mechanism for the enhanced conversion to cell pluripotency in this model.
Literature	GENERIF_SUMMARY	KRAS, BRAF, and PIK3CA mutations in colorectal cancer have sustained prevalence rate in the Taiwanese population.
Literature	GENERIF_SUMMARY	In the present study, we did not find any significant correlations between KRAS, BRAF and PIK3CA mutations and the loss of PTEN expression and various clinicopathological features in Chinese patients with colorectal cancer.
Literature	GENERIF_SUMMARY	Data suggest that phosphoinositide-3-kinase/Akt and mammalian target of rapamycin (mTOR) pathway signaling differentially regulate phospho-Mst1-Thr-120/Thr-183.
Literature	GENERIF_SUMMARY	Our findings implicate amplifications of 3q26 focused on PIK3CA and 8q24 focused on MYC in chronic lymphocytic leukemia.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphism in PIK3CA is associated with synchronous endometrial and ovarian cancer.
Literature	GENERIF_SUMMARY	Breast cancer patients with tumors harboring a PIK3CA mutation have a better clinical outcome than those with a wild-type PIK3CA gene.
Literature	GENERIF_SUMMARY	Angiosarcoma (AS) is characterized by a complex pattern of alterations of the TP53 and PIK3CA/AKT/mTOR pathways that does not depend on MYC amplification status.
Literature	GENERIF_SUMMARY	In conclusion, SPRY2 is an important tumour suppressor in prostate cancer since its loss drives the PI3K/AKT pathway via functional interaction with the ErbB system.
Literature	GENERIF_SUMMARY	CLOVES syndrome is caused by postzygotic activating mutations in PIK3CA
Literature	GENERIF_SUMMARY	Our data highlights a new functional synergism between oncogenic EGFR and PIK3CA with WNT/beta-catenin conferring high tolerance to oxidative stress generated

Main Category	Category	Terms
		by nutrient deprivation.
Literature	GENERIF_SUMMARY	KRAS and PIK3CA mutations frequently coexist in patients with colorectal cancer, and are associated with clinical characteristics and outcome.
Literature	GENERIF_SUMMARY	The interaction between HGF and c-Met increases OPN expression in human osteoblasts via the PI3K, Akt, c-Src, c-Jun, and AP-1 signaling pathway.
Literature	GENERIF_SUMMARY	several Rho family small GTPases activate PI3K by an indirect cooperative positive feedback that required a combination of Rac, CDC42, and RhoG small GTPase activities
Literature	GENERIF_SUMMARY	PIK3CA H1074R mutation and PTEN loss differentially regulate LIM-kinase1 and slingshot-1L.
Literature	GENERIF_SUMMARY	Cyclic hydrodynamic pressure stimulating the roliferation of human bladder smooth muscle cells cultured in scaffolds. The signal transduction mechanism for this process is involved with the PI3K/SGK1 and not the PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	PIK3CA mutations were significantly associated with a favorable overall survival of patients with ovarian clear cell carcinoma
Literature	GENERIF_SUMMARY	Mucinous breast carcinomas lack PIK3CA mutations.
Literature	GENERIF_SUMMARY	Using high throughput mutation screening platform, we identified that PIK3CA mutations were the most frequently observed target for gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	Sequencing of PIK3CA in ten individuals with fibroadipose hyperplasia syndromes identified either the p.His1047Leu alteration or a second cancer-associated alteration, p.His1047Arg, in nine cases.
Literature	GENERIF_SUMMARY	Exome sequencing and mass spectrometry analysis in paired brain-blood samples from individuals with hemimegalencephaly (20 cases) identified de novo somatic mutations in 30% of affected individuals in the PIK3CA, AKT3 and MTOR genes.
Literature	GENERIF_SUMMARY	identified mutations in AKT3, PIK3R2 and PIK3CA in 11 unrelated families with megalencephaly-capillary malformation and megalencephaly-polymicrogyria-polydactyly-hydrocephalus syndromes
Literature	GENERIF_SUMMARY	A high copy number of MET or PIK3CA was

Main Category	Category	Terms
		found to be associated with poorer prognostic features and triple receptor-negative disease
Literature	GENERIF_SUMMARY	analysis of multiorgan metastasis of human HER-2+ breast cancer in Rag2-/-;Il2rg-/- mice and treatment with PI3K inhibitor
Literature	GENERIF_SUMMARY	this study evaluated whether activating mutations of PI3KCA or complete loss of PTEN is associated with response to anti-human Her2 treatment in breast cancer
Literature	GENERIF_SUMMARY	Epithelial CEACAMs associate in cis with other membrane receptor(s) via their extracellular domains to trigger bacterial uptake in a PI3K-dependent manner.
Literature	GENERIF_SUMMARY	These data support a strong biological-link between NFkB and the PI3-kinase/AKT pathway in the modulation of anti-apoptotic effects in primary effusion lymphoma cells.
Literature	GENERIF_SUMMARY	study demonstrated that the somatic mutations of PIK3CA and JAK2 occurred in a small fraction of sarcomas and that these mutations may not play a principal role in the development of sarcomas
Literature	GENERIF_SUMMARY	This study demonstrates the presence of somatic mutations and amplifications of the PIK3CA gene in a second series of pituitary adenomas, corroborating the previously described involvement of the PI3K/Akt signaling pathway in this tumorigenic process.
Literature	GENERIF_SUMMARY	In CRC patients treated with cetuximab, activating mutation signatures for PIK3CA (49 genes) were developed.
Literature	GENERIF_SUMMARY	PIK3CA-targeted short hairpin RNAs can block the phosphoinositide 3-kinase-Akt signaling pathway and inhibit cell growth, increase apoptosis, and induce cell cycle arrest in PIK3CA-mutant colon cancer SW948 cells
Literature	GENERIF_SUMMARY	Data provide information for developing new therapeutics targeting PI3K/Akt/mTOR signaling pathway in autoimmune diseases like psoriasis and rheumatoid arthritis.
Literature	GENERIF_SUMMARY	Electroacupuncture profoundly activated PI3K/Akt signaling in ischemic cerebral tissues.
Literature	GENERIF_SUMMARY	PI3K expression and PIK3CA mutations are related to colorectal cancer metastases
Literature	GENERIF_SUMMARY	LOX-1 up-regulation induced by AGE-BSA was a receptor mediated through RAGE and

Main Category	Category	Terms
		is via the PI3K/PDK1/mTORC2 pathway
Literature	GENERIF_SUMMARY	High-throughput genotyping in metastatic esophageal squamous cell carcinoma identifies phosphoinositide-3-kinase and BRAF mutations.
Literature	GENERIF_SUMMARY	Data suggest that targeting Hedgehog (Hh) pathway alone or in combination with PI3K/AKT pathway could be a novel therapeutic option in treating endocrine-resistant breast cancer.
Literature	GENERIF_SUMMARY	The Epstein-Barr virus protein LMP1, which lacks the canonical ITAM required for Syk activation, can nevertheless activate Syk, and the Src kinase Fyn, resulting in downstream c-Cbl and PI3K/Akt activation.
Literature	GENERIF_SUMMARY	Deregulated c-MYC and PI3K activity cooperate in Burkitt lymphoma pathogenesis.
Literature	GENERIF_SUMMARY	These results demonstrated that hepatitis B virus X protein could upregulate LASP-1 through PI3-K pathway to promote the proliferation and migration of hepatoma cells.
Literature	GENERIF_SUMMARY	Grape proanthocyanidin inhibit pancreatic cancer cell growth in vitro and in vivo through induction of apoptosis and by targeting the PI3K/Akt pathway
Literature	GENERIF_SUMMARY	Data indicate that mutation frequency in malanoma patients was found witih BRAF(V600) in 51%, NRAS in 19%, PI3K pathway in 41% and PTEN in 22%.
Literature	GENERIF_SUMMARY	Molecular genetic aberrations involving the p53, cyclin E-FBXW7, and PI3K pathways represent major mechanisms in the development of uterine serous carcinoma.
Literature	GENERIF_SUMMARY	Class IA phosphatidylinositol 3-kinase p110alpha is required for vacuole maturation
Literature	GENERIF_SUMMARY	heterogeneity in frequencies of KRAS/PIK3CA somatic mutations is consistent with already-reported discrepancies in distribution of germline mutations for other malignancies within Sardinian population.
Literature	GENERIF_SUMMARY	Data indicate that resveratrol suppressed IL-1beta-induced activation of NF-kappaB and PI3K in a dose- and time-dependent manner.
Literature	GENERIF_SUMMARY	Data indicate that KRAS, BRAF, PIK3CA, and AKT1 mutations can be rapidly and accurately detected for cancer diagnosis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	these results demonstrate that miR-124 functions as a growth-suppressive miRNA and plays an important role in inhibiting the tumorigenesis through targeting PIK3CA.
Literature	GENERIF_SUMMARY	results suggest that PIK3CA gene mutations may play a key role in tumourigenesis and aggressiveness of bronchopulmonary neuroendocrine tumours
Literature	GENERIF_SUMMARY	diverse PIK3CA mutations stimulate lipid kinase activity by facilitating allosteric motions required for catalysis on membranes
Literature	GENERIF_SUMMARY	The subgroup of ovarian clear cell carcinomas harboring activated PIK3CA seems to have better prognosis possibly due to more indolent biological property compared to tumors without PIK3CA activation.
Literature	GENERIF_SUMMARY	Results indicate that CIB1 is uniquely positioned to regulate PI3K/AKT and MEK/ERK signaling and that simultaneous disruption of these pathways synergistically induces a nuclear GAPDH-dependent cell death.
Literature	GENERIF_SUMMARY	the Ras-PI3K signaling pathway targets specific epigenetic modifications in tumor cells
Literature	GENERIF_SUMMARY	PI3K has a crucial role in regulating Brn-2 and Pax3 expression and neoplasm invasiveness, and it is a key determinant of melanoma subpopulation diversity.
Literature	GENERIF_SUMMARY	Data indicate that copy number amplification of the PIK3CA gene is associated with poor prognosis in head and neck squamous cell carcinoma (HNSCC) patients without lymph node metastasis.
Literature	GENERIF_SUMMARY	LY294002 prevented phosphorylation of protein kinase B (PKB/Akt) by inhibition of PI3K phosphorylation activity.
Literature	GENERIF_SUMMARY	Investigated PIK3CA mutations in ILBC with special reference to late stage tumor progression. PIK3CA mutations were detected in 32/88 (36%) specimens.
Literature	GENERIF_SUMMARY	PI3K is a mediator of stretch-induced COX-2 expression in urothelial cells.
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with early-stage low-grade endometrial cancers.
Literature	GENERIF_SUMMARY	EGFR and downstream genetic alterations in KRAS/BRAF and PI3K/AKT pathways have roles in colorectal cancer and treatment

Main Category	Category	Terms
		[review]
Literature	GENERIF_SUMMARY	PIK3CA mutations lead to a shift towards a highly glycolytic phenotype.
Literature	GENERIF_SUMMARY	p110alpha was activated at mitosis entry and regulated early mitotic events, such as PIP(3) generation, prometaphase progression, and spindle orientation.
Literature	GENERIF_SUMMARY	Src family kinases, PI3 kinase and protein kinase C synergize to mediate Gq-dependent platelet activation
Literature	GENERIF_SUMMARY	PIK3CA H1047R mutation was the only independent factor predicting response to PI3K/AKT/mTOR inhibitors in patients with advanced cancers
Literature	GENERIF_SUMMARY	Data indicate that resistance to UVB induced apoptosis in the alpha6 integrin(high+)/CD44(+) cells involved increased expression of TAp63 and was overcome by PI-3 kinase inhibition.
Literature	GENERIF_SUMMARY	PI3K/Akt signaling is utilized by alphaIIb-mediated outside-in signaling to activate platelets.
Literature	GENERIF_SUMMARY	either mutant PIK3CA expression or PTEN loss, but not mutant AKT1 E17K, cause disrupted epithelial architecture.
Literature	GENERIF_SUMMARY	activation of the PI3K-AKT pathway through loss of PTEN or PIK3CA mutation was frequently observed in trastuzumab refractory breast cancer.
Literature	GENERIF_SUMMARY	Use of aspirin after diagnosis was associated with longer survival among patients with mutated-PIK3CA colorectal cancer. Suggests that the PIK3CA mutation in colorectal cancer may serve as a predictive molecular biomarker for adjuvant aspirin therapy.
Literature	GENERIF_SUMMARY	identified somatic gain-of-function mutations in PIK3CA in the affected nerve that caused macrodactylous overgrowth
Literature	GENERIF_SUMMARY	PI3K/AKT/mTOR signalling components appear to be differentially implicated in urothelial tumorigenesis and, with the exception of p85aPI3K, are unrelated to the PIK3CA or AKT1 mutational status.
Literature	GENERIF_SUMMARY	Data indicate that cells grown in three-dimensional (3D) gels showed reduced proliferation and migration as well as reduced PI3K pathway activation when compared to cells grown in 2D.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	In the present study, the authors provide evidence that the West Nile virus capsid protein blocks apoptosis through a phosphatidylinositol (PI) 3-kinase-dependent pathway.
Literature	GENERIF_SUMMARY	One hundred and ten patients (51%) were identified who were potentially nonresponders to anti-EGFR therapy
Literature	GENERIF_SUMMARY	PIK3CA amplification was detected in thirty-seven percent of lung squamous tumors and five percent of lung adenocarcinomas.
Literature	GENERIF_SUMMARY	Molecular characterization of single CTCs demonstrated considerable intra- and interpatient heterogeneity of EGFR expression and genetic alterations in EGFR, KRAS, and PIK3CA
Literature	GENERIF_SUMMARY	Targeting the phosphoinositide 3-kinase p110-alpha isoform impairs cell proliferation, survival, and tumor growth in small cell lung cancer.
Literature	GENERIF_SUMMARY	Data show that a photodynamic therapy (PDT) dose-dependent upregulation of CRP gene, as well as of PTX3 and ficolin 1 genes in lung tumor A549 cells, and indicate critical role played by PI3K/Akt/AP-1 pathway.
Literature	GENERIF_SUMMARY	Our finding that PIK3CA exon 20 mutations were associated with more aggressive breast cancer and poor outcomes, regardless of the treatment regimen, has important clinical implications.
Literature	GENERIF_SUMMARY	A dramatically lower prevalence of the heterozygous genotype G/T at rs17849071 of PIK3CA was found in patients with follicular thyroid cancer.
Literature	GENERIF_SUMMARY	In conclusion, authors have found that the varicella-zoster virus ORF12 protein activates the PI3K/Akt pathway to regulate cell cycle progression.
Literature	GENERIF_SUMMARY	in primary endometrial carcinoma, mutations in PIK3CA, TP53, PTEN and FBXW7 correlate with high tumor grade, endometrial cancer type and lymph node status; PIK3CA H1047R mutations serve as prognostic markers for relapse-free survival in endometrial cancer patients
Literature	GENERIF_SUMMARY	Mutations affecting BRAF, EGFR, PIK3CA, and KRAS are not associated with sporadic vestibular schwannomas.
Literature	GENERIF_SUMMARY	miR-148b controls malignancy by coordinating a novel pathway involving over

Main Category	Category	Terms
		130 genes and, in particular, it directly targets players of the integrin signaling, such as ITGA5, ROCK1, PIK3CA/p110alpha, and NRAS, as well as CSF1
Literature	GENERIF_SUMMARY	Findings indicate driver mutations in AKT1, and extend the number and type of mutations that activate the PI3-kinase pathway in human breast cancers.
Literature	GENERIF_SUMMARY	Findings suggest that inhibiton of Bcl-2, Bcl-xL and PI3K, and release of Bim from Bcl-2/Bcl-xL and GSK3alpha/beta culminating in Bax/Bak activation and apoptosis.
Literature	GENERIF_SUMMARY	8 of 91 (8.8%) unrelated CS individuals without germline PTEN mutations carried 10 germline PIK3CA mutations
Literature	GENERIF_SUMMARY	PIK3CA mutations in diverse cancers were not associated with clinical characteristics, but were correlated with MAPK mutations. PIK3CA mutations, especially, H1047R, were associated with attaining a PR/CR to PI3K/AKT/mTOR pathway inhibitors.
Literature	GENERIF_SUMMARY	Linoleic acid induces opening of connexin26 hemichannels through a PI3K/Akt/Ca(2+)-dependent pathway
Literature	GENERIF_SUMMARY	In cervical cancer patients treated with CRT, tumor PIK3CA mutation status was associated with overall survival in FIGO stage IB/II cervix cancers.
Literature	GENERIF_SUMMARY	Coincided with the activation of the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Selective inhibition of phosphoinositide 3-kinase p110alpha preserves lymphocyte function
Literature	GENERIF_SUMMARY	Data indicate protein secretion pathways activated by monosodium urate (MSU) in macrophages, and reveal a novel role for cathepsin B and Src, Pyk2, PI3 kinases in the activation of unconventional protein secretion.
Literature	GENERIF_SUMMARY	FGFR2, KRAS and PIK3CA are frequently mutated in primary and metastatic endometrial lesions.
Literature	GENERIF_SUMMARY	PIK3CA has a role in response to everolimus and letrozole in estrogen receptor positive breast cancer
Literature	GENERIF_SUMMARY	No correlation was observed between PIK3CA exon 9 and 20 mutations and age at breast cancer diagnosis, histological type, hormone receptor and HER2 status.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data indicate that phosphatidylinositol 3'-kinase (PI3K) alpha and beta p110 isoforms of PI3K are broadly expressed in myeloma.
Literature	GENERIF_SUMMARY	PIK3CA amplifications in large cell and squamous cell carcinomas were significantly higher compared with adenocarcinomas.
Literature	GENERIF_SUMMARY	High PIK3CA/PIK3CD ratio identified a subset of primary mantle cell lymphomas resistant to GS-1101 and this ratio increased significantly with relapse.
Literature	GENERIF_SUMMARY	Results indicate a key role of class I PI3K in IL-2R endocytosis that creates a link with IL-2 signalling.
Literature	GENERIF_SUMMARY	The rate of PIK3CA mutations in this series of micropapillary carcinomas is similar to invasive ductal carcinomas; however, there may be an enrichment of AKT1 mutations.
Literature	GENERIF_SUMMARY	only 5 tumours with exon 20 PIK3CA mutations were detected. Only PIK3CA mutations occasionally coexisted with other gene mutations in colorectal cancer.
Literature	GENERIF_SUMMARY	PI3K abrogation interferes with key T cell activation processes through both differential expression and alternative splicing, which together actively contribute to T cell suppression.
Literature	GENERIF_SUMMARY	14-3-3 negatively regulates the RGC downstream of the PI3-kinase/Akt signaling pathway
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with lung cancer.
Literature	GENERIF_SUMMARY	PIK3CA-mutant lung cancers are dependent on this driver oncogene, whereas in other patient cases, the mutation may modulate the effect of another oncogenic process.
Literature	GENERIF_SUMMARY	down regulation of fibroblast growth factor receptor 3 (FGFR3) and PI3K/Akt signaling mechanisms leading to inhibition of cell proliferation and induction of molecular mechanisms of apopto
Literature	GENERIF_SUMMARY	Studies suggest that adipokines, AMPK, PI3K and the PPAR signaling pathways could be targets of therapeutic strategies for the ovarian cancers and reproductive pathologies.
Literature	GENERIF_SUMMARY	Results indicate that the induction of MCP-2/CCL8 by mycobacteria is dependent on the activation of TLR2/PI3K/Akt signaling pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The correlation between T stage of gastric cancer and the higher expression of CD40, VEGF, AKT, and PI3K, along with lower S100 expression in DC, may provide insights into future targets for more effective immunotherapy for cancer.
Literature	GENERIF_SUMMARY	PIK3CA mutation is a promising predictive biomarker for poor survival in mCRC patients treated with anti-EGFR mAbs, particularly in KRAS wild-type patients.
Literature	GENERIF_SUMMARY	cell-autonomous phosphoinositide 3-kinase and 3-phosphoinositide-dependent protein kinase 1 are key effectors of oncogenic Kras in the pancreas, mediating cell plasticity, acinar-to-ductal metaplasia, and pancreatic ductal adenocarcinoma formation
Literature	GENERIF_SUMMARY	these data suggest the efficacy of agents that target the MAPK and PI3K pathways can be improved by combination with a Bcl-2 family inhibitor.
Literature	GENERIF_SUMMARY	PI3K is involved in CKII inhibition-mediated cellular senescence in HCT116 cells.
Literature	GENERIF_SUMMARY	PIK3CA mutations in esophageal squamous cell carcinoma are associated with longer survival.
Literature	GENERIF_SUMMARY	High phospho-stathmin(S38) is associated with PI3K pathway alterations
Literature	GENERIF_SUMMARY	Data suggest that 17 beta-estradiol enhances the the glucose uptake, activates PI3K/Akt signaling pathway, leading to translocation of glucose transporter 4 to the plasma membrane in an ERalpha-dependent manner.
Literature	GENERIF_SUMMARY	About one fourth of mCRC cases wild-type for KRAS codons 12 and 13 present other mutations either in KRAS, BRAF, or PIK3CA, many of which may explain the lack of response to anti-EGFR therapy observed in a significant proportion of these patients.
Literature	GENERIF_SUMMARY	Data indicate that concurrent BRAF and PI3K/mTOR blockade results in induction of apoptosis.
Literature	GENERIF_SUMMARY	APG dramatically reduced Nrf2 expression at both the messenger RNA and protein levels through downregulation of PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	PIK3CA mutations are not associated with tyrosine kinase inhibitors treatment efficacy in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	p110alpha acts as a negative regulator of

Main Category	Category	Terms
		beta cell exocytosis and insulin secretion, while p110beta is a positive regulator of insulin secretion through a mechanism separate from its catalytic activity
Literature	GENERIF_SUMMARY	Data show that somatic cell knockin of both KRAS G12V and oncogenic PIK3CA mutations in human breast epithelial cells results in cooperative activation of the phosphoinositide 3-kinase (PI3K) and mitogen-activated protein kinase (MAPK) pathways.
Literature	GENERIF_SUMMARY	mutational analysis of TP53 (exons 4-9) and CTNNB1 (exon 3) as well as PIK3CA (exon 9) genes in hepatocellular carcinoma from Southern Italy
Literature	GENERIF_SUMMARY	PI3K signaling contributed to tumor growth and vasculogenic mimicry of gallbladder carcinomas.
Literature	GENERIF_SUMMARY	PIK3CA mutation status was also not associated with treatment outcome after first-line tamoxifen
Literature	GENERIF_SUMMARY	the review provides a complete picture of the PIK3CA oncogene and its deregulation in oral cancer (Review)
Literature	GENERIF_SUMMARY	These results confirm that the outcome of HER2-positive patients treated with trastuzumab is significantly worse in patients with PIK3CA-mutated compared with wild-type tumours.
Literature	GENERIF_SUMMARY	Results show that stromal cell-derived factor-1 (SDF-1) enhanced ovarian cancer cell invasion through alphavbeta6 integrin-mediated urokinase-type plasminogen activator (uPA) expression via the p38 MAPK and PI3 K/Akt pathway.
Literature	GENERIF_SUMMARY	Data indicate that the PI3K/Akt pathway plays an important role in the regulation of telomerase reverse transcriptase (hTERT) activity in response to histone deacetylases (HDACs) inhibitor.
Literature	GENERIF_SUMMARY	Results indicate that ADAM17 is upregulated by hypoxia and contributes to hypoxia-induced cisplatin resistance via EGFR/PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	the PI3K/AKT/mTOR signaling pathway, which was overactivated in APA and IHA compared with normal zona glomerulosa, may mediate aldosterone hypersecretion and participate in the development of PA.
Literature	GENERIF_SUMMARY	PIK3CA exon 20 and PTEN mutations associated with features of colon cancer

Main Category	Category	Terms
		tumorigenesis
Literature	GENERIF_SUMMARY	our results suggest that CAPE suppresses phosphoinositide 3-kinase/AKT/XIAP pathway leading to apoptosis in melanoma tumor cells in vitro and in vivo
Literature	GENERIF_SUMMARY	Disruption of the IRS1-p110alpha E545K interaction destabilizes the p110alpha protein, reduces AKT phosphorylation, and slows xenograft tumor growth of a cancer cell line expressing p110alpha E545K.
Literature	GENERIF_SUMMARY	Aberrations in PIK3CA and PTEN did not add significant prognostic information on colorectal carcinoma patients.
Literature	GENERIF_SUMMARY	These results reveal a previously unknown, essential role of PI3K in the life cycle of human astrovirus type 1.
Literature	GENERIF_SUMMARY	Report a synthetic lethal interaction between PI3K/mTOR and MEK inhibitors in rhabdomyosarcoma.
Literature	GENERIF_SUMMARY	Mitotic Plk1 activity is regulated not only by Plk1-Thr210 phosphorylation, but also by Plk1 binding to 14-3-3gamma following Plk1-Ser99 phosphorylation downstream of the PI3K-Akt signalling pathway.
Literature	GENERIF_SUMMARY	Loss of ARID1A expression, PI3K-Akt pathway alterations, TP53 status and microsatellite instability are involved in endometrial cancer.
Literature	GENERIF_SUMMARY	Effectively inhibited the PI3K signaling pathway.
Literature	GENERIF_SUMMARY	no specific relationship between mutation in the exon of PIK3CA and clinical characteristics of the hepatocellular carcinoma in korean patients
Literature	GENERIF_SUMMARY	Data indicate that NVP-BKM120 inhibited pan-class I phosphatidyl-inositol-3-kinase and decreased cell proliferation and increased apoptotic cell death in diffuse large B-cell lymphoma (DLBCL).
Literature	GENERIF_SUMMARY	A downregulation of FASN effectively inhibits the activity of ""HER2-PI3K/Akt axis"" and alters the malignant phenotype in colorectal cancer cells."
Literature	GENERIF_SUMMARY	We also examined 61 of our cases for the presence of PIK3CA, AKT1, PTEN and K-RAS mutations.
Literature	GENERIF_SUMMARY	PIK3CA-mutated colorectal cancer appears to have a distinct epidemiologic profile that is of clinical significance.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data indicate that the activation of MAPK and PI3K pathways resulted in TGF-beta1 signaling by down-regulating Nm23-H1 expression and up-regulating the expression of TbetaRI and TbetaRII, favoring further activation of multiple signaling pathways.
Literature	GENERIF_SUMMARY	PIK3CA mutations were associated with a better outcome, however this effect disappeared after 3 years. There were no statistically significant associations with trastuzumab benefit.
Literature	GENERIF_SUMMARY	The PI3K/Akt inhibitor LY294002 can enhance chemosensitivity of human gastric cancer to VCR.
Literature	GENERIF_SUMMARY	PIK3CA mutation also contributes to significantly decreased survival for patients with wild-type BRAF tumors.
Literature	GENERIF_SUMMARY	4 of 41 HPV-negative tumors (10%) demonstrated PIK3CA hot spot mutations, 3 at codon 1047 and 1 at codon 542. Of 46 HPV-positive tumors, 13 (28%) demonstrated activating PIK3CA mutations, including 7 at codon 542, 5 at codon 545, and 1 at codon 1047.
Literature	GENERIF_SUMMARY	These results suggest that the phosphatidylinositol 3-kinase/Akt signaling pathway plays an important role in beta1 integrin-mediated internalization of Staphylococcus aureus by alveolar epithelial cells.
Literature	GENERIF_SUMMARY	Activation of the PI3K/Akt pathway with expression of survivin identified a subgroup of acute myeloid leukemia patients with an excellent outcome.
Literature	GENERIF_SUMMARY	PIK3CA mutation and copy number gain were detected in 3 (4.9%) and 8 (13.1%), respectively, of 61 gastric cancer samples while PTEN loss was detected in 24 (39%) of the tumours. Two tumours had both PTEN loss and PIK3CA copy number gain.
Literature	GENERIF_SUMMARY	This study investigated the impact of KRAS, NRAS, BRAF, PI3KCA and TP53 status on outcome of elderly metastatic colorectal cancer patients
Literature	GENERIF_SUMMARY	Study developed a cancer stem cell model from a colorectal cancer patient-derived xenograft tumor and identified a PIK3CA (H1047R) mutation in the CSC population as a predictive biomarker to guide targeted therapy.
Literature	GENERIF_SUMMARY	Single-nucleotide polymorphism in PIK3CA gene is associated with Insulin Resistance.

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Literature	GENERIF_SUMMARY	Data indicate that increased activation of PI3K/AKT signaling was observed in the cholangiocarcinoma (CCA) tissues, and suggest that inhibition of PI3K and mTOR activity by the inhibitor NVP-BEZ235 has anti-cancer activity against CCA.
Literature	GENERIF_SUMMARY	PIK3CA mutations and/or PTEN loss were identified in the wide morphologic spectrum of salivary duct carcinomas.
Literature	GENERIF_SUMMARY	frequent PIK3CA mutations in patients with high-risk HNSCC confined predominantly to the oropharyngeal and sinonasal subsites; for the first time, mutation in AKT1 has been identified in HNSCC
Literature	GENERIF_SUMMARY	we discuss the preclinical and clinical experience with the rapalogues in RCC, potential mechanisms of resistance to the rapalogues, and the progress in the clinical development of novel agents directed against the phosphatidylinositol 3-kinase/ Akt/mTOR
Literature	GENERIF_SUMMARY	This study shows that PI3K inhibitors only transiently inhibit AKT which limits their antitumor activities.
Literature	GENERIF_SUMMARY	Among 30 breast tumors, 10(33%) had BCSCs with AKT1, HRAS, or PIK3CA mutations. 3 mutations were detected in PIK3CA.Tumors in which BCSCs have defects in PI3K/Akt signaling are significantly more likely to manifest nodal metastases.
Literature	GENERIF_SUMMARY	Increase in the expression of PI3K was associated with colorectal carcinoma.
Literature	GENERIF_SUMMARY	our findings suggest that loss of CDKN2A and, to a lesser extent, KRAS and PIK3CA somatic mutations have a role in the progression of a benign to an atypical proliferative Brenner tumor
Literature	GENERIF_SUMMARY	AKT1 E17K is a bona fide oncogene in a human luminal breast cancer context. Distinct PI3K pathway mutations confer differential sensitivity to drugs targeting the pathway at different points and by distinct mechanisms.
Literature	GENERIF_SUMMARY	Results suggest that IFN-gamma-mediated IL-27 protein but not IL-27p28 gene expression is positively regulated by the C-Jun N-terminal kinases (JNK), mitogen-activated protein kinases (MAPKs) and the phosphoinositide 3-kinase (PI3K).
Literature	GENERIF_SUMMARY	Uterine serous carcinoma cell lines with oncogenic PIK3CA mutations were

Main Category	Category	Terms
		significantly more sensitive to GDC-0980 exposure.
Literature	GENERIF_SUMMARY	In PIK3CA-mutant cancer cell lines with persistent mTORC1 signaling, the addition of the allosteric mTORC1 inhibitor RAD001 to the cells along with BYL719 resulted in reversal of resistance in vitro and in vivo.
Literature	GENERIF_SUMMARY	Oncogenic mutations of p110alpha isoform of PI 3-kinase upregulate its protein kinase activity.
Literature	GENERIF_SUMMARY	PIK3CA mutations are frequently observed in BRCAX but not BRCA2-associated male breast cancer.
Literature	GENERIF_SUMMARY	Data indicate that inflammatory cytokine-induced increases in chronic lymphocytic leukemia (CLL) cell adhesion to stromal cells are correlated with ZAP-70 expression and blocked by PI3K inhibitor.
Literature	GENERIF_SUMMARY	PIK3CA mutations are significantly associated with lower grade and hormone receptor positivity in Arab breast cancer patients.
Literature	GENERIF_SUMMARY	PI3K p85alpha-depletion results in activated expression of apoptosis-associated genes.
Literature	GENERIF_SUMMARY	Multivariate analyses revealed that PTEN expression (HR, 0.48; p = 0.02) and absence of PI3K expression (HR, 0.2; p = 0.001) were independent prognostic factors for increased PFS
Literature	GENERIF_SUMMARY	Data indicate that PYK2 N-terminal domain interacting receptor 1 (Nir1) could induce epithelial-mesenchymal transition by stabilising Snail via the PI3K/Akt/GSK3beta/Snail signalling pathway through binding to CCL18.
Literature	GENERIF_SUMMARY	mutations in KRAS codon 61, KRAS codon 146, BRAF, NRAS, or PIK3CA detected in Asian patients were not predictive of clinical benefits from cetuximab treatment.
Literature	GENERIF_SUMMARY	PIK3CA gene mutations have been shown to predict a better otcome in ER/HER2 negative breast cancer, treated with endocrine therapy.
Literature	GENERIF_SUMMARY	Data indicate that dual targeted inhibitors of PI3K/mTOR in combination with inhibitors of RAS/ERK signalling as a potentially effective approach to treating ovarian cancer.
Literature	GENERIF_SUMMARY	PIK3CA mutation does not predict benefit from rofecoxib therapy, but it does predict

Main Category	Category	Terms
		reduced colorectal carcinoma recurrence in regular users of low-dose aspirin after diagnosis.
Literature	GENERIF_SUMMARY	suggesting that the changes in AKT/BCL2 signaling molecules might mediate growth and anti-apoptosis processes at least in part, thus increasing the survival potential of Congenital Giant Nevi melanocytes and maintaining their integrity
Literature	GENERIF_SUMMARY	Some mutations in the gene encoding for the catalytic subunit of phosphatidyl-inositol-3-kinase, the PIK3CA gene, plays an important role, especially in the resistance to targeted therapies.
Literature	GENERIF_SUMMARY	Heavy metals in cigarette smoke induce PI3K expression in epithelial cells and tumor cells.
Literature	GENERIF_SUMMARY	Induced apoptosis mainly via the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Its signal plays a crucial role in tumor genesis and tumor progression by promoting cell proliferation and inhibiting apoptosis.
Literature	GENERIF_SUMMARY	Phosphoinositide 3-kinase (PI3K) is required for PSGL-1-induced beta1 integrin clustering which ultimately regulates beta1 integrin-mediated Jurkat cell adhesion to fibronectin.
Literature	GENERIF_SUMMARY	PIK3CA gene mutations were found in 13 (16%), of 28 patients with squamous cell anal cancer
Literature	GENERIF_SUMMARY	Results show PI3KCA mutations present in 34% of human breast cancer patients and significantly higher PIK3CA expression in breast cancer tissues in comparison with control tissues.
Literature	GENERIF_SUMMARY	In a Chinese cohort, rs7646409 genotype distributions between the osteosarcoma and control groups was significantly different (P = 0.032). Taking genotype TT as a reference, the risk of osteosarcoma increased three fold in patients with genotype CC.
Literature	GENERIF_SUMMARY	The frequently discordant genotypes of usual ductal hyperplasia/columnar cell change and concurrent carcinoma support a role for PIK3CA-activating point mutations in breast epithelial proliferation, perhaps more so than transformation.
Literature	GENERIF_SUMMARY	A screen of a cohort of radial scars for a large panel of activating point mutations,

Main Category	Category	Terms
		revealing PIK3CA mutations in over 60% of specimens.
Literature	GENERIF_SUMMARY	In chemotherapy-refractory colorectal cancer, neither PIK3CA mutation status nor PTEN expression were prognostic, nor were they predictive of benefit from cetuximab
Literature	GENERIF_SUMMARY	PIK3CA mutations and PIK3R1 underexpression show opposite effects on patient outcome and could become useful prognostic and predictive factors in breast cancer.
Literature	GENERIF_SUMMARY	Overall tumor PIK3CA mutation status is not associated with stage III colon cancer prognosis. PIK3CA mutation does not appear to serve as a predictive tumor molecular biomarker for response to irinotecan-based adjuvant chemotherapy.
Literature	GENERIF_SUMMARY	PI3KCA mutation was associated with the absence of Her2/neu amplification/ overexpression and a low level of MIB1/ Ki-67 labelling
Literature	GENERIF_SUMMARY	PIK3CA gene hypomethylation plays a key role in the tumorigenesis and development of esophageal cancer in Chinese patients, while the mutations of PIK3CA gene have little effect on the development of esophageal cancer
Literature	GENERIF_SUMMARY	Knockdown of RPS7 resulted in increased expression of P85alpha, P110alpha, and AKT2.
Literature	GENERIF_SUMMARY	This allows the usage of p110alpha-specific PI3K inhibitors and thus may spare patients the toxicities of pan-PI3K inhibition to allow maximal dosage and efficacy.
Literature	GENERIF_SUMMARY	PIK3CA exon 9 or exon 20 missense mutations were more commonly observed in cases that were deficient for mismatch repair protein expression (p=0.0058) and showed loss of ARID1A expression
Literature	GENERIF_SUMMARY	Data iindicate that he combination therapy of mTOR inhibitor RAD001 and PI3K (phosphatidylinositol 3-kinase) inhibitor LY294002 decreased the cell growth in vitro and progression of liver metastasis in vivo.
Literature	GENERIF_SUMMARY	Findings show that PPP1R1B-STARD3 fusion transcript has a key role in subsets of gastric cancers through the activation of PI3K/AKT signaling.
Literature	GENERIF_SUMMARY	Our study demonstrates relatively low frequency of PIK3CA mutations and amplifications in ovarian clear cell

Main Category	Category	Terms
		carcinoma.
Literature	GENERIF_SUMMARY	Dengue virus 2 subgenomic RNA induces apoptosis through the Bcl-2-mediated PI3k/Akt signaling pathway.
Literature	GENERIF_SUMMARY	The Pik3ca(H1047R) mutation promotes the development of highly aggressive and invasive adenocarcinomas in intestine.
Literature	GENERIF_SUMMARY	Data indicate that responses were noted in patients with HER2-amplified breast cancer resistant to trastuzumab, HER2-mutant non-small-cell lung cancer, and tumor types without identified mutations in the HER-PI3K-mTOR pathway.
Literature	GENERIF_SUMMARY	Data indicate that seven genes showed statistical enrichment for mutation
Literature	GENERIF_SUMMARY	Low expression of miR-1 and high expression of PIK3CA are closely related to the invasion, progression and metastasis of non-small-cell lung cancer.There is a negative correlation between the expression of miR-1 and PIK3CA in non-small-cell lung cancer.
Literature	GENERIF_SUMMARY	PIK3CA mutations/amplification, HRAS mutation, or PTEN loss that likely lead to PI3K pathway activation were identified in 34 of 75 cases (45%) and did not correlate with disease specific survival.
Literature	GENERIF_SUMMARY	PIK3CA gene alteration in non-muscle invasive bladder tumors predicts reduced recurrence.
Literature	GENERIF_SUMMARY	blocking PI3K/Akt signaling pathway by LY294002 or Akt siRNA could remarkably inhibit the PAK1 activation and cell invasion has to
Literature	GENERIF_SUMMARY	patients with PIK3CA-mutated tumors showed trend towards favorable overall survival (P = .085) but not disease-free survival (P = .238), suggesting that PIK3CA gene status may be favorable predictive marker in esophageal squamous cell carcinoma patients
Literature	GENERIF_SUMMARY	This paper presents an overview of PIK3CA mutations in the path PI3K/PTEN/Akt of the most common cancers.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with non--small-cell lung cancer.
Literature	GENERIF_SUMMARY	Affected tissue from individuals with facial infiltrating lipomatosis contains PIK3CA mutations that have previously been reported in cancers and in affected tissue

Main Category	Category	Terms
		from other nonheritable, overgrowth disorders.
Literature	GENERIF_SUMMARY	AFP can promote the proliferation of hepatoma cells via activation of PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	Data indicate that PI3K signaling pathway was critical for the anti-myeloma activity of BENC-511.
Literature	GENERIF_SUMMARY	Data indicate that siRNA-mediated silencing of PIK3CA blocked the inhibitory effect of miR-375 on Colorectal cancer (CRC) cell growth.
Literature	GENERIF_SUMMARY	PIK3CA mutations showed a trend toward an association with an increased risk of death compared to absence of those mutations in African American colorectal cancer patients.
Literature	GENERIF_SUMMARY	Te structure and function of the p110alpha enzyme and two of its most common oncogenic mutations. [Review]
Literature	GENERIF_SUMMARY	These results suggest that the combination of PI3K inhibition with dual HER2 blockade is necessary to circumvent the resistance to HER2 inhibitors conferred by PIK3CA mutation and also provides benefit to HER2+ tumors with wild-type PIK3CA tumors.
Literature	GENERIF_SUMMARY	this is the first reported case to our knowledge of a patient with both a possible germline EGFR T790M mutation and a somatic PIK3CA mutation.
Literature	GENERIF_SUMMARY	PIK3CA mutations do not have clinical validity to predict intrinsic resistance to adjuvant tamoxifen and may therefore be unsuitable as companion diagnostic for PI3K/AKT/mTOR inhibitors in ERalpha-positive, postmenopausal, early breast cancer patients.
Literature	GENERIF_SUMMARY	mutations may confer resistance to epidermal growth factor receptor-tyrosine kinase inhibitors in lung adenocarcinoma patients with mutated EGFR
Literature	GENERIF_SUMMARY	molecular or pharmacologic inhibition of the Lyn-PI3K/AKT pathway, markedly increased the sensitivity of the otherwise chemoresistant Cbl mutant-JMML cells to chemotherapeutic agents currently used in the treatment of JMML patients
Literature	GENERIF_SUMMARY	PIK3CA mutation was found to be a frequent genetic change in all breast cancer subtypes but occurred with the highest rate

Main Category	Category	Terms
		in HR(+)/HER2(-) tumors
Literature	GENERIF_SUMMARY	miR-1 may play an important role in the pathogenesis of NSCLC by regulating PIK3CA through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	frequent PIK3CA amplification and promoter methylation of RASSF1A and PAX6 genes in gastric cancers
Literature	GENERIF_SUMMARY	Studied the expression of phosphatidylinositol 3 kinase (PI3-K), protein kinase B (PKB) and glycogen synthase kinase 3 beta (GSK-3 beta) in skeletal muscle tissue of gestational diabetes mellitus patients.
Literature	GENERIF_SUMMARY	Studied the PIK3CA gene expression status and hotspot mutations in six cases of SDC by immunohistochemistry and genomic DNA sequencing.
Literature	GENERIF_SUMMARY	Data indicate that oncogenic activation of the phosphoinositide 3-kinase (PI3K) pathway involving phosphoinositide-3-kinase, catalytic, alpha polypeptide (PIK3CA) gene mutations can contribute to the progression of mucin-producing neoplasms
Literature	GENERIF_SUMMARY	PIK3CA mutations frequently coexist with EGFR/KRAS mutations and may have a role in survival and prognosis
Literature	GENERIF_SUMMARY	Data show the expression of N-myc downstream-regulated gene 2 (NDRG2) is frequently downregulated in adult T-cell leukaemia-lymphoma, resulting in enhanced phosphorylation of PTEN and activation of the phosphatidylinositol 3-kinase (PI3K)-AKT pathway.
Literature	GENERIF_SUMMARY	These results reveal betaig-h3 enhances the metastasis potentials of human osteosarcoma cells via integrin alpha2beta1-mediated PI3K/AKT signal pathways.
Literature	GENERIF_SUMMARY	PIK3CA mutations are more common in HPV(+) OPSCC and are associated with activation of mTOR, but not AKT. These data suggest that inhibitors for mTOR may have activity against HPV(+) PIK3CA mutant oropharyngeal cancers.
Literature	GENERIF_SUMMARY	Data suggest that the expression of ATP-binding cassette sub-family G member 2 (ABCG2) and the side population may be regulated by phosphatase and tensin homologue deleted on chromosome -10 (PTEN) through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	NGF regulates the expression of VEGF

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		through the PI3K/mTOR signaling pathway in human HaCaT keratinocytes.
Literature	GENERIF_SUMMARY	Our data indicate that activation of the PI3K/AKT pathway evidenced on the protein level might be a valuable prognostic marker to stratify patients for their risk of tumour recurrence.
Literature	GENERIF_SUMMARY	Inhibition of PI3K (and mTOR) leads to an increase in catalase expression in MCF-7 breast cancer cells.
Literature	GENERIF_SUMMARY	PIK3CA mutations play a critical role in the genesis of rhabdomyosarcoma
Literature	GENERIF_SUMMARY	PIK3CA gene was found to be mutated in 11 (22%) squamous cell anal carcinoma cases.
Literature	GENERIF_SUMMARY	The present experiments demonstrated a critical role for the GPRC6A-ERK1/2 and PI3K/Akt signaling pathway in arginine-mediated fibroblast survival.
Literature	GENERIF_SUMMARY	In this review, we discuss the emerging evidence for a functional relationship between the IGF-I/PI3K/AKT/mTOR pathway and ASDs, as well as a possible role of this signaling pathway in the diagnosis and treatment of ASDs. [review]
Literature	GENERIF_SUMMARY	Mutations in PIK3CA predict resistance to anti-EGFR therapies.
Literature	GENERIF_SUMMARY	Among the 42 patients who were seropositive for hepatitis B, 17 (40.5%) had HBx mutations and 4 (9.52%) had mutations in PIK3CA.
Literature	GENERIF_SUMMARY	Increased PTGS2 expression or the presence of mutated PIK3CA did not predict benefit from aspirin in colon cancer.
Literature	GENERIF_SUMMARY	the outcome of 68 patients with advanced colorectal cancer and RAS, BRAF and PI3KCA status according to ALK gene status, is reported.
Literature	GENERIF_SUMMARY	We examined the response of 80 samples of primary cells from acute myeloid leukemia patients to selective inhibitors of the phosphatidylinositol 3 kinase (PI3K)/ Akt/mammalian target of rapamycin (mTOR) axis.
Literature	GENERIF_SUMMARY	Data indicate PI3K/Akt signaling as an essential pathway regulating the distribution of nucleolin.
Literature	GENERIF_SUMMARY	Consistent expression of the PI3K catalytic isoform PI3K p110alpha was detected in a panel of glioblastoma patient samples.

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		Inhibition of PI3K p110alpha activity impaired the anchorage-dependent growth of glioblastoma cells and induced tumor regression in vivo.
Literature	GENERIF_SUMMARY	Protein expression of PI3K, Akt and survivin were significantly associated with the development, progression and metastasis of gastric cancer and may have value as diagnostic and prognostic markers in gastric cancer.
Literature	GENERIF_SUMMARY	Aberrant activation of the PI3K/Akt/mTOR axis acts in parallel to RCAN1 in phosphorylating tau, in DS and DS/AD.
Literature	GENERIF_SUMMARY	Increased SOX2 gene copy number is associated with FGFR1 and PIK3CA gene gain in non-small cell lung cancer and predicts improved survival in early stage disease
Literature	GENERIF_SUMMARY	PI3K-dependent binding of FOXO to the HBP1 promoter is part of a pathway that controls cell proliferation.
Literature	GENERIF_SUMMARY	PIK3CA isoform is of paramount importance for constitutive Akt activity in multiple myeloma , and - in contrast to inhibition of other class I isoforms - only the blockade of PIK3CA was sufficient to induce cell death in a sizeable subgroup of MM samples.
Literature	GENERIF_SUMMARY	TEIF is a downstream effector in EGF/PI3K/Akt signaling.
Literature	GENERIF_SUMMARY	Data demonstrate that the BRAF(V600E) cooperates with either PIK3CA(H1074R) or with silencing of the tumor-suppressor PTEN, to promote development of anaplastic thyroid carcinoma.
Literature	GENERIF_SUMMARY	This study indicated that mutations of KRAS, PIK3CA and NRAS were rare in advanced gastric cancer.
Literature	GENERIF_SUMMARY	We have identified no significant relationship between PIK3CA amplification, race (p= 0.630) and histological type (p=0. 731) in breast carcinoma, but correlation of PIK3CA amplification and age showed a significant relationship (p=0. 003) between them
Literature	GENERIF_SUMMARY	These data indicate that HIV-1 might also promote PI3K signalling to balance processing of its transcripts by regulating phosphorylation of splicing regulatory proteins.
Literature	GENERIF_SUMMARY	E4-ORF1 interacts with Dlg1 and PI3K to assemble a ternary complex where E4-ORF1

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		hijacks the Dlg1 oncogenic function to relocate cytoplasmic PI3K to the membrane for constitutive activation.
Literature	GENERIF_SUMMARY	FAM3A plays crucial roles in the regulation of glucose and lipid metabolism in the liver, where it activates the PI3K-Akt signaling pathway by way of a Ca(2+) /CaM-dependent mechanism
Literature	GENERIF_SUMMARY	Our results indicate that, besides HIF-1alpha, several other pathways, including PI3K/Akt, may be involved in the response to hypoxia of immature and, more specifically, of mature DC to sustain their trafficking and functions within hypoxic microenvironments.
Literature	GENERIF_SUMMARY	Dual MAPK and PI3K/Akt inhibition was detected.
Literature	GENERIF_SUMMARY	The expression of PIK3CA and survivin had a significant correlation with pathological grade and lymph node metastasis in the primary adenosquamous lung carcinoma patients.
Literature	GENERIF_SUMMARY	Chinese colorectal cancer patients with either KRAS or PIK3CA mutations are more susceptible to distant metastasis.
Literature	GENERIF_SUMMARY	We identified differential therapeutic vulnerabilities that emerge upon the loss of both PTEN and p53, and observed that combined inhibition of PARP and PI3K provides increased efficacy in hormone-insensitive advanced prostate cancer.
Literature	GENERIF_SUMMARY	MCP-1 was shown to have a role in activation of the ER1 by phosphorylating Ser167 via the PI3K/Akt/mTORC1 signaling pathway
Literature	GENERIF_SUMMARY	Further experiments demonstrated that phosphatidylinositide 3-kinases/protein kinase B/mammalian target of rapamycin/ p70S6 kinase (PI3K/AKT/mTOR/p70S6K) and extracellular regulated protein kinases (ERK).
Literature	GENERIF_SUMMARY	Data indicate that upregulation of TRPM cation channel TRPM8 promoted epithelial-mesenchymal transition of breast cancer cells via activating c-akt protein/glycogen synthase kinase 3 beta (GSK-3beta) pathway.
Literature	GENERIF_SUMMARY	Data indicate that Src kinase participates in the regulation of apoptosis through Src/PI-3 K/Akt signaling pathway in Hep-2 cell line.
Literature	GENERIF_SUMMARY	PIK3CA single nucleotide polymorphism

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		rs17849071 GT+GG genotype was inversely associated with P53 protein accumulation in breast cancer samples.
Literature	GENERIF_SUMMARY	PIK3CA mutations were observed in 11 of 79 cases (13.92%), including a rare mutation, Q546 P, that had not previously been reported in OSCC.
Literature	GENERIF_SUMMARY	Data indicate that combined PI3K inhibitor LY294002 and mitogen activated protein kinase kinase (MEK) inhibitor AZD6244 showed synergistic effect on inhibition of SKOV3/DDP cell growth by inducing apoptosis and blocking cell cycle.
Literature	GENERIF_SUMMARY	Findings implicate APOBEC3B activity as a key driver of PIK3CA mutagenesis and HPV-induced transformation.
Literature	GENERIF_SUMMARY	sequencing analysis revealed missense mutations in PIK3CA, and TP53 genes in the breast cancer samples of various histologic types
Literature	GENERIF_SUMMARY	Shorter GT dinucleotide repeat in the intron 1 of PIK3CA gene is associated with increased risk of breast cancer in Iranian women.
Literature	GENERIF_SUMMARY	the transcriptional activity of nuclear beta-Catenin depends on PI3K activity and both factors synergize for full WNT signaling activity and are associated with distant metastasis in colon cancer
Literature	GENERIF_SUMMARY	our results demonstrate that activation of the PI3K/Akt/mTOR/p70S6K signaling pathway is involved in S100A4-induced viability, migration, upregulation of VEGF and downregulation of E-cadherin in CRC cells.
Literature	GENERIF_SUMMARY	Results show that thioridazine regulates endothelial cell function and subsequent angiogenesis by inhibiting VEGFR-2/PI3K/mTOR signal transduction.
Literature	GENERIF_SUMMARY	Data indicate that malic enzyme 2 knockdown impacts phosphatidylinositol 3-kinases/proto-oncogene protein akt (PI3K/AKT) signaling.
Literature	GENERIF_SUMMARY	Results show that cell stimulation promotes p110alpha and p110beta association, demonstrating oligomerization of PI3K catalytic subunits within cells.
Literature	GENERIF_SUMMARY	Data indicate that combination of epidermal growth factor (EGFR) inhibitors with either the phosphatidylinositol 3-kinases inhibitor ZSTK474 or the mTOR serine-threonine

Main Category	Category	Terms
		kinases inhibitor sirolimus showed increased activity.
Literature	GENERIF_SUMMARY	Knockdown of AKT3 and/or PI3KCA genes in T98G cells led to a significant reduction in cell viability, the accumulation of subG1-phase cells and, a reduced fraction of cells in the S and G2/M phases.
Literature	GENERIF_SUMMARY	High expression of PI3K is associated with sinonasal and laryngeal small-cell neuroendocrine carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with classical markers of good prognosis and signatures of PI3K pathway activity. The presence of a PIK3CA mutation does not preclude a response to neoadjuvant anastrozole treatment.
Literature	GENERIF_SUMMARY	The combination of PI3K and CDK 4/6 inhibitors overcomes intrinsic and adaptive resistance leading to tumor regressions in PIK3CA mutant xenografts.
Literature	GENERIF_SUMMARY	Different targeted therapies may be necessary for Triple-negative breast cancer that involves activation of PIK3CA and AKT1 .
Literature	GENERIF_SUMMARY	In tDNA, 50 mutations (36 EGFR, 5 ERBB2, 4 KRAS, 3 BRAF, and 2 PIK3CA) were identified, of which 26 were detected in cfDNA. These data demonstrate the feasibility and potential utility of mutation screening in the detection of a range of tumor biomarkers
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with good response to docetaxel plus epirubicin neoadjuvant chemotherapy in breast cancer.
Literature	GENERIF_SUMMARY	We detected a mosaic disease-causing mutation in PIK3CA by sequencing of DNA extracted from cultured amniotic cells.
Literature	GENERIF_SUMMARY	Studies indicate the importance of the EGFR/PI3K/PTEN/Akt/mTORC1/GSK-3 pathway in breast cancer.
Literature	GENERIF_SUMMARY	PIK3CA mutations were not correlated with clinicopathological characteristics. However in the esophageal squamous cell carcinoma patients with family cancer history, PIK3CA mutations were independently correlated with worse overall survival.
Literature	GENERIF_SUMMARY	PTEN was neither a significant prognostic nor predictive factor. PIK3CA mutations were an adverse prognostic factor for survival but not predictive for lapatinib

Main Category	Category	Terms
		benefit.
Literature	GENERIF_SUMMARY	we have found that the PI3Ka/b isoforms and AKT are involved in the downstream mechanisms leading to actin polymerization, ATP secretion and aggregation of human platelets stimulated by 2-AG.
Literature	GENERIF_SUMMARY	KRAS, BRAF, and PIK3CA mutations were evaluated in 204 colorectal carcinoma samples; the frequency of KRAS, BRAF, and PIK3CA mutations was 23.5, 9.8, and 5.9 %, respectively.
Literature	GENERIF_SUMMARY	A significant percentage of Uterine malignant mesodermal mixed tumors, Uterine clear cell carcinomas, and Uterine serous carcinomas have mutations in PIK3CA.
Literature	GENERIF_SUMMARY	The presented data subclassified colorectal cancers (CRCs) based on their activated signaling pathways and identify a role for c-MET and IGF1R-driven PI3K signaling in CRCs, which is superior to KRAS mutational tests alone.
Literature	GENERIF_SUMMARY	The presence of PIK3CA and RAS mutations and other alterations affecting the mTOR pathway activity in HNSCC could be exploited to predict the potential resistance to cetuximab
Literature	GENERIF_SUMMARY	PIK3CA mutations in androgen receptor-positive triple negative breast cancer confer sensitivity to the combination of PI3K and androgen receptor inhibitors.
Literature	GENERIF_SUMMARY	PIK3CA mutation status are not associated with recurrence, metastatic disease or progression in endocrine-treated breast cancer.
Literature	GENERIF_SUMMARY	Germline SNPs in the PI3 K-AKT-mTOR pathway were associated with breast cancer.
Literature	GENERIF_SUMMARY	It is anticipated that through an improved understanding of the biology of the PI3K/Akt pathway in prostate cancer, relevant biomarkers and rationale combination therapies will optimize targeting of this pathway
Literature	GENERIF_SUMMARY	other mutation was in exon 19 of the phosphatidylinositol-4,5-bisphosphate 3-kinase, catalytic subunit alpha (PIK3CA) gene, resulting in a G-A substitution at position 1624 (E542K)
Literature	GENERIF_SUMMARY	The PI3K/AKT/mTOR pathway is

Main Category	Category	Terms
		distinguishable in SCLC genomic alterations. Therefore, a sequencing-based comprehensive analysis could stratify SCLC patients by potential therapeutic targets.
Literature	GENERIF_SUMMARY	Studies indicate that the PI3K/PTEN/Akt/mTOR pathway is in the pathogenesis of malignancies and development of resistance to anticancer therapies, and inhibitors are a promising therapeutic option.
Literature	GENERIF_SUMMARY	These results provide a rationale for targeting mutant RAS or PI3K/AKT/mTOR signaling in the subset of Erdheim Chester disease patients with NRAS or PIK3CA mutations.
Literature	GENERIF_SUMMARY	E2-induced Nrf2 activation was completely suppressed by the PI3K inhibitors.
Literature	GENERIF_SUMMARY	PIK3CA mutations were significantly related to ER/PR expression.
Literature	GENERIF_SUMMARY	Studies indicate that phosphatidylinositol 3-kinase (PI3K) pathway involved in glioblastoma (GBM) pathogenesis, and PI3K pathway inhibitors can help optimize GBM vaccine immunotherapy.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with better outcomes in primary estrogen receptor positive disease and have been reported to be associated with resistance to human epidermal growth factor receptor 2 and endocrine therapies.
Literature	GENERIF_SUMMARY	Data show that IL-27 on IFN-gamma mediated autophagy perpetuates by simultaneous induction of kinases mTOR via JAK/PI3 K/Akt cascade and increment in myeloid cell leukemia sequence 1 (Mcl-1) level through PI3 K pathway.
Literature	GENERIF_SUMMARY	Data indicarte that Helicobacter pylori infection promotes protein synthesis through phosphatidylinositol 3 kinase/mTOR serine-threonine kinases(PI3K/Mtor).
Literature	GENERIF_SUMMARY	KRAS (41%) and PIK3CA (8%) mutations in Colorectal cancer in patient-derived xenograft models closely paralleled patient tumors (35 to 51% KRAS, 12 to 21% PIK3CA).
Literature	GENERIF_SUMMARY	HER2-positive breast carcinomas with a PIK3CA mutation are less likely to achieve a pCR after neoadjuvant anthracycline-taxane-based chemotherapy plus anti-HER2 treatment, even if a dual anti-HER2 treatment is given.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	In this review we present a state of art review on various classes of small molecule inhibitors selectively involving PI3K/Akt/mTOR pathway and their clinical potential in this disease.
Literature	GENERIF_SUMMARY	Mutations in the TP53 gene and to lesser extent in the PIK3CA gene seem more frequent in cervical adenocarcinoma than in squamous cell carcinoma and CIN3
Literature	GENERIF_SUMMARY	Aberrant activation of PI3K pathway leads to increased phosphorylation of a surprisingly wide variety of kinases and downstream signalling networks.
Literature	GENERIF_SUMMARY	Impaired EPCs proliferation could be reversed by BBR via the PI3K/AKT/eNOS signaling pathway.
Literature	GENERIF_SUMMARY	RI represents an important predictor of progression in patients with CRC and suppresses proliferation and metastasis in CRC cells through inhibiting PI3K/AkT pathway.
Literature	GENERIF_SUMMARY	High PI3K is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	The phosphoinositide 3-kinase/Akt signaling pathway was activated in cells overexpressing Sox2.
Literature	GENERIF_SUMMARY	Data indicate that quercetin induces apoptosis via phosphatidylinositol 3-kinase PI3k/Akt pathways, suggesting quercetin has potential to be used as an anti-tumor agent against human cervix cancer.
Literature	GENERIF_SUMMARY	These results suggest that the combination of a selective MEK inhibitor and a PI3K/mTOR inhibitor was effective in CRC harboring with KRAS and PIK3CA mutations.
Literature	GENERIF_SUMMARY	Report PIK3CA mutation status in colorectal neoplasms and neuroendocrine differentiation in primary/metastatic colorectal neoplasms.
Literature	GENERIF_SUMMARY	Studied the structure and dynamics of the PIK3CA wild-type and H1047R oncogenic mutant.
Literature	GENERIF_SUMMARY	HER3 knock down decreases cell proliferation preferentially in PIK3CA kinase domain mutation (H1047R) cells.
Literature	GENERIF_SUMMARY	Classification by PIK3CA and SOX2 protein expression is useful for predicting the prognosis of Asian patients with lung SQCC with stage I.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Studies indicate that phosphatidylinositol 3-kinases (PI3Ks) and phosphatidylinositol 3-kinase-related protein kinases (PIKKs) are two related families of kinases and share very similar kinase domains.
Literature	GENERIF_SUMMARY	PIK3CA hotspot mutations are present at a relatively high frequency in CTCs and their presence is associated with worse survival in patients with breast cancer with metastasis.
Literature	GENERIF_SUMMARY	PIK3CA gene mutation is a good indication of breast cancer development in early diagnosis of cancer.
Literature	GENERIF_SUMMARY	MALAT1 might suppress the tumor growth and metastasis via PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	Breast cancers harboring oncogenic PIK3CA activate SGK3 signaling while suppressing Akt, indicative of oncogenic functions for both INPP4B and SGK3 in these tumors.
Literature	GENERIF_SUMMARY	certain Akt/mTOR/4E-BP1 pathway signals could be novel therapeutic targets for Merkel cell carcinomas regardless of Merkel cell polyomavirus infection status
Literature	GENERIF_SUMMARY	Patients with HER2-amplified tumors were more likely to present PIK3CA mutations compared to patients with triple negative tumors.
Literature	GENERIF_SUMMARY	Data show that the combination of PI3 kinase inhibitors with ErbB receptors HER2/HER3 targeting antibodies may constitute a promising therapeutic strategy for prostate cancer.
Literature	GENERIF_SUMMARY	Data indicate that phosphatidylinositol-45-bisphosphate 3-kinase catalytic subunit alpha (PIK3CA) mutations have no negative impact on survival after surgery or systemic therapy. However
Literature	GENERIF_SUMMARY	Tim-3 triggers growth factor type responses in acute myeloid leukaemia cells by activating a phosphatidylinositol-3 kinase (PI-3K)/mammalian target of rapamycin (mTOR) pathway.
Literature	GENERIF_SUMMARY	Combination therapy of PI3Kis with IGF1R inhibitors for treating IGF1R-positive human cancers.
Literature	GENERIF_SUMMARY	Study sequenced a large clinical series of patients with metastatic colorectal cancer (mCRC) for RAS and PIK3CA mutations. RAS alterations occurred in 48% of mCRC cases, with mutations within KRAS exon 2 in

Main Category	Category	Terms
		43%, KRAS exons 3 and 4 in 3%, and NRAS in 2% of cases. PIK3CA mutations occurred in about 10% of mCRC and were significantly associated with concurrent RAS mutation.
Literature	GENERIF_SUMMARY	While P7170 inhibits PI3K and mTOR in ER+/HER2- human breast cancer cells and tumors ex vivo, in vivo data indicate that the primary mechanism of P7170 anti-tumor action is inhibition of mTOR and cell proliferation
Literature	GENERIF_SUMMARY	Data indicate that pretreatment of BEZ235 can confer the radiation-inducing phosphoinositide 3-kinase/mammalian target of rapamycin/AKT signaling pathway, and further inhibit the cell viability.
Literature	GENERIF_SUMMARY	Data indicate that ibrutinib significantly reduced PI3K-Akt activity in cardiac cells.
Literature	GENERIF_SUMMARY	AKT3, PI3KCA, and VEGFR2 silencing reduces the invasiveness of glioblastoma multiforme T98G cells.
Literature	GENERIF_SUMMARY	Drugs interfering with expression and activity of XIAP re-sensitize mutant PIK3CA colorectal cancer cells to TRAIL-induced cell death.
Literature	GENERIF_SUMMARY	self-renewal versus differentiation of human ESCs (hESCs) in response to Wnt signaling is predominantly determined by a two-layer regulatory circuit involving beta-catenin, E-cadherin, PI3K/Akt, and Slug in a time-dependent manner.
Literature	GENERIF_SUMMARY	The purpose of the study was to investigate the heterogeneity of PIK3CA mutational status within single circulating Tumor Cells isolated from individual metastatic breast cancer patients.
Literature	GENERIF_SUMMARY	Combined analysis of PIK3CA mutation and PTEN expression may serve as critical indicators to identify patients unlikely to respond to trastuzumab.
Literature	GENERIF_SUMMARY	The expression of both PIK3CA and PIK3CB is increased and related to the development and progress of colorectal carcinoma and multidrug resistance
Literature	GENERIF_SUMMARY	we will summarize the current knowledge about the role of PAM (PI3K/AKT/mTOR) signaling in medulloblastoma and discuss the strategies that are currently being evaluated with targeted agents against this pathway.
Literature	GENERIF_SUMMARY	High tumor PI3K signaling was correlated

Main Category	Category	Terms
		with adverse survival outcome in patients with rectal cancer.
Literature	GENERIF_SUMMARY	suggests that PIK3CA mutations account for a small fraction of PI3K pathway activation and have a limited impact in interfering with the BRAF/NRAS-driven growth in melanoma
Literature	GENERIF_SUMMARY	our results unveil a functional cross-talk between YB-1, DeltaNp63alpha and the PI3K/AKT pathway critically governing survival of squamous carcinoma cells.
Literature	GENERIF_SUMMARY	PIK3CA mutations predict recurrence in localized microsatellite stable colon cancer
Literature	GENERIF_SUMMARY	The present study emphasizes that the treatment for breast cancer should include molecular analysis of relevant driver mutations with priority to PIK3CA gene while taking clinical decisions in breast cancer patients.
Literature	GENERIF_SUMMARY	Somatic PIK3CA mutations are the most common cause of isolated LMs and disorders in which LM is a component feature.
Literature	GENERIF_SUMMARY	PIK3CA amplifications and mutations in Non-small cell lung cancer have been reported in 37% and 9%, respectively. Both are poor prognostic factors in Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA, SMAD4, and TP53 are most often involved in clonal divergence, providing potential targets that may help guide the clinical management of tumor progression or metastases.
Literature	GENERIF_SUMMARY	Lastly, 17.5% of hepatocellular carcinoma paradoxically had diffuse loss of the oncoprotein p110alpha compared with strong expression in background cirrhotic liver
Literature	GENERIF_SUMMARY	The PIK3CA hotspot mutations E542K, E545K, and H1047R were found to be significantly more prevalent in PIK3CA-mutant MSS (21/58, 36%) compared to PIK3CA-mutant MSI-H EECs (5/37, 13.5%; p=0.019).
Literature	GENERIF_SUMMARY	Data show that licochalcone A (LicoA) suppresses solar UV-induced cyclooxygenase (COX-2) expression by acting as a potent inhibitor of enzymes PI3K, MEK1, and B-Raf.
Literature	GENERIF_SUMMARY	findings identify PI3K/AKT pathway mutations as an important cause of

Main Category	Category	Terms
		epileptogenic brain malformations and establish megalencephaly, hemimegalencephaly, and focal cortical dysplasia as part of a single pathogenic spectrum.
Literature	GENERIF_SUMMARY	Mutations in the PI3K pathway seem therefore to be an important parameter to predict absence of response to cetuximab.
Literature	GENERIF_SUMMARY	Circulating tumor DNA containing PIK3CA mutations is highly predictive of shorter recurrence-free survival and overall survival in primary breast cancer patients.
Literature	GENERIF_SUMMARY	Neither all RAS gene mutation status nor PIK3CA mutation status was prognostic for PFS or OS.
Literature	GENERIF_SUMMARY	Overexpression of STAT1 or of a constitutively active PI3Kalpha mutant failed to induce basal IDO production
Literature	GENERIF_SUMMARY	PI3K is activated after imatinib secondary resistance in gastrointestinal stromal tumors.
Literature	GENERIF_SUMMARY	PIK3CA and AKT2 mutations occurred at low frequency in gastric cancer.
Literature	GENERIF_SUMMARY	Somatic mutations and copy number amplification of PIK3CA were associated with response to CCRT in patients with cervical squamous cell carcinoma.
Literature	GENERIF_SUMMARY	The frequency of mutations in either KRAS or PIK3CA were significantly higher in patients without lymph node metastasis than those with.
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	Study demonstrates the oncogenic cooperation between PI3K and Yap pathways along liver carcinogenesis. The PIK3CA/Yap mouse represents an important preclinical liver tumor model.
Literature	GENERIF_SUMMARY	These findings provide evidence that a miR-520a binding site polymorphism rs141178472 in the PIK3CA 3'-UTR may play a role in the etiology of colorectal cancer
Literature	GENERIF_SUMMARY	This result suggests polymorphic GT repeat of PIK3CA gene may be a potential predictive marker of colorectal cancer risk in Iranian population
Literature	GENERIF_SUMMARY	PIK3CA hotspot mutations (c.1624G>A [p.Glu542Lys] and c.1633G>A [p.Glu545Lys]) were enriched in APOBEC-signature

Main Category	Category	Terms
		esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	in a high number of HNSCC tumours, PI3K/Akt signalling is activated by different molecules or by the combination of these molecules.
Literature	GENERIF_SUMMARY	PIK3CA mutations often affect a limited fraction of transformed metastatic colorectal cancer cells.
Literature	GENERIF_SUMMARY	Data indicate that high-grade serous ovarian carcinoma (HGSC) cell lines with high GRB2 associated binding protein 2 (GAB2) expression are more responsive to phosphatidylinositol 3-kinases (PI3K) inhibitor PF-04691502.
Literature	GENERIF_SUMMARY	The PIK3CA mutations within our engineered cell model did not lead to enhanced oncogene-dependent cell death when treated with direct inhibition of the PI3K enzyme yet did show increased sensitivity compared with control with dual PI3K/mTOR inhibition.
Literature	GENERIF_SUMMARY	Despite the rare prevalence of the 5 most common PIK3CA mutations in Langerhans cell histiocytosis, this report highlights that the PI3K-AKT pathway could be implied in the pathogenesis of Langerhans cell histiocytosis.
Literature	GENERIF_SUMMARY	Among 264 patients, mutations in KRAS exon 2, KRAS exons 3 or 4, NRAS, BRAF and PIK3CA were detected in 34.1%, 3.8%, 4.2%, 5.4% and 6.4%, respectively. Thus, a total of 12.1% of patients without KRAS exon 2 mutations had other RAS mutations
Literature	GENERIF_SUMMARY	MiR-10b directly targets the 3'-UTRs of ZEB1 and PIK3CA.
Literature	GENERIF_SUMMARY	Data show that microRNA miR-21 plays an oncogenic role in diffuse large B-cell lymphoma (DLBCL) by modulating the PI3-Kinase/AKT protein/mTOR protein/forkhead box O1 protein (FOXO1) pathway at multiple levels resulting in strong prognostic implication.
Literature	GENERIF_SUMMARY	Data show that microRNA miR-451 target tuberous sclerosis 1 (TSC1) gene and activates the PI3-kinase/Akt protein/mTOR protein signaling in multiple myeloma (MM) Side population (SP) cells.
Literature	GENERIF_SUMMARY	Proteins induced by PIK3CA mutations correlated with EGFR signaling and reduced relapse-free survival in basal-like breast cancer.

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Literature	GENERIF_SUMMARY	PI3KCA gene polymorphism is associated with poor drug response in metastatic renal cell carcinoma.
Literature	GENERIF_SUMMARY	treatment with butein significantly increased reactive oxygen species (ROS) generation and reduced the phosphorylation of PI3K, AKT and mTOR expression, which contributes to the inhibition of the tumor growth of cervical cancer
Literature	GENERIF_SUMMARY	The prevalence of PIK3CA mutation status in colorectal cancer does not differ according to race/ethnicity, but may vary according to other relevant clinicopathologic and etiologic factors
Literature	GENERIF_SUMMARY	PIK3CA amplification may be an early and important event in colorectal carcinogenesis and may drive the development of left-side tubular adenomas independently with KRAS mutation.
Literature	GENERIF_SUMMARY	KRAS and PIK3CA genotype may be considered as negative predictive markers and should carry out the analysis expeditiously.
Literature	GENERIF_SUMMARY	our results demonstrate that miR-375 functions as a growth-suppressive miRNA and plays an important role in inhibiting the tumorigenesis through targeting PIK3CA in osteosarcoma.
Literature	GENERIF_SUMMARY	Data show that PI3 kinase is an attractive therapeutic target in malignancies induced by proto-oncogene protein c-Kit mutations.
Literature	GENERIF_SUMMARY	The aim of the present study was to obtain the first data for the phosphatidylinositol-4,5-bisphosphate 3-kinase (PIK3CA) mutation frequency among nonsmall-cell lung cancer (NSCLC) patients in Turkey.
Literature	GENERIF_SUMMARY	This is the first study identifying the PIK3CA mutation specifically in sporadic cribriform-morular variant of papillary thyroid carcinoma
Literature	GENERIF_SUMMARY	we also explored the downstream molecules of miR-19a/PIK3CA signaling. We found that Notch signaling was induced by upregulation of miR-19a, and inactivation of Notch signaling attenuated the cell proliferation, migration and invasion promoted by miR-19a
Literature	GENERIF_SUMMARY	observations during clinical tumor genotyping suggest that when other pro-oncogenic pathways are mutated along with PIK3CA, then, PIK3CA inhibition alone may

Main Category	Category	Terms
		not be effective and combination therapy may be warranted
Literature	GENERIF_SUMMARY	Loss of PIPKIgamma or its focal adhesion-targeting variant, PIPKIgammai2, impaired PI3K/Akt activation upon stimulation with growth factors or extracellular matrix proteins in different tumor cells.
Literature	GENERIF_SUMMARY	PIK3CA Mutation and Amplification is not associated with Clinicopathological Characteristics of Gastric Cancer.
Literature	GENERIF_SUMMARY	ata show that activation of EphB2 receptor kinase arrests tau protein hyperphosphorylation through phosphatidylinositol 3-kinase (PI3K)/Akt protein-mediated glycogen synthase kinase-3beta (GSK-3beta) inhibition.
Literature	GENERIF_SUMMARY	Analysis of interdomain interactions of wild-type and mutants at the p110alpha-p85alpha interface with molecular dynamics simulations suggest that tumor-associated mutations weaken interactions between p110alpha and p85alpha by disrupting key stabilizing interactions.
Literature	GENERIF_SUMMARY	It also inhibited extrinsic pathway protein caspase 8, growth-related protein PI3K and further activated PTEN in combination with cisplatin.
Literature	GENERIF_SUMMARY	Mutations of PIK3CA occur in approximately 20% of urothelial carcinoma ,and often include gain-of-function mutations that involve the helical and kinase domain.
Literature	GENERIF_SUMMARY	Genomic amplifications through copy gain of the PI3KCA have been frequently found in Follicular thyroid carcinoma (in 24% of tumors) and Anaplastic thyroid carcinoma (in 42% of tumors).
Literature	GENERIF_SUMMARY	Data show that immunoglobulin-like transcript 4 (ILT4) increases the expression of the co-inhibitory molecule B7-H3 through PI3K/AKT/mTOR signalling.
Literature	GENERIF_SUMMARY	we also observed frequent PIK3CA mutations in Tiawanese oral squamous cell carcinoma patients.
Literature	GENERIF_SUMMARY	HSP60 overexpression increases the protein levels of the p110alpha subunit of phosphoinositide 3-kinase and c-Myc
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	Studied how rapamycin affects CCN2 expression and migration of lung fibroblasts in vitro; PI3K and inhibitors were used to

Main Category	Category	Terms
		determine rapamycin's mechanism of action.
Literature	GENERIF_SUMMARY	the current study illustrated that downregulation of miR203 and overexpression of PIK3CA were novel potential prognosis predictors of gastric cancer patients
Literature	GENERIF_SUMMARY	Found no association between PIK3CA mutational status and other clinical/ biological parameters in breast cancer treated with adjuvant trastuzumab.
Literature	GENERIF_SUMMARY	there is a key effect of PIK3CA(H1047R) on mammary cell fate in the pre-neoplastic mammary gland and the cell of origin of PIK3CA(H1047R) tumours dictates their malignancy, thus revealing a mechanism underlying tumour heterogeneity and aggressiveness
Literature	GENERIF_SUMMARY	oncogenic Pik3ca(H1047R) activates a multipotent genetic program in normally lineage-restricted populations at the early stage of tumour initiation, setting the stage for future intratumoural heterogeneity
Literature	GENERIF_SUMMARY	Studied the global phosphotyrosine proteome of isogenic knock-in cell lines containing two hotspot mutations in PIK3CA (E545K and H1047R); 824 unique phosphopeptides were identified.
Literature	GENERIF_SUMMARY	Findings indicate inositol polyphosphate 5-phosphatase PIPP (INPP5J) as a suppressor of oncogenic phosphoinositide 3-kinase (PI3K)/AKT signaling in breast cancer.
Literature	GENERIF_SUMMARY	PIK3CA Mutations are associated with Head and Neck Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	Distinct beta-catenin and PIK3CA mutation are linked in endometriosis-associated ovarian endometrioid and clear cell carcinomas
Literature	GENERIF_SUMMARY	It plays a role in cell survival, cell differentiation and cell proliferation in neoplasms.
Literature	GENERIF_SUMMARY	Data show that leukemia inhibitory factor (LIF) signaling promote chemoresistance in cholangiocarcinoma by up-regulating myeloid cell factor-1 (Mcl-1) via phosphatidylinositol 3-kinase (PI3K)/c-akt protein (AKT)-dependent pathway.
Literature	GENERIF_SUMMARY	assessed the clinical impact of changes in the frequency and topology of PIK3CA mutation and HER2 (ERBB2) amplification

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		within HER2-positive breast cancer during neoadjuvant the
Literature	GENERIF_SUMMARY	Results show that oncogenic PIK3CA mutations in HER2/neu overexpressing uterine serous carcinoma cells are common and may constitute a major mechanism of resistance to trastuzumab.
Literature	GENERIF_SUMMARY	Data show that dual epidermal growth factor receptor 2 HER2/phosphatidylinositol 3-kinase PIK3CA blockade may be a drug therapy for uterine serous carcinoma (USC) with HER2 gene amplification and mutated wild-type PIK3CA resistant to chemotherapy.
Literature	GENERIF_SUMMARY	analysis of ARID1A, CDH1, cMET and PIK3CA expression and target-related microRNA expression in gastric cancer
Literature	GENERIF_SUMMARY	A concurrent PIK3CA mutation is a poor prognostic factor in patients with advanced EGFR-mutant or KRAS-mutant lung adenocarcinomas.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	Somatic mutations in PIK3CA have roles in PIK3CA-related somatic overgrowth spectrum
Literature	GENERIF_SUMMARY	We describe the case of a patient in whom we were able to confirm a causative PIK3CA mutation.
Literature	GENERIF_SUMMARY	the presence of PIK3CA major mutations of cell-free DNA could be a discriminatory predictor of survival in early-stage triple negative breast cancer
Literature	GENERIF_SUMMARY	Data show that rapamycin-insensitive companion of mTOR protein (RICTOR) plays a central role in phosphatidylinositol 3-kinase (PI3K)/proto-oncogene protein c-akt (AKT) pathway in melanocytes and its deregulation could be involved in melanoma development.
Literature	GENERIF_SUMMARY	patients with mutant PIK3CA had distinct characteristics in age, menopausal status, and histological subtype and have better treatment outcome and less distant metastasis after surgery-based multimodal therapy.
Literature	GENERIF_SUMMARY	Overexpression of PI3K-p110alpha is associated with progression of uterine cervical neoplasia, and the expression of pAkt and DJ-1 is positively correlated with PI3K-p110alpha expression in this process.
Literature	GENERIF_SUMMARY	Down-regulation of PIK3 is associated with

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		osteoporosis.
Literature	GENERIF_SUMMARY	29b lowers postoperative IOP and sustains the function of filtering bleb. It inhibits the proliferation of fibroblasts and reduces collagen deposition by repressing the PI3K/Akt/Sp1 pathway in rabbits subjected to GFS.
Literature	GENERIF_SUMMARY	Data show that apoptotic cell death by esculetin was associated with inhibition of the EGFR/PI3K/Akt signaling pathway and the decreased expression of nucleophosmin (NPM).
Literature	GENERIF_SUMMARY	no prognostic significance of specific PIK3CA mutations in breast cancer treated with adjuvant chemotherapy
Literature	GENERIF_SUMMARY	We have observed a 4% oncogenic mutation frequency of the PIK3CA gene, which plays a minor role in the development of OSCC in an Indian population.
Literature	GENERIF_SUMMARY	Data show that RNA silencing of CCN family member 1 protein (CCN1) inhibits umbilical vein endothelial cell (HUVEC) proliferation under hypoxic conditions by inhibiting phosphoinositide 3-kinase (PI3K)/AKT protein signaling.
Literature	GENERIF_SUMMARY	Results indicate that resveratrol could increase the anti-proliferative activity of bestatin through downregulating P-gp expression via suppressing the PI3K/Akt/mTOR signaling pathway.
Literature	GENERIF_SUMMARY	Data show that synergistic apoptotic effect of the combination treatment is correlated with the block of the PI3K/Akt signal pathway in breast cancer cells.
Literature	GENERIF_SUMMARY	Data show that microRNA miR-221 targets PTEN protein, leading to activation of the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Somatic mutations affecting PI3K pathway genes were found to be highly prevalent in infiltrating epitheliosis, suggesting that these lesions may be neoplastic rather than hyperplastic.
Literature	GENERIF_SUMMARY	Data show that low-level phosphoinositide-3-kinase, catalytic, alpha polypeptide (PIK3CA) mutations were detectable in the serum indicating the utility of cell free DNA (cfDNA) as a DNA source to detect mutations in metastatic biliary cancers.
Literature	GENERIF_SUMMARY	Data show that puquitinib mesylate (XC-302) induced autophagy in

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		nasopharyngeal cancer cells CNE-2 by promoting cell autophagy and inhibits the phosphatidylinositol 3-Kinase (PI3K)/AKT protein/TOR serine-threonine kinases (mTOR) signaling pathway.
Literature	GENERIF_SUMMARY	Data suggest that the overexpression of microRNA miR16 inhibits the proliferation, invasion and metastasis of hepatocellular carcinoma cells associated with the
Literature	GENERIF_SUMMARY	Results indicate that p110alpha primarily mediates PI3K/AKT signaling in HER2+ PTEN deficient breast tumors whereas p110beta conditionally mediates HER2/PI3K signaling in the absence of p110alpha.
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with response to therapy in Breast Cancer.
Literature	GENERIF_SUMMARY	Data show that the combination of a phosphatidylinositol 3-Kinase (PI3K)/mTOR protein inhibitor with an androgen receptor {AR) inhibitor results in a synergistic anti-tumor response in non-castration-resistant prostate cancer (CRPC) and CRPC models.
Literature	GENERIF_SUMMARY	Data indicate the role of tropomyosin-related kinase-B (TrkB) in activation of the interleukin-6/Janus kinase 2/STAT3 transcription factor and PI3 kinase/c-AKT proto-oncogene protein pathway in breast cancer.
Literature	GENERIF_SUMMARY	Mutations in PIK3CA gene are associated with esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA mutations and PI3K/AKT pathway activation are common events in TNBC.
Literature	GENERIF_SUMMARY	High PI3K p110alpha expression contributes to MET inhibitor resistance in gastric carcinoma xenografts.
Literature	GENERIF_SUMMARY	Cytokines IL-2/7/9 stimulation activates PI3K/Akt pathways downstream of Ras in RasGRP1 T-cell acute lymphoblastic leukemia (T-ALL).
Literature	GENERIF_SUMMARY	miR-490-5p functions as a tumour suppressor in renal carcinoma by targeting PIK3CA
Literature	GENERIF_SUMMARY	Findings indicate that Epstein-Barr virus(EBV)-encoded latent membrane protein 1 (LMP1), PI3K/AKT, miR-21 and PTEN constitute a positive feedback loop and have a key role in LMP1-induced cancer stem cells (CSCs) in nasopharyngeal carcinoma (NPC).
Literature	GENERIF_SUMMARY	Increased Expression of

Main Category	Category	Terms
		Phosphatidylinositol 3-Kinase p110alpha and Gene Amplification of PIK3CA is associated with Nasopharyngeal Carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA mutations were observed in 32% (8) of the 25 breast cancer tissues examined, Sequencing of exon 2 of KRAS suggested that 20% (5) of the 25 samples harbored a mutation and 16% (4) of BRAF harbored a mutation.
Literature	GENERIF_SUMMARY	miR-203 represses NF-kappaB signaling via targeting TAK1 and PI3KCA and miR-203 overexpression may contribute to the COPD initiation.
Literature	GENERIF_SUMMARY	Somatic activating PIK3CA mutations cause venous malformation.
Literature	GENERIF_SUMMARY	Inducible expression of mutant PIK3CA resulted in a robust activation of phosphatidylinositol-3-kinase (PI3K)/AKT signaling but attenuation of Her2/Her3 signaling, and this can be reversed by deinduction of PIK3CAH1047R expression.
Literature	GENERIF_SUMMARY	These results indicate an association between PIK3CA gene polymorphism on the rs7621329(C/T) site and the risk of schizophrenia.
Literature	GENERIF_SUMMARY	As depicted in Figure 3, APC (25%) followed by KRAS (20%), TP53 (14%), and PIK3CA (11%) were the most frequently mutated genes.
Literature	GENERIF_SUMMARY	Findings of frequent mutations in CDKN2A, NOTCH1, PIK3CA, and EGFR (all in excess of 20%) point to potential therapeutic avenues. Trials of targeted therapies directed toward these mutations should be explored.
Literature	GENERIF_SUMMARY	Study shows that high EGFR protein expression and exon 9 PIK3CA activating mutations are independent prognostic factors in triple negative breast cancers.
Literature	GENERIF_SUMMARY	High PI3K expression is associated with prostate cancer.
Literature	GENERIF_SUMMARY	These results revealed differences in the genetic profiles of KRAS, NRAS, PIK3CA and BRAF at mutation hotspots between Chinese CRC patients and those of Western countries.
Literature	GENERIF_SUMMARY	Patients with PIK3CA amplifications were more likely to have diffuse-type and poorly differentiated gastric cancers and were more likely to experience peritoneal recurrence compared with those without

Main Category	Category	Terms
		PIK3CA amplifications.
Literature	GENERIF_SUMMARY	PIK3CA Mutation is associated with Colorectal Cancer.
Literature	GENERIF_SUMMARY	It was suggested that somatic point mutations in BRAF, CDKN2A and PI3KCA do not participate in the oncogenesis of Medullary thyroid carcinoma
Literature	GENERIF_SUMMARY	none of the nine tested SNPs of PIK3CA was associated with susceptibility to oral squamous cell carcinoma in the studied population.
Literature	GENERIF_SUMMARY	Data show that 14-3-3beta protein augmented the expression of matrix metalloproteinasea MMP2 and MMP9 through PI3 kinase/Akt protein/NF-kappaappa B pathway, thereby enhancing the invasiveness of hepatocellular carcinoma (HCC) cells.
Literature	GENERIF_SUMMARY	Continuous and metronomic PI3K inhibition elicits robust anticancer effects in ER(+), PIK3CA-mutant breast cancer.
Literature	GENERIF_SUMMARY	PIK3CA and PIK3CB significantly increase multidrug resistance of CRC cells in vivo.
Literature	GENERIF_SUMMARY	effects. Further studies indicated that PRKCI knockdown-mediated autophagy was associated with the inactivation of phosphatidylinositol 3-kinase alpha/AKT-mammalian target of rapamycin (PIK3CA/AKT-MTOR) signaling.
Literature	GENERIF_SUMMARY	we reported the rare occurrence of PIK3CA mutations in imatinib-naive gastrointestinal stromal tumors
Literature	GENERIF_SUMMARY	Therefore, we identified a tumorigenic role of Id-1 in OS and suggested a potential therapeutic target for OS patients.
Literature	GENERIF_SUMMARY	Based on the collective findings, we suggest that vIRF2 acts as an activator in PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	KRAS and PIK3CA mutations can influence the response of DLD1 colorectal cancer cell lines to MEK and PI3K inhibitors, but that the effect is dependent on the experimental model used to assess drug sensitivity.
Literature	GENERIF_SUMMARY	Suggest that copy number loss of Variation_91720 in PIK3CA predicts risk of esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	PIK3CA rs17849071 and rs17849079 polymorphisms may increase the risk of hepatocellular carcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	This work identifies PI3K signaling as an important target for precision medicine trials in meningioma patients
Literature	GENERIF_SUMMARY	Though Wortmannin was found to nullify the effect of H1047R over the PIK3CA, further studies are required for designing a better compound.
Literature	GENERIF_SUMMARY	Activation of the NF-kappaB-JNK pathway is independent of the PIK3CA-Rac1-JNK pathway involved in the bFGF-regulated human fibroblast cell migration and wound healing.
Literature	GENERIF_SUMMARY	PIK3CA and MET may play an important role in oncogenesis of certain specific subtypes of head and neck cancer
Literature	GENERIF_SUMMARY	Results indicate that eIF4B integrates the signals from Pim and PI3K/Akt/mTOR pathways in Abl-expressing leukemic cells.
Literature	GENERIF_SUMMARY	Here we report a Hungarian girl with macrodactyly and syndactyly. Genetic screening at hotspots in the PIK3CA gene identified a mosaic mutation (c.1624G > A, p.Glu542Lys) in the affected tissue, but not in the peripheral blood
Literature	GENERIF_SUMMARY	Results suggest, a potential role for PIK3CA, VEGFR2, RET and FGFR2 as therapeutic targets in EGFR non-mutated NSCLC that requires further clinical validation.
Literature	GENERIF_SUMMARY	our results offer significant insight into how PIK3CA overexpression drives squamous cell carcinoma (HNSCC) invasion and metastasis, providing a rationale for targeting PI3K/PDK1 and TGFb signaling in advanced HNSCC patients with PIK3CA amplification
Literature	GENERIF_SUMMARY	Combined detection of p110alpha and p110beta is helpful to determine the neoplastic potential of colorectal villous adenoma and serrated adenoma.
Literature	GENERIF_SUMMARY	PIK3CA gene plays a crucial role in carcinogenesis in general and head and neck squamous cell carcinoma in particular.
Literature	GENERIF_SUMMARY	The siRNA-induced AKT3 and PI3KCA mRNA knockdown.
Literature	GENERIF_SUMMARY	Data suggest that class I phosphatidylinositol 3-kinase (PI3K) inhibitor ZSTK474 induced autophagy might contribute to the antitumor activity.
Literature	GENERIF_SUMMARY	PIK3CA gene mutations are associated with breast cancer

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Literature	GENERIF_SUMMARY	PI3KCA mutations were extremely rare in pancreatic cancer, suggesting that they play a limited role in PC development.[review]
Literature	GENERIF_SUMMARY	No relation was found between PIK3CA expression and clinical features of ovarian clear cell carcinoma. PIK3CA expression had no impact on survival.
Literature	GENERIF_SUMMARY	Fulvestrant plus palbociclib was associated with significant and consistent improvement in progression-free survival compared with fulvestrant plus placebo, irrespective of the degree of endocrine resistance, hormone-receptor expression level, and PIK3CA mutational status. The combination could be considered as a therapeutic option for patients with HER2-negative breast cancer
Literature	GENERIF_SUMMARY	investigated mutations in exons 9 and 20 of the PI3K gene in gallbladder cancer (GBC) ; results show the PI3K gene to be mutated in approximately 17% of GBC
Literature	GENERIF_SUMMARY	Pharmacophore model suggested the mechanism of the inhibition of PI3Kalpha and mTOR by the compounds were mostly the same, which was supported by the docking study showing similar docking modes. The analysis also suggested the importance of the flat plane shape of the ligands, the space surrounding the ligands in the binding pocket, and the slight difference in the shape of the binding sites between PI3Kalpha and mTOR.
Literature	GENERIF_SUMMARY	Data underscore the importance of phosphatidylinositol 3-kinase (PI3K) signalling in myeloid cells in the context of chronic inflammatory diseases.
Literature	GENERIF_SUMMARY	PIK3CA was overexpressed in Epstein Barr virus-associated gastric adenocarcinoma and was associated with a less aggressive phenotype and better prognosis.
Literature	GENERIF_SUMMARY	PIK3CA mutations detected at high frequencies in breast cancer and mostly associated with better or insignificant outcomes in estrogen receptor-positive (ER+) early stage breast cancer and tend to be with worse prognosis in ER- disease.
Literature	GENERIF_SUMMARY	Low incidence of PIK3CA mutations in adenocarcinomas of the ampulla of Vater suggest no major role in tumorigenesis.
Literature	GENERIF_SUMMARY	PIK3CA plays a key role in promoting the proliferation and motility of esophageal squamous carcinoma cells.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with

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		metastatic lung adenocarcinomas.
Literature	GENERIF_SUMMARY	Data indicate that PIK3CA copy number gain was an independent poor prognostic factor for disease-free survival (DFS).
Literature	GENERIF_SUMMARY	A low prevalence of PIK3CA mutation exists in the distal extrahepatic bile duct (EBD) carcinoma of Korean patients, indicating that mutation screening may not be useful in determining prognosis or in formulating a treatment response to targeted inhibition in Korea.
Literature	GENERIF_SUMMARY	Data show that the elevated activities of PI3Ks, AKt and GSK-3beta were effectively suppressed by exogenous carbon monoxide (CO), leading to the improvement of platelet function.
Literature	GENERIF_SUMMARY	activating PIK3CA mutations gives rise to sporadic venous malformations in mice, which closely resemble the histology of the human disease.
Literature	GENERIF_SUMMARY	These data demonstrate a causal relationship between activating Pik3ca mutations and the genesis of vascular malformations.
Literature	GENERIF_SUMMARY	The authors present three clinically challenging cases with congenital mosaic tissue overgrowth caused by a heterozygous missense mutation in PIK3CA in the affected tissue.
Literature	GENERIF_SUMMARY	Our study demonstrates that the IGF1R/p110b/AKT/mTOR axis confers resistance to BYL719 in PIK3CA mutant breast cancers
Literature	GENERIF_SUMMARY	Mutations of PIK3CA were found in our cohort in 21 cases (28 %), 10 (13 %) in exon 9 and 11(15 %) in exon 20. Mutation frequency of AKT1 and PTEN genes was 4 and 3 %, respectively
Literature	GENERIF_SUMMARY	A recursively partitioned mixture model of all 3 somatic alteration burden measures resulted in classes of tumors whose epigenetic and genetic burden profile were associated with the PAM50 subtype and mutations in TP53, PIK3CA, and CDH1
Literature	GENERIF_SUMMARY	Data indicate that EPHA3 is involved in regulating the multidrug resistance (MDR) of small cell lung cancer (SCLC) via PI3K/BMX/STAT3 signaling and may be a therapeutic target in SCLC.
Literature	GENERIF_SUMMARY	The results suggest PIK3CA H1047R mutant cells have a selective advantage in breast, contribute to breast cancer susceptibility,

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		and drive tumor progression during breast carcinogenesis, even when present as only a subpopulation of tumor cells.
Literature	GENERIF_SUMMARY	Concomitant inhibition of PIM kinase and p110alpha activities inhibits glioma stem cell function and survival.
Literature	GENERIF_SUMMARY	PIK3CA gene mutations were sought in 145 tissue samples from central nervous system metastases of NSCLC using 3 PCR techniques. One E542K and two H1047R substitutions were found in male patients.
Literature	GENERIF_SUMMARY	Data show that long noncoding RNA RMEL3 is required for MAPK and PI3K signaling, and its knockdown decrease BRAFV600E melanoma cell survival and proliferation.
Literature	GENERIF_SUMMARY	Results indicate that des-gamma-carboxy prothrombin (DCP) antagonizes the inhibitory effects of Sorafenib on hepatocellular carcinoma (HCC) through activation of the Raf/MEK/ERK and PI3K/Akt/mTOR signaling pathways.
Literature	GENERIF_SUMMARY	PI3K isoforms p110a, p110b and p110g were selectively inhibited and effects on proliferation and migration assessed. Inhibition increased transcription of cancer stem cell genes in both models, suggesting de-differentiation in response to blockade of the PI3K/AKT/mTOR signalling axis.
Literature	GENERIF_SUMMARY	Overall PIK3CA mutant/HER2+ tumours had significantly lower pCR rates compared with wild-type tumours, however mainly confined to the HR+/PIK3CA mutant population. No definite conclusions can be drawn regarding survival
Literature	GENERIF_SUMMARY	Like the mammary intraductal papillomas, activating mutations in PIK3CA or AKT1 are present in most hidradenoma papilliferums.
Literature	GENERIF_SUMMARY	Single TOR serine-threonine kinases (mTOR) and dual phosphatidylinositol 3-kinases (PI3K)/mTOR inhibitors promote Treg cells immunosuppressive activity.
Literature	GENERIF_SUMMARY	we report a fourth patient with asymmetric overgrowth due to a somatic PIK3CA mutation who had nephroblastomatosis or Wilms tumor. Similar to two of three reported patients with a somatic PIK3CA mutation and renal tumors, he shared a PIK3CA mutation affecting codon 1047, presented at birth with asymmetric overgrowth, and had fibroadipose overgrowth.
Literature	GENERIF_SUMMARY	PIK3CA gene mutation is associated with

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		breast Cancers.
Literature	GENERIF_SUMMARY	Study identified PIK3CA mutation as an independent prognostic factor in patients who underwent APR for ASCC recurrence.
Literature	GENERIF_SUMMARY	that the spindle cell type of Mmtaplastic breast carcinoma with myoepithelial features exhibits a higher frequency of PIK3CA mutation than other types of metaplastic or basal-like breast cancer
Literature	GENERIF_SUMMARY	Identify mutually exclusive activating hotspot mutations in FGFR1 and related PI-3K/RAS signaling genes in malignant phyllodes tumors which are implicated in tumor pathogenesis and/or progression.
Literature	GENERIF_SUMMARY	Mutations in PIK3CA Sensitize Breast Cancer Cells to Physiologic Levels of Aspirin.
Literature	GENERIF_SUMMARY	to our knowledge, this is the largest study to evaluate the PIK3CA mutation in Indian breast cancer patients. The frequency and distribution pattern of PIK3CA mutations is similar to global reports
Literature	GENERIF_SUMMARY	While TP53 mutation and MYC amplification were synergistic in promoting tumor progression, PIK3CA mutation was found to have alleviated the oncogenic effect of either the TP53 mutation or MYC amplification, and was associated with a significant reduction in mitotic activity in TP53 mutated and/or MYC amplified breast cancer
Literature	GENERIF_SUMMARY	We revealed that activation of the PI3K/Akt pathway was the main cause of impaired autophagy in ELK3 KD. Our results suggest that targeting ELK3 may be a potential approach to overcome doxorubicin resistance in breast cancer therapeutics.
Literature	GENERIF_SUMMARY	Data indicated next-generation sequencing (NGS) for the identification of somatic variation in the phosphoinositide-3-kinase, catalytic, alpha polypeptide (PIK3CA)-related overgrowth spectrum (PROS) among 14 patients submitted for clinical testing.
Literature	GENERIF_SUMMARY	MARCH2 has a role in regulating autophagy by promoting CFTR ubiquitination and degradation and PIK3CA-AKT-MTOR signaling
Literature	GENERIF_SUMMARY	Study shows that PIK3CA is required for the proliferation, migration, invasion and clone formation of gallbladder carcinoma (GBC) cells in vitro and that the E545K mutation of PIK3CA promotes GBC progression through

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		its enhanced binding to EGFR.
Literature	GENERIF_SUMMARY	The data establish oncogenic PIK3CA mutations as a cause of glutamine dependency in colorectal cancer and suggest that targeting glutamine metabolism may be an effective approach to treat patients harboring PIK3CA mutations.
Literature	GENERIF_SUMMARY	AQP9 overexpression decreased the protein levels of phosphatidylinositol-3-kinase (PI3K), leading to reduced phosphorylation of Akt, and subsequently the protein levels of forkhead box protein O1 (FOXO1) were increased.
Literature	GENERIF_SUMMARY	This meta-analysis proved that detecting PIK3CA gene mutation by cfDNA has high diagnostic accuracy in breast cancer, especially for metastatic breast cancer. It may serve as a reliable non-invasive assay for detecting and monitoring PIK3CA mutation status in order to deliver personalized and precise treatment.
Literature	GENERIF_SUMMARY	NEDD4L negatively regulates PIK3CA protein levels via ubiquitination and is required for the maintenance of PI3K-AKT signaling pathway.
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with endometrial carcinoma progression and abdominopelvic metastasis.
Literature	GENERIF_SUMMARY	TP53 and PIK3CA mutations appear to have diverse effects on the outcome of early breast cancer patients, according to whether these genes are co-mutated or not, and for TP53 according to tumor-infiltrating lymphocyte density and ER/PgR-status
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with salivary duct carcinoma.
Literature	GENERIF_SUMMARY	Mutations in PIK3CA, AKT1, and PTEN also were less frequently found in cancers with mutations in the DNA guardian, TP53. This observation might seem counterintuitive because increased genome instability might facilitate development and selection of advantageous mutations.
Literature	GENERIF_SUMMARY	Results showed that mutations in PIK3CA exons 9 and 20 were observed in 25 of the 208 (12 %) gastric cancer patients but found that no correlation with prognosis.
Literature	GENERIF_SUMMARY	PIK3CA, PTEN, and AKT1 mutations occurred more frequently in the presence of hormone receptor overexpression (androgen, progesterone, or estrogen

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		receptor). PIK3CA mutations were also more common in the HER2-positive than in the HER2-negative group; the opposite pattern was seen for PTEN mutation or PTEN loss
Literature	GENERIF_SUMMARY	Combined inhibition of PI3K and PARP effectively synergized to block the growth of three wild-type PIK3CA ovarian cancer cell lines and explants of a primary ovarian tumor specimen.
Literature	GENERIF_SUMMARY	CLOVES syndrome is an overgrowth syndrome caused by mosaic activating mutation in phosphoinositide-3-kinase, catalytic, alpha polypeptide (PIK3CA) gene.
Literature	GENERIF_SUMMARY	Our findings suggest that PIK3CA mutation has the neutral prognostic effects on colorectal cancers (CRC) OS and progression-free survival. Evidence was accumulating for the establishment of CRC survival between PIK3CA mutations and patient-specific clinical or molecular profiles.
Literature	GENERIF_SUMMARY	Our data indicate that, similar to other benign proliferative lesions occurring in the breast parenchyma, activating PIK3CA mutations are very common in nipple adenomas
Literature	GENERIF_SUMMARY	Combined treatment with gamma-irradiation (gammaIR) and a dual PI3K/mTOR inhibitor causes loss of stemness and of FoxO1/3 proteins in p53-proficient glioblastoma multiforme stem cells (GBM-SCs).
Literature	GENERIF_SUMMARY	Data show that zinc finger protein 143 (ZNF143) promotes epithelial-mesenchymal transition (EMT) through regulating PI3K/AKT pathway in gastric cancer (GC) cells.
Literature	GENERIF_SUMMARY	The FGFR3, PIK3CA, TERT mutations, along with aberrant DNA methylation patterns, can serve as reliable genetic markers for the diagnosis of urinary bladder cancer. (Review)
Literature	GENERIF_SUMMARY	Mutant PIK3CA is a potent oncogenic driver in many urothelial carcinoma cell lines
Literature	GENERIF_SUMMARY	Cervical cancer patients whose tumours are positive for the PIK3CA-E545K mutation may benefit from PI3K inhibitor therapy in concert with standard cisplatin and radiation therapy.
Literature	GENERIF_SUMMARY	HPV16 E6-E7 promotes cancer stem-like cells phenotype in esophageal squamous cell carcinoma cells through the activation of PI3K/Akt signaling pathway

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Literature	GENERIF_SUMMARY	relevant genomic aberrations such as mutations in the hotspot regions of exon 9 and 20 of the PIK3CA gene can be detected in single circulating tumor cells and might provide insights into mechanisms of resistance to HER2-targeted therapies.
Literature	GENERIF_SUMMARY	the expression of 3 miRs and 9 mRNAs associate with the PIK3CA status. Expression of LRG1 is independent of luminal (A or B) subtype, decreased after neo-adjuvant aromatase inhibitor treatment.
Literature	GENERIF_SUMMARY	the majority of cells within the tumor appears to express predominantly TAp63 isoform. While DeltaNp63 exerts its effects by regulating a PI3K/CD44v6 pathway.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	A dual PI3K and mTOR inhibitor, BEZ235 with the HDAC inhibitor Trichostatin A (TSA) significantly increased anti-tumor activities compared with BEZ235 and TSA alone, suggesting a combination treatment for non-small-cell lung cancer (NSCLC).
Literature	GENERIF_SUMMARY	In the very high-risk Bladder Cancer patients , several genes had a higher frequency of mutations than reported in the The Cancer Genome Atlas database, including PIK3CA . Mutation associations with receipt of neoadjuvant chemotherapy, nodal involvement, metastatic disease development, and survival were analyzed.
Literature	GENERIF_SUMMARY	PTEN and PIK3CA were shown to be upregulated by 89 and 76% (western blot), respectively, in the pemetrexed-resistant cell line compared with control cells.
Literature	GENERIF_SUMMARY	Study found that PIK3CA mutation occurred in approximately 2.8% of patients with lung adenocarci- noma in a cohort of Chinese patients.
Literature	GENERIF_SUMMARY	Findings indicate that PTEN and its downstream PI3K/AKT signaling targets were modulated by JQ1 compound.
Literature	GENERIF_SUMMARY	The primary end point was progression-free survival (PFS) in the intention-to-treat (ITT) population and patients with PIK3CA-mutated tumours.In patients with PIK3CA-mutated tumours, the median PFS was 7.3 months for pictilisib (n = 32) versus 5.8 months with placebo (n = 30) (HR, 1.06; 95% CI 0.52-2.12; P = 0.88)
Literature	GENERIF_SUMMARY	E545K missense mutation of PIK3CA is associated with loss of protein stability and development of breast cancer.

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Literature	GENERIF_SUMMARY	Results indicate that specific inhibition of PI3K p110alpha could provide an alternative therapeutic approach for colorectal cancer (CRC) patients, particularly those harboring KRAS mutations.
Literature	GENERIF_SUMMARY	Bis(acetylacetonato)-oxidovanadium(IV) and sodium metavanadate induced the activation of both PI3K/AKT and MAPK/ERK signaling pathways dose- and time-dependently.
Literature	GENERIF_SUMMARY	Studies suggest to maximize therapeutic benefit, drug combinations and schedules for Inhibition of the PI3K/AKT/mTOR Pathway must be explored to identify those with the highest efficacy and lowest toxicity overlap.
Literature	GENERIF_SUMMARY	Expression of LINC00520 is regulated by oncogenic Src, PIK3CA and STAT3, and may contribute to the molecular etiology of breast cancer.
Literature	GENERIF_SUMMARY	Platelet-derived growth factor (PDGF) activates the PI3K/Akt/mTOR pathway, and suppresses apoptosis.
Literature	GENERIF_SUMMARY	Data show that the incidence of phosphatidylinositol 3-kinase catalytic 110-KD alpha (PIK3CA) mutation is low in small cell lung cancer (SCLC) and the pseudogene-positive alteration of A1634C is prone to occur in exon 9 of PIK3CA mutations in cancers.
Literature	GENERIF_SUMMARY	PI3K signaling enhanced SOX2 translation, which in turn induced the expression of genes regulating the stemness characteristics of cancer stem cells .
Literature	GENERIF_SUMMARY	Sixteen of 20 patients treated at the MTD had reduced (18)FDG-PET uptake; 7 (33%) had a reduction >25%. One patient achieved a complete response (CR; endometrial carcinoma exhibiting both PIK3CA and PTEN mutations and complete PTEN loss) and two had a partial response (PR; both metastatic breast cancer).
Literature	GENERIF_SUMMARY	High expression of PI3K is associated with glioma.
Literature	GENERIF_SUMMARY	miR-130a could alleviatehuman coronary artery endothelial cells (HCAECs) injury and inflammatory responses by down-regulating PTEN and activating PI3K/Akt/eNOS signaling pathway.
Literature	GENERIF_SUMMARY	PIK3CA mutation may not be associated with primary resistance to EGFR TKI among lung adenocarcinoma patients

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Literature	GENERIF_SUMMARY	No significant impact on prognosis was observed for mutated KRAS, NRAS, and PIK3CA genes or combined RAS mutations
Literature	GENERIF_SUMMARY	PIK3CA and AKT1 are frequently mutated in Hidradenoma papilliferum (HP) tumours (29% and 14%, respectively). PIK3CA/AKT1 pathway alterations in HP further support the hypothesis that HP is the vulvar (anogenital mammary-like gland) analogue of breast intraductal papilloma.
Literature	GENERIF_SUMMARY	Gains of PIK3CA is associated with hypopharyngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Report outcome of PIK3CA/mTOR/MEK co-targeted inhibition strategy in the treatment of non-small cell lung carcinomas.
Literature	GENERIF_SUMMARY	Mutations in PIK3CA gene is associated with breast cancer.
Literature	GENERIF_SUMMARY	Our study did not demonstrate a significant survival advantage from aspirin use in stage 2 PIK3CA mutated colon cancer (CC). The 'real-world' nature of our cohort and the subsequent uncontrolled differences in age and fitness in aspirin users are likely to have contributed to this result. In the PIK3CA mutated group, there was a trend towards improved recurrence-free survival of CC.
Literature	GENERIF_SUMMARY	This study, for the first time segregates atypical verruciform lesions by virtue of a unique genotype (PIK3CA mutant/TP53 wild type) illustrating an example of progression to a TP53-mutated keratinizing carcinoma.
Literature	GENERIF_SUMMARY	these results suggest that PIK3CA mutation may be a new predictive biomarker for celecoxib efficacy treatment in breast cancer
Literature	GENERIF_SUMMARY	We find that both primary and recurrent Squamous cell carcinoma of the anal canal (ASCC) tumors harbor mutations in genes, such as PIK3CA and FBXW7, that are also mutated in other HPV-associated cancers. Overall mutational burden was not significantly different in pre- versus post-CRT tumors, and several examples of shared clonal driver mutations were identified
Literature	GENERIF_SUMMARY	Gli1 promotes colorectal cancer cells metastasis in a Foxm1-dependent manner by activating EMT and PI3K-AKT signaling.
Literature	GENERIF_SUMMARY	Report targeted analysis of KRAS, BRAF and PIK3CA mutations in circulating tumor cells captured by a label-free platform and compare to expression in primary tumor

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		tissue and metastases.
Literature	GENERIF_SUMMARY	PI3K catalytic-alpha(PIK3CA) mutations assessed from circulating tumor DNA represent the first potentially viable serum biomarker for the selection of ET combinations, and new data demonstrate the feasibility of this minimally invasive technique as an alternative to traditional tissue analysis
Literature	GENERIF_SUMMARY	Study shows that PIK3CA mutation occurs approximately in 45% of breast cancer patients, and are frequent in exon 20 in comparison with exon 9 of the gene. Also, PIK3CA mutations were found significantly associated with low-grade tumors.
Literature	GENERIF_SUMMARY	Because apoptosis and tumor regression occurred early but not late in the treatment course, and proliferative arrest was maintained throughout treatment, p110a/b inhibitors may be considered short-term cytotoxic agents and long-term cytostatic agents
Literature	GENERIF_SUMMARY	Functional studies demonstrated that IDH2 and PIK3CA hotspot mutations are likely drivers of solid papillary carcinoma with reverse polarity
Literature	GENERIF_SUMMARY	Data show that acid sensing ion channel 1a (ASIC1a) mediated calcium influx, which resulted in the activation of PI3K/AKT signaling and increased drug resistance, suggesting that ASIC1a/Ca(2+)/PI3K/AKT signaling represents a pathway that regulates drug resistance, thus offering a potential target for chemotherapy of hepatocellular carcinoma (HCC).
Literature	GENERIF_SUMMARY	FBS. Following stimulation of RTKs (receptor tyrosine kinases), microinjected PI3Kalpha was recruited to the PM, but oncogenic forms of PI3Kalpha were not recruited to the PM to a greater extent and did not reside at the PM longer than the wild-type PI3Kalpha. Instead, the E545K mutant specifically bound activated Cdc42
Literature	GENERIF_SUMMARY	Findings show that calmodulin (CaM) stimulates phosphoinositide-3-kinase (PI3K) lipid kinase activity by binding MARCKS and displacing it from phosphatidylinositol 4,5-bisphosphate (PIP2) headgroups, thereby releasing free PIP2 that recruits active PI3K to the membrane and serves as the substrate for the generation of phosphatidylinositol 3,4,5-trisphosphate (PIP3).

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Literature	GENERIF_SUMMARY	High PI3K expression is associated with liver cancer.
Literature	GENERIF_SUMMARY	PIK3CA mutations have a role in breast tumor response to targeting mTOR-mediated MCL-1 translation
Literature	GENERIF_SUMMARY	the most frequent genomic abnormality in patients with primary pulmonary neuroendocrine carcinoma in China
Literature	GENERIF_SUMMARY	Report KRAS, NRAS, PIK3CA, and BRAF mutational profile in poorly differentiated clusters of KRAS-mutated colon cancer may depend on tumor histology.
Literature	GENERIF_SUMMARY	Somatic PIK3CA variants in breast cancers associate with rs252913 and rs331499 SNPs in the locus 5q11.2 locus.
Literature	GENERIF_SUMMARY	The combination of SAR256212 and SAR245408 resulted in stable disease as the best response. Side effects seen in combination were similar to the profiles of each individual drug. Patient outcome was the same regardless of tumor PI3KCA mutation status
Literature	GENERIF_SUMMARY	Cytoplasmic retention of FOXO3a may represent a potential biomarker for response to combined treatment with inhibitors of PI3K and autophagy in PIK3CA-mutant cervical cancer cells.
Literature	GENERIF_SUMMARY	Report a functional interaction between EZH2- and PIK3CA-dependent signaling pathways in non-muscle invasive bladder cancer cells.
Literature	GENERIF_SUMMARY	Despite a significant prevalence of BRAF mutation, more than 70% of hobnail variant of papillary thyroid carcinomas (HPTCs) in our series showed concurrent mutations of other genes such as TP53, PIK3CA, CTNNB1 and hTERT, in contrast to classic PTC.
Literature	GENERIF_SUMMARY	PIK3CA mutations are associated with esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Increased levels of FGFR3 and PIK3CA mutated DNA in urine and plasma are indicative of later progression and metastasis in bladder cancer.
Literature	GENERIF_SUMMARY	The expression of ribosomal and proteasomal components was significantly up-regulated in primary human colorectal tumors harboring PI3K pathway activation
Literature	GENERIF_SUMMARY	PIK3CA mutations are common in lobular carcinoma in situ, but are not a biomarker of progression.

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Literature	GENERIF_SUMMARY	Upregulation of PI3K is associated with thyroid cancer.
Literature	GENERIF_SUMMARY	Upregulation of PI3K is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	MicroRNA miR-155-5p promoted proliferation, invasion and migration, but inhibited apoptosis in hepatocellular carcinoma (HCC) by directly targeting the 3'-UTR of PTEN through the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	High expression of PI3KCA is associated with drug resistance and proliferation of breast cancer.
Literature	GENERIF_SUMMARY	We identified a de novo phosphatidylinositol-4,5- bisphosphate 3-kinase catalytic subunit alpha gene (PIK3CA) mutation as a potential drug resistance mechanism.
Literature	GENERIF_SUMMARY	Derlin-1 is overexpressed in bladder cancer and promotes malignant phenotype through ERK/MMP-2/9 and PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	PIK3CA gene mutations were not found; however, a synonymous single nucleotide polymorphism (SNP) [rs17849079] was observed frequently [35/96 (36.4%)] in oral cancer samples.
Literature	GENERIF_SUMMARY	PIK3CA E545K/E542K mutations are associated with everolimus resistance in breast cancer.
Literature	GENERIF_SUMMARY	KRAS mutations and AKT activation are present in Wilms tumors (WT) and may represent novel therapeutic targets for this tumor.
Literature	GENERIF_SUMMARY	KRAS and PIK3CA genetic variants are associated with aggressive clinicopathological features in colorectal adenocarcinoma.
Literature	GENERIF_SUMMARY	Compared to single-gene-targeted cells and parental controls, cells with both a PIK3CA mutation and TP53 alteration had increased cancerous phenotypes including cell proliferation, soft agar colony formation, aberrant morphology in acinar formation assays, and genomic heterogeneity.
Literature	GENERIF_SUMMARY	Phosphatidylinositol-3-kinase signaling is regulated by INPP4B and PTEN in the triple-negative breast cancer cells.
Literature	GENERIF_SUMMARY	Data indicate that PI3K pathway mutation predicts longer time to local progression after radioembolization of colorectal liver

Main Category	Category	Terms
		metastases.
Literature	GENERIF_SUMMARY	Recurrent PIK3CA mutations associated with Epstein-Barr virus - infected Gastroesophageal junction tumor.
Literature	GENERIF_SUMMARY	abundance is stimulated, in part, through AKT- and nuclear factor kappaB (NF-kappaB)-dependent pathways and that presence of PTX3 is necessary for PI3K-induced stem cell-like traits.
Literature	GENERIF_SUMMARY	This study provides insights as to how mutationally activated PIK3CA acts in concert with MEK1/2 signaling to cooperatively regulate mTORC1/2 to sustain PIK3CA-mutated melanoma proliferation.
Literature	GENERIF_SUMMARY	Data show that cancer-associated fibroblasts (CAFs)-derived hepatocyte growth factor (HGF) or recombinant HGF activated c-Met/phosphoinositide 3-kinase (PI3K)/Akt and glucose-regulated protein 78 (GRP78) signalling pathways in ovarian cancer cells.
Literature	GENERIF_SUMMARY	Psgl-1 deficiency accelerates bleomycin-induced lung fibrosis and inflammation in mice through activating the PI3K/AKT axis.
Literature	GENERIF_SUMMARY	This study identified PIK3CA mutations in breast carcinoma patients of Western China that will enable a more rapid molecular diagnosis, and provide a stronger rationale evidence for development of precision therapeutic approaches as well as promising therapeutic targets for breast cancer treatment or patient management
Literature	GENERIF_SUMMARY	Out of 171 EGFR mutated tumor-patient cases, 20 were sequenced using at least a limited comprehensive genomic profiling platform. 50% harbored concurrent TP53 mutation, 10% PIK3CA mutation, 5% PTEN mutation, among others.
Literature	GENERIF_SUMMARY	in series of benign and malignant neoplasms thought to be associated with anogenital mammary-like glands, PIK3CA mutation was detected in extramammary Paget disease, mammary-type invasive ductal carcinoma and tubulolobular carcinoma
Literature	GENERIF_SUMMARY	Somatic PIK3CA mutations in seven patients with PIK3CA-related overgrowth spectrum have been reported.
Literature	GENERIF_SUMMARY	Preliminary data consistent with preclinical data indicate increased antitumor activity of taselisib in patients with PIK3CA-mutant tumors (in comparison with patients with

Main Category	Category	Terms
		tumors without known activating PIK3CA hotspot mutations) starting at the lowest dose tested of 3 mg, thereby supporting higher potency for taselisib against PIK3CA-mutant tumors.
Literature	GENERIF_SUMMARY	A combination of CAL-101, a p110delta-specific inhibitor, with bortezomib treatment of HT/EBV cells synergistically suppressed proliferation, reduced levels of drug resistance-related proteins, activated caspase cleavage and recovered expression of p110a/p110b.
Literature	GENERIF_SUMMARY	Tetraarsenic hexoxide (As4O6) induced G2/M arrest, apoptosis and autophagic cell death through PI3K/Akt and p38 MAPK pathways alteration in SW620 colon cancer cells.
Literature	GENERIF_SUMMARY	p110alpha subunit of PI3K and PKD mediate YAP activation in response to insulin and neurotensin in pancreatic cancer cells. Inhibitors of PI3K or PKD disrupt crosstalk between insulin receptor and GPCR signaling systems by blocking YAP/TEAD-regulated gene expression in pancreatic cancer cells
Literature	GENERIF_SUMMARY	APC and PIK3CA Mutational Cooperativity Predicts Pathologic Response and Survival in Patients Undergoing Resection for Colorectal Liver Metastases.
Literature	GENERIF_SUMMARY	Data, including data from hydrogen/deuterium exchange mass spectrometry studies using full-length constructs for both catalytic/p110alpha and regulatory/p85alpha subunits of phosphoinositide 3-kinase isoform, suggest that the catalytic subunit exhibits binding sites for four inhibitor compounds studied, including a low-affinity binding site for CAL-101.
Literature	GENERIF_SUMMARY	PI3K protein was expressed in 81.3% (91/112) of all ESCC samples, whereas it was found in only 4.9% (5/56) of adjacent normal cells.PIK3CA gene mutations were highly correlated with age, tobacco use, and clinical stage (p
Literature	GENERIF_SUMMARY	phenotypic changes in metabolism following a single copy knock-in of mutant PIK3CA (H1047R) in the MCF10A cell line, an important cell model for studying oncogenic transformation in breast tissues, were examined.
Literature	GENERIF_SUMMARY	because PIK3CA mutant cells exhibit a profound reliance on glucose metabolism, malate-aspartate shuttle deregulation leads

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		to a specific proliferative block due to the inability to maintain NAD(+)/NADH homeostasis. Together these observations define a precise metabolic vulnerability imposed by a recurrently mutated oncogene.
Literature	GENERIF_SUMMARY	High Expressions of PI3K is associated with metastasis of esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Data suggest that the phosphoinositide-3-kinase, catalytic, alpha polypeptide (PIK3CA) mutations may potentially contribute to acquired cetuximab resistance in patients with metastatic colorectal cancer (mCRC).
Literature	GENERIF_SUMMARY	there was no association between PIK3CA mutation and expression, or of other related genes, namely PIK3CB, PIK3CD, PIK3CG, and PTEN. Mutation of PIK3CA might alter calmodulin expression and other genetic pathways
Literature	GENERIF_SUMMARY	Oncogenic beta-catenin and PI3K activities interact in the acquisition of multiple cancer-related phenotypes in organoids grown in Matrigel.
Literature	GENERIF_SUMMARY	In vivo single-agent mTORC inhibition inhibited growth of one PIK3CA-mutant cancer, but had little effect on any PIK3CA(WT) or a second PIK3CA-mutant model. In all models, the combination therapy showed greater growth delay than monotherapy. The uniform ability of PI3K and mTORC inhibition to suppress the growth of Head and neck squamous cell carcinomas (HNSCC) cells highlights the pathway's role in driving proliferation
Literature	GENERIF_SUMMARY	PIK3CA mutant and -wildtype gastric tumor subclones are skilled to metastasize independently to different regional lymph nodes.
Literature	GENERIF_SUMMARY	Suggest that PIK3CA gene mutations could act as a prognostic biomarker in Northwest Chinese esophageal squamous cell carcinoma patients.
Literature	GENERIF_SUMMARY	we demonstrate that oncogenic p110alphaH1047R, K-RasG12V, H-RasG12V, and Her2 use a common Akt1-FOXO3a pathway in suppression of DeltaNp63alpha expression and, consequently, promote cell migration/invasion and tumor metastasis. Thus, DeltaNp63alpha may function as a critical integrator of oncogenic signals in cell motility and cancer metastasis.

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Literature	GENERIF_SUMMARY	our data point to a targetable Axl-PI3 kinase-PD-L1 axis that is highly associated with radiation resistance
Literature	GENERIF_SUMMARY	In this study of 71 patients with advanced solid tumors who had received prior treatments (86% of whom were PIK3CA mutated), 3 different dose schedules of TAK-117 were evaluated. The MTD of TAK-117 was established as 150 mg once daily and 900 mg for both intermittent (MWF/MTuW) schedules.
Literature	GENERIF_SUMMARY	Data suggest HRas activates both p110alpha and p110delta isoforms; membrane-resident HRas, not soluble HRas, increases membrane recruitment of both p110alpha and p110delta. (HRas = v-Ha-ras Harvey rat sarcoma viral oncogene homolog; p110alpha = class IA phosphoinositide 3-kinases, subunit p110alpha; p110delta = class IA phosphoinositide 3-kinases, subunit p110delta)
Literature	GENERIF_SUMMARY	PIK3CA mutations can lead to resistance to MET inhibition, supporting future clinical evaluation of combinations of PI3K and MET inhibitors in common scenarios of malignant neoplasms featuring aberrant MET expression and PIK3CA mutations.
Literature	GENERIF_SUMMARY	PI3K overexpression is associated with colorectal cancer metastasis.
Literature	GENERIF_SUMMARY	Expression of Phosphoinositide 3-Kinase p110alpha and p110beta Subunits and PIK3CA Mutation in Patients With Advanced Gastric Carcinoma.
Literature	GENERIF_SUMMARY	This study revealed that (1) PIK3CA hotspot mutations occurred with low frequency in gastric cancer; (2) PIK3CA hotspot mutations were not directly associated with PI3K pathway activation; and (3) p-AKT (+) may be a biomarker for better outcomes for gastric cancer patients undergoing gastrectomy regardless of the PIK3CA mutation status.
Literature	GENERIF_SUMMARY	The lack of KRAS, NRAS, BRAF, and PIK3CA mutation in our study may suggest that a subset of eyelid sebaceous carcinomas is unlike that of eyelid sebaceous carcinomas of western countries.
Literature	GENERIF_SUMMARY	Mosaic mutations in PIK3CA or PIK3R2 which activate class 1A PI3K cause severe non-ketotic hypoglycaemia in a subset of patients, with the metabolic phenotype presumably related to the extent of

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		mosaicism within the liver.
Literature	GENERIF_SUMMARY	Mosaic gain of function mutation in PIK3CA gene leads to abnormal AKT-mTOR pathway activation and is responsible of the clinical manifestations.
Literature	GENERIF_SUMMARY	The findings suggest that PIK3CA mutation has no significant effects on overall survival and on disease-free survival in esophageal squamous cell carcinoma patients. (Meta-analysis)
Literature	GENERIF_SUMMARY	High expression of PI3K is associated with nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	p85alpha plays a tumor-suppressive role in transformation; p110alpha-selective therapeutics may be effective in the treatment of breast cancer patients with PIK3R1 loss
Literature	GENERIF_SUMMARY	Results show that head and neck squamous cell carcinoma tumors with low P120CTN and PI3K pathway mutations have higher levels of MMP1 compared to tumors with high P120CTN and no PI3K pathway mutations demonstrating that P120CTN downregulation and PIK3CA mutations promote MMP1-driven invasion.
Literature	GENERIF_SUMMARY	Case Reports
Literature	GENERIF_SUMMARY	Data suggest that Lys-776 (in the P-loop of PIK3CA) is critical for autophosphorylation in PIK3CA/PIK3R1 heterodimer; both His-936 and His-917 (in the activation and catalytic loops of PIK3CA) are important catalytically, but appear to have different functions; bisubstrate kinetic analysis of PIK3CA/PIK3R1 heterodimer indicates an ordered sequential mechanism.
Literature	GENERIF_SUMMARY	Knowing the mutation status of KRAS, BRAF or PIK3CA in stage II colorectal cancer can significantly improve the accuracy of prognoses.
Literature	GENERIF_SUMMARY	Data suggest a critical role for KDM3A in the PI3K/AP-1 oncogenic axis and propose a novel strategy for inhibition of KDM3A against liver tumor development under PI3K pathway activation.
Literature	GENERIF_SUMMARY	An increase in the frequency of PIK3CA gene mutations with disease progression (from 2.4 to 28.7% with tumor progression from I-IIa to III-IV stage; p=0.0001) and a trend towards its increase in the tumors with unfavorable prognostic characteristics (high histological grade, triple negative phenotype) were demonstrated.

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Literature	GENERIF_SUMMARY	PIK3CA and/or ERBB family mutations were detected in 23 (31.1%) tumour samples tested, whereas PTEN expression was low in 31.1% of cases tested
Literature	GENERIF_SUMMARY	Variation in organ-specific PIK3CA and KRAS mutant levels in normal human tissues correlates with mutation prevalence in corresponding carcinomas.
Literature	GENERIF_SUMMARY	High PI3K expression is associated with melanoma.
Literature	GENERIF_SUMMARY	The bladder cancer cell line TCCSUP, which harbors a PIK3CA E545K mutation, was sensitive to pictilisib compared to cell lines with wild-type PIK3CA Pictilisib exhibited stronger antitumor activity in bladder cancer patient-derived xenografts models with PI3KCA H1047R mutation or amplification than the control patient-derived xenografts model
Literature	GENERIF_SUMMARY	RAS signaling through PI3K is necessary for normal lymphatic vasculature development and for RAS-induced transformation in vitro and in vivo, especially in lung cancer, where it is essential for tumor initiation and necessary for tumor maintenance. [review]
Literature	GENERIF_SUMMARY	these results suggest that aspirin inhibited human uterine leiomyoma cell growth by regulating KRasp110alpha interaction. Aspirin which targeting on interaction between K-Ras and p110alpha may serve as a new therapeutic drug for uterine leiomyoma treatment.
Literature	GENERIF_SUMMARY	while telomere length did not correlate with the presence of a mutation in BRAF (V-raf murine sarcoma viral oncogene homolog B), PIK3CA (phosphatidylinositol 3-kinase catalytic subunit), or MSI status, it was significantly associated with the occurrence of a mutation in KRAS
Literature	GENERIF_SUMMARY	While PIK3CA mutations generally had no impact on survival, and were not associated with clinicopathological variables, patients with exon 9 charge-changing mutations were associated with poor survival in endometrial primary tumors. PIK3CA mRNA levels were increased in metastases compared to the primary tumors, independent of PIK3CA mutation status, which rather associated with reduced PIK3CA mRNA expression.
Literature	GENERIF_SUMMARY	These data indicate that increased irisin levels may have protective roles in liver cancer cells through partial activation of the

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		PI3K/AKT pathway, which may facilitate liver cancer progression and decrease the sensitivity to chemotherapy.
Literature	GENERIF_SUMMARY	These results indicated that dual targeting of PI3K and PERK pathways might improve clinical prognosis and enhance the treatment of ESCC patients.
Literature	GENERIF_SUMMARY	A mosaic gain-of-function mutation in PIK3CA was identified in a patient with gait disturbance, leg pain, isolated macrodactyly of the foot, and mild intellectual disability.
Literature	GENERIF_SUMMARY	results suggest that c-Jun, p38 MAPK, PIK3CA/Akt, and GSK3 signaling involved in the effect of miR-203 on the proliferation of hepatocellular carcinoma cells.
Literature	GENERIF_SUMMARY	Study showed for the first time that co-occuring oncogenic HER2 and mutant PIK3CA could induce replication stress in mammary epithelial cells and drives breast cancer progression.
Literature	GENERIF_SUMMARY	PIK3CA mutation was associated with treatment response in breast cancer.
Literature	GENERIF_SUMMARY	Combined PI3Kalpha and CDK4/6 inhibition significantly improved disease control in human xenograft models compared with either monotherapy.
Literature	GENERIF_SUMMARY	Targeting activating mutations in PIK3CA assay was used for screening tumor DNA from Metastatic Relapse in Advanced Bladder Cancer patients; at least one mutation was detected in 19 of the 60 patients.
Literature	GENERIF_SUMMARY	Ehrlichia chaffeensis infection activated the phosphatidylinositol 3-kinase (PI3K)-Akt-mTOR (mechanistic target of rapamycin) pathway, and activation was induced by three ehrlichial tandem repeat protein (TRP) effectors, with TRP120 inducing the strongest activation.
Literature	GENERIF_SUMMARY	Overexpression of KAT6A or TRIM24 promoted PIK3CA expression, AKT phosphorylation, and cell proliferation.
Literature	GENERIF_SUMMARY	Studies indicate the role of the insulin-like growth factor 1 (IGF1) - phosphoinositide 3-kinase (PI3K)-Akt signaling pathway in regulating exercise-induced physiological cardiac hypertrophy and cardiac protection [Review].
Literature	GENERIF_SUMMARY	When considered together (43 cases), 1/25 cases (4%) with a PIK3CA mutation and/or low PTEN expression levels had a

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		pathologic complete response (pCR)compared to 7/18 cases (39%) with wild-type PI3KCA and high PTEN expression levels (p = 0.006).
Literature	GENERIF_SUMMARY	Together, dual inhibition of BRD4 and PI3K by SF2523 suppresses human prostate cancer cell growth in vitro and in vivo.
Literature	GENERIF_SUMMARY	The majority of PIK3CA H1047R mutations in the breast cancer cohort precede genome doubling.
Literature	GENERIF_SUMMARY	PIK3CA mutation is a distinctive genetic feature of NSCLC with COPD, regardless of age, smoking dose, pathological stage, and histology.
Literature	GENERIF_SUMMARY	PI3K is a signal linker between L-selectin and PSGL-1 in IL-18 transcriptional activation at the promoter level.
Literature	GENERIF_SUMMARY	The role of PI3K in cancer has been well established, and mutations of PIK3CA, the gene coding for catalytic subunit p110alpha of PI3K, are found in approximately 30% human cancers. we review the structures and activation modes of PI3Ks and its implications in angiogenesis, extracellular matrix remodeling and tumor immunity.
Literature	GENERIF_SUMMARY	CTNNB1 mutations were found in 60% of Basal cell adenoma but not in basal cell adenocarcinoma. None of the tested cases had PIK3CA mutations. CTNNB1 mutation trended to be more common in those cases having a predominant tubular or tubulotrabecular patterns.
Literature	GENERIF_SUMMARY	We detected the same PIK3CA mutation in her affected tissue, urine, and tumor, indicating Wilms tumors probably arise from PIK3CA mutant cells in patients with CLOVES. No urine sample from a participant with Klippel-Trenaunay (KTS) had detectable PIK3CA mutations.
Literature	GENERIF_SUMMARY	By monitoring single-cell dynamics in each of these contexts, the authors identified PI3K/Akt regulation of glycolysis as a multifaceted modulator of single-cell metabolic dynamics that is required to maintain metabolic stability in proliferating cells.
Literature	GENERIF_SUMMARY	PIK3CA gene fusion and expression is associated with poor outcome in the hormone receptor-positive breast cancer.
Literature	GENERIF_SUMMARY	Oncogenic PIK3CA alters methionine and cysteine utilization, partly by inhibiting xCT to contribute to the methionine dependency

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		phenotype in human breast cancer cells.
Literature	GENERIF_SUMMARY	High PIK3CA expression is associated with metastasis in colon cancer.
Literature	GENERIF_SUMMARY	High PI3K expression is associated with cervical cancer.
Literature	GENERIF_SUMMARY	The pooled analysis confirmed that the presence of a PIK3CA mutation represents an independent negative prognostic factor (HR = 1.67, 95%CI
Literature	GENERIF_SUMMARY	Inflammatory stress displayed by endothelial cells derived from lymphatic malformation is consistent with errors in lipid metabolism that may be linked to PIK3CA acquired mutations.
Literature	GENERIF_SUMMARY	High Resolution Melting Analysis can be used as a rapid and sensitive method for mutation screening. Dysregulation of PIK3CA gene in bladder cancer reveals its potentials as a mechanistic link for cancer development, which in turn suggests its special use in interventional studies for targeted therapy
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with decreased risk of peritoneal metastases in chemo-resistant metastatic colorectal cancer.
Literature	GENERIF_SUMMARY	KRAS wild-type metastatic colorectal cancer patients with mutant type phosphatidylinositol 3-kinases receiving an epidermal growth factor receptor monoclonal antibody (anti-EGFR) inhibitor as first-line therapy have significantly lower overall response rate, shorter progression-free survival and overall survival.
Literature	GENERIF_SUMMARY	Double mutation of PIK3CA and TP53 is an independent predictive factor for overall survival in stage II/III colorectal cancer patients receiving 5-FU-based chemotherapy.
Literature	GENERIF_SUMMARY	We identified a total of five distinct mutations in PIK3CA (NM_006218.2), including one hotspot mutation (c.1624G>A;p.Glu542Lys); two recurrent, strong (gain of function) mutations (c.3140A>T;p.His1047Leu, c.1258T>C;p.Cys420Arg); one previously described mutation in patients with macrodactyly (c.344G>C;p.Arg115Pro); and one novel somatic PIK3CA mutation (c.248T>C; p.Phe83Ser) not previously described
Literature	GENERIF_SUMMARY	In this large pooled analysis, PIK3CA

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		mutations were significantly associated with a better IDFS, DDFS, and OS, but had a lesser prognostic effect after adjustment for other prognostic factors.
Literature	GENERIF_SUMMARY	We report the first characterization of clinical acquired resistance to MEKi + CDK4i, identifying a rare preexisting PIK3CA(E545K) subpopulation that expands upon therapy and exhibits drug resistance.
Literature	GENERIF_SUMMARY	High PIK3CA expression is associated with head and neck carcinoma.
Literature	GENERIF_SUMMARY	biomarkers that are predictive of a response, such as PIK3CA mutations for inhibitors of the PI3K catalytic subunit alpha isoform, must be identified and analytically and clinically validated.. we review the current experience with anticancer therapies that target the PI3K-AKT-mTOR pathway.
Literature	GENERIF_SUMMARY	These results suggest that in addition to overgrowth of bone and soft tissue, gain of function variants in PIK3CA may predispose to overgrowth of scar tissue, resulting in the formation of abnormal scars such as keloids or hypertrophic scars.
Literature	GENERIF_SUMMARY	Our results indicate that low-grade adenosquamous carcinoma of the breast of the breast is a low-grade triple-negative breast cancer that harbours a basal-like phenotype with no androgen receptor expression, and shows a high rate of PIK3CA mutations
Literature	GENERIF_SUMMARY	High frequency of somatic PIK3CA mutations was detected in Chinese breast cancer patients and associated with poor prognosis. The prevalence of the PIK3CA mutation was 46.5% (236/507), and 35 patients carried two or three variants of PIK3CA gene.
Literature	GENERIF_SUMMARY	Study shows that ARQ092 counteracts overactivation of PI3K signal pathway in primary fibroblast cells derived from patients with PIK3CA-related overgrowth spectrum.
Literature	GENERIF_SUMMARY	Six mutations located at the BRAF, KRAS, and PIK3CA genes have been detected in colorectal cancer patients using a novel multiplex surface enhanced Raman spectroscopy-multiplex PCR method.
Literature	GENERIF_SUMMARY	In early breast cancer, PIK3CA mutations seem to identify HER2+ patients who are less likely to reach pCR. The clinical implications of PIK3CA mutations tend to

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		vary between exon 9 and exon 20. This mechanism should be explored in further studies.
Literature	GENERIF_SUMMARY	Herein, we present a case of PROS that reflects the great variability of cutaneous manifestations associated with this disease spectrum, many of which were found in our patient.
Literature	GENERIF_SUMMARY	Results show PIK3CA mutation correlates with poor prostate cancer prognosis and causes prostate cancer in mice. Moreover, PIK3CA mutation and PTEN loss coexist in prostate cancer and can cooperate in vivo to accelerate tumorigenesis and facilitate castration-resistant prostate cancer.
Literature	GENERIF_SUMMARY	High expression of PIK3CA in HNSCC primary tumors identifies patients at high risk for recurrence. In these tumors, progression could rely on the Hippo-YAP pathway instead of the canonical Akt/mTOR pathway
Literature	GENERIF_SUMMARY	We validated the presence of IDH2 R172 hotspot mutations and PIK3CA hotspot mutations in 100% and 67% of solid papillary breast carcinoma with reverse polarity tested, respectively
Literature	GENERIF_SUMMARY	The PIK3CA and PIK3R1 impactful mutations exhibit a mutually exclusive pattern, leading to oncogenesis and hyperactivity of PI3K pathway in breast cancer.
Literature	GENERIF_SUMMARY	Our results indicate that a high proportion of isolated macrodactyly patients carry a pathogenic PIK3CA mutation.
Literature	GENERIF_SUMMARY	This study determined the mutation frequencies and prognostic values of KRAS exon 2, 3 and 4, NRAS exon 2 and 3, PIK3CA exon 9 and 20, and BRAF exon 15 by PCR and direct sequencing in 353 colorectal cancer patients from two Chinese clinical centers.
Literature	GENERIF_SUMMARY	These results showed high concordance between detection of PIK3CA mutations in tumor tissue and in corresponding serum circulating tumor DNA.
Literature	GENERIF_SUMMARY	higher frequency of ESR1 and PIK3CA mutations in the plasma than in the serum in 33 MBC patients; therefore, serum samples should not be considered the preferred source of cfDNA.
Literature	GENERIF_SUMMARY	This study detected gain-of-function mutations in exon 20 of PI3KCA gene in 11%

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		of gastric cancer patients.
Literature	GENERIF_SUMMARY	9p21.3 (CDKN2A) loss and PIK3CA mutation characterize a subgroup of metaplastic breast cancers with myoepithelial and spindle cell differentiation.
Literature	GENERIF_SUMMARY	In patients without PIK3CA alteration, TP53 nonfunctional mutations are associated with poor prognosis.
Literature	GENERIF_SUMMARY	TRAF6 is a novel PIK3CA regulator whose deregulated overexpression represents a mechanism for PI3K overactivation in tumors.
Literature	GENERIF_SUMMARY	High PI3K expression is associated with metastasis in ovarian cancer.
Literature	GENERIF_SUMMARY	Linc-ROR is highly expressed in the ectopic endometrium of adenomyosis, and it can promote the proliferative activity of endometrial cells by activating the PI3K-Akt pathway.
Literature	GENERIF_SUMMARY	Taselisib plus fulvestrant had clinical activity irrespective of PIK3CA mutation status, with numerically higher objective response rate and clinical benefit rate in patients with PIK3CA-mutated (vs. -MND) locally advanced or metastatic HER2-negative, HR-positive breast cancer. No new safety signals were reported.
Literature	GENERIF_SUMMARY	AR + /FOXA1 + expression defines a luminal-like TNBC subgroup affected with a worse outcome compared to other TNBC and a higher risk of late recurrences. This subgroup appears enriched in PIK3CA mutations, suggesting a role for PI3K inhibitors in this subgroup.
Literature	GENERIF_SUMMARY	PIK3CA copy number aberration and activation of the PI3K-AKT-mTOR pathway in varied disease states of penile cancer
Literature	GENERIF_SUMMARY	The crucial role of p110beta and the more subtle role of p110alpha in the production of PIP3 molecular species following platelet stimulation has been demonstrated.
Literature	GENERIF_SUMMARY	PIK3CA mutation in gastric cancer is a rare finding. It is strongly associated with the microsatellite instability (MSI) molecular subgroup, presenting a worse outcome than other MSI patients. A completely different outcome is associated with the mutation in exon 9 compared to the mutation in exon 20, with the latter being more favorable.
Literature	GENERIF_SUMMARY	IL1B promotes the nuclear translocaiton of

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		S100A4 protein in gastric cancer cells and the cell's stem-like properties through PI3K pathway.
Literature	GENERIF_SUMMARY	These results suggest that the classical metabolic PI3K pathway and not the canonical proliferation ERK pathway is involved in the insulin/insulin-like growth factor-1-induced increase in crypt proliferation in obese humans, which may contribute to abnormal tissue renewal and function.
Literature	GENERIF_SUMMARY	pK15-dependent signaling may occur from intracellular vesicles and rely on the endocytotic machinery. Specifically, a class II PI3K, PI3K-C2alpha, is recruited by pK15 and involved in pK15-dependent intracellular signaling and viral reactivation from latency.
Literature	GENERIF_SUMMARY	Data suggest that receptor-tyrosine kinase HEK2 (EphB3) and phosphatidylinositol 3-kinase catalytic alpha polypeptide (PIK3CA) are cooperating oncogenes that contribute toward its pathogenesis.
Literature	GENERIF_SUMMARY	PIK3CA mutations confer resistance to first-line chemotherapy in colorectal cancer.
Literature	GENERIF_SUMMARY	a subset of PIK3CAmut promote tumorigenesis
Literature	GENERIF_SUMMARY	LASP1 may function as an oncogene in glioblastoma and regulate cell proliferation and chemosensitivity in a PI3K/AKT-dependent mechanism.
Literature	GENERIF_SUMMARY	Results show that activating PIK3CA mutations in lymphatic malformations are specifically localized in lymphatic endothelial cells (LECs); identified a new mutation in exon 10 and a Glu109 deletion of the gene and show that several kinase inhibitors have the potential to block LEC proliferation.
Literature	GENERIF_SUMMARY	in Stage I colorectal cancer presence of KRAS mutations, that of simultaneous mutations in PIK3CA gene, or that of multiple KRAS mutations was significantly associated with shorter cancer specific survival; PIK3CA or multiple KRAS mutations were associated with nodal micrometastases and poorly differentiated clusters G3 as well
Literature	GENERIF_SUMMARY	To investigate the reasons for which miR-375 holds a special place among circulating miRNAs in Acute myocardial infarction (MI), enrichment and network analyses of miR-375 target genes and their

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		interactions were carried out. PIK3CA and TP53 genes, regulated by miR-375, were identified as the key players of MI disease module.
Literature	GENERIF_SUMMARY	A high proportion of isolated macrodactyly patients carry a PIK3CA mutation.
Literature	GENERIF_SUMMARY	PIK3CA overexpression is associated with esophageal squamous cell cancer.
Literature	GENERIF_SUMMARY	High PIK3CA expression is associated with metastasis via epithelialmesenchymal transition carcinoma in colorectal cancer.
Literature	GENERIF_SUMMARY	Survival analyses revealed that PIK3CA mutation was a significant prognostic factor for poor overall survival [multivariate adjusted hazard ratio (HR), 3.9; 95% confidence interval (95% CI), 1.3-11.8; P = .017] and cancer-specific survival (multivariate adjusted HR, 3.6; 95% CI, 1.2-11.0; P = .024) in stage IIB to IVA cervical cancers treated by concurrent chemoradiotherapy with weekly cisplatin.
Literature	GENERIF_SUMMARY	miR152 may have an important role in pancreatic beta cell function, and established an association between miR152 and the PI3Kalpha axis.
Literature	GENERIF_SUMMARY	Study identified a novel F977Y kinase domain PIK3CA mutation exclusively present in a recurrent cell line of head and neck squamous cell carcinoma (HNSCC), potentially conferring resistance to PI3K inhibitors.
Literature	GENERIF_SUMMARY	Multivariate analyses revealed that the PIK3CA mutation and clinical T stage were independent favorable prognostic factors (hazard ratio 0.34, 95% confidence interval
Literature	GENERIF_SUMMARY	This overgrowth syndrome is unique and is caused by the somatic PIK3CA mutation c.3140A>G.
Literature	GENERIF_SUMMARY	PIK3CA mutations are found in benign overgrowth syndromes, collectively known as PIK3CA-related overgrowth spectrum. (Review)
Literature	GENERIF_SUMMARY	findings provided a comprehensive mutation profiling of AKT1, PIK3CA, PTEN and TP53 genes in Chinese breast cancer patients, which have potential implications in clinical management.
Literature	GENERIF_SUMMARY	High PIK3CA expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	High PI3K expression is associated with

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		periodontitis.
Literature	GENERIF_SUMMARY	In this study, we used the Ion Personal Genome Machine (PGM) and Ion Torrent Ampliseq Cancer panel to sequence hotspot regions from PIK3CA, AKT and PTEN genes to identify genetic mutations in 39 samples of TNBC subtype from Moroccan patients and to correlate the results with clinical-pathologic data
Literature	GENERIF_SUMMARY	These results demonstrate the dynamical and structural effects induced by the p110alpha E545K mutation in atomic level detail and indicate a possible mechanism for the loss of regulation that E545K confers on PI3Kalpha.
Literature	GENERIF_SUMMARY	Letter
Literature	GENERIF_SUMMARY	Chromatin immunoprecipitation and luciferase reporter gene assays together demonstrated that SOX2 trans-activates PIK3CA and OPA1.
Literature	GENERIF_SUMMARY	these associations may reflect features of the somatic copy-number alteration-high subgroup of uterine corpus endometrial carcinoma rather than effects of PIK3CA amplification itself
Literature	GENERIF_SUMMARY	there is a difference in PIK3CA mutation, COX-2 expression, and paraneoplastic thromboembolism between ovarian clear cell carcinomas with a different stroma.
Literature	GENERIF_SUMMARY	H2O2 induces oxidative stress to breast cancer tumor cell line MCF-7 in a dose dependent manner. Over-expression of p110alpha protein restores pro-survival effects against H2O2 induced apoptosis and increase phosphorylated proto-oncogene proteins c-akt levels.
Literature	GENERIF_SUMMARY	Via the miR-183-5p/PIK3CA regulatory axis.
Literature	GENERIF_SUMMARY	Levels of miR381 were decreased in clear cell and endometrioid carcinoma ovarian cancer. Experimentally induced upregulation of miR381 led to a decrease in the level of PIK3CA in ovarian cancer cells and inhibition of ovarian cancer cells proliferation and migration.
Literature	GENERIF_SUMMARY	Findings provide evidence that the PI3K E542K and E545K/beta-catenin/SIRT3 signaling axis regulates glucose metabolism and proliferation in cervical cancers with PIK3CA mutations.
Literature	GENERIF_SUMMARY	Data suggest that overexpression of phosphatidylinositol 3-kinase (PI3K), protein

Main Category	Category	Terms
		kinase B (PKB), and mammalian target of rapamycin (mTOR) proteins is an important indicator of poor survival in nasopharyngeal carcinoma (NPC).
Literature	GENERIF_SUMMARY	High PIK3CA expression is associated with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	The RAS-PI3K interaction is an important signaling node and potential therapeutic target in EGFR-mutant lung cancer.
Literature	GENERIF_SUMMARY	Somatic activating mutations in PIK3CA cause generalized lymphatic anomaly.
Literature	GENERIF_SUMMARY	Authors used six whole-exome-sequenced primary HGSOC/USC cell-lines and three xenografts overexpressing HER2/neu and harboring mutated or wild-type PIK3CA/PIK3R1 genes to evaluate the role of PI3K-mutations as potential mechanism of resistance to afatinib.
Literature	GENERIF_SUMMARY	In ER+/HER2- invasive ductal carcinomas of the breast, glucose uptake is independently associated with PIK3CA mutations.
Literature	GENERIF_SUMMARY	PIK3CA silencing inhibited Nalm-6 cell proliferation and invasion, and promoted their apoptosis and sensitivity to chemotherapeutic drugs, potentially through regulation of the PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	Authors found that continuously activating PIK3CA(H1047R) conferred human mammary epithelial MCF-10A cells to cellular senescence upon serum-starvation. Similarly, breast cancer T47D and HCC1954 cells harboring H1047R mutation were senescent when cells were deprived of serum.
Literature	GENERIF_SUMMARY	Use of nonsteroidal anti-inflammatory drugs predicts improved patient survival for PIK3CA-altered head and neck cancer.
Literature	GENERIF_SUMMARY	The expression levels of ARID1A and PIK3CA in gastric cancer were significantly associated with the depth of invasion of gastric cancer.
Literature	GENERIF_SUMMARY	Gene alterations, such as TP53 mutation, PIK3CA mutation, ERBB2 amplification and CCND1 amplification, and MAPK pathway alteration were associated with pCR in our study.
Literature	GENERIF_SUMMARY	Compared with invasive ductal breast cancer , mucinous breast cancer had a lower genomic instability and a decreased prevalence of PIK3CA mutations

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Patients with somatic co-mutation of TP53 and PIK3CA were associated with unfavorable survival compared with non-carriers. Co-mutation of TP53 and PIK3CA could be used as a potential prognosis marker in post-neoadjuvant chemotherapy breast cancer patients.
Literature	GENERIF_SUMMARY	Involving MNK targeting to sensitize medulloblastoma CSCs to PI3Kalpha inhibition.
Literature	GENERIF_SUMMARY	PIK3CA mutations could be essential specifically in early stages of breast carcinogenesis.
Literature	GENERIF_SUMMARY	The ARID1A and PIK3CA were the most frequently mutated genes. Specifically, ARID1A mutations and PIK3CA mutations were detected in 77.8% and 66.7% of ovarian clear cell carcinoma patients, respectively. Mutations in other genes, including MLH1 (6.3%) and CREBBP (6.3%), were detected in the Taiwanese population.
Literature	GENERIF_SUMMARY	Study found that PIK3CA mutations and copy-number gains of chromosome 22q11 were more frequent in a Chinese cohort than in The Cancer Genome Atlas.
Literature	GENERIF_SUMMARY	Oncogenic mutations on PIK3CA causing lymphatic malformations and venous malformations are also frequently found in epithelial cancer.
Literature	GENERIF_SUMMARY	Neither PIK3CA mutations nor Fc-gamma-receptor polymorphisms were associated with sTILs. sTILs may have utility as a prognostic biomarker identifying HER2-positive early breast cancer at low recurrence risk.
Literature	GENERIF_SUMMARY	GORASP2/GRASP55 collaborates with the PtdIns3K UVRAG complex to facilitate autophagosome-lysosome fusion.
Literature	GENERIF_SUMMARY	Early activating somatic PIK3CA mutations promote ectopic muscle development and upper limb overgrowth.
Literature	GENERIF_SUMMARY	PIK3CA mutations might be involved in GC development and might be used as favorable prognostic factor in GC population with low PIK3CA mutations prevalence.
Literature	GENERIF_SUMMARY	Authors propose that inhibition of the WNT pathway, particularly beta-catenin, may bypass resistance to MEK inhibition in human PIK3CA mt colon cancer.
Literature	GENERIF_SUMMARY	Increased PIK3CA expression is associated with Colorectal Cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Investigated the role of mutational activation of PI3-kinase p110 subunit alpha (PIK3CA) in cancer and in overgrowth disorders known as PIK3CA-related overgrowth spectrum (PROS) by engineering isogenic human induced pluripotent stem cells (iPSCs) with heterozygous or homozygous knockin of PIK3CA.
Literature	GENERIF_SUMMARY	The current data show that PIK3CA mutations appear to play an important role in carcinogenesis and tumor aggressiveness in Epstein-Barr virus-associated gastric cancer (EBV-GC), and also support the concept that exon 9 mutation of PIK3CA is a prognostic indicator for predicting patient outcomes and a rationale for therapeutic targeting in EBV-GC.
Literature	GENERIF_SUMMARY	Study data from human hepatocellular carcinoma (HCC) liver samples, human HCC cells, and Sgk3 knockout mice model suggest that SGK3 plays a role in transducing helical domain mutant PIK3CA signaling during liver tumor development.
Literature	GENERIF_SUMMARY	Data show that most of the cell lines which exhibited enhanced dependency on RNA guanine-7 methyltransferase (RNMT) harboured oncogenic mutations in phosphatidylinositol 3-kinase catalytic 110-KD alpha (PI3Kalpha).
Literature	GENERIF_SUMMARY	The data suggest that tumourigenic alterations in the PI3K/AKT-pathway are indispensable in hidradenoma papilliferum (HP) and establish a homogenous morphomolecular entity with a functionally converging and selecting tumourigenic mechanism.
Literature	GENERIF_SUMMARY	PIK3CA activating mutations are associated with more disseminated disease at presentation and earlier recurrence in glioblastoma.
Literature	GENERIF_SUMMARY	Breast adenomyoepitheliomas are genetically heterogeneous exhibiting recurrent mutually exclusive mutations of PIK3CA and AKT1 in a majority of cases.
Literature	GENERIF_SUMMARY	The study shows a high concordance of KRAS, NRAS, BRAF and PIK3CA gene mutations in plasma against mutation status in tumor tissue.
Literature	GENERIF_SUMMARY	results indicate that tumor cell PIK3CA-AKT signaling limits T cell recognition and clearance of pancreatic cancer cells. Strategies that target this pathway may yield

Main Category	Category	Terms
		an effective immunotherapy for this cancer.
Literature	GENERIF_SUMMARY	Multiplex PCR analysis of apocrine lesions shows frequent PI3K-AKT pathway mutations in both benign and malignant apocrine breast tumors.
Literature	GENERIF_SUMMARY	Establishment and genomic characterization of gingivobuccal carcinoma cell lines with smokeless tobacco associated genetic alterations and oncogenic PIK3CA mutation.
Literature	GENERIF_SUMMARY	TERT and concurrent PIK3CA mutations predict worse overall survival in poorly differentiated thyroid carcinoma (PDTC)/ anaplastic thyroid carcinoma (ATC) patients.
Literature	GENERIF_SUMMARY	Rosette-forming glioneuronal tumors share a distinct DNA methylation profile and mutations in FGFR1, with recurrent co-mutation of PIK3CA and NF1.
Literature	GENERIF_SUMMARY	PIK3CA hotspot mutations in circulating tumor cells and paired circulating tumor DNA in breast cancer
Literature	GENERIF_SUMMARY	PIK3CA expression in triple negative breast cancer is not associated with specific clinical or diagnostic features. Further molecular studies and meta-analysis are warranted to clarify the prognostic and predictive role of PIK3CA protein expression.
Literature	GENERIF_SUMMARY	Interaction between PIK3CA rs2699887 SNP and age, number of liveborn, tobacco, alcohol, a family history of ovarian cancer and other factors are associated with ovarian cancer risk. Interaction between PIK3CA rs3976507 and rs6443626 loci, and factors such as BMI, number of liveborn, tobacco, alcohol, and family history of ovarian cancer are associated with ovarian cancer risk.
Literature	GENERIF_SUMMARY	Our study provides a proof of concept and attests of the feasibility of cfDNA analysis for patients with ER+ locally advanced or metastatic BC. Early identification of this population with cfDNA PIK3CA mutations allows the identification of a population affected by an unfavorable prognosis.
Literature	GENERIF_SUMMARY	PIK3CA gene mutation is associated with cervical Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	Study utilized human endometrial epithelial cells to elucidate the consequences of ARID1A loss and PIK3CAH1047R in vitro and discover a mechanism by which ARID1A and PIK3CA mutations result in a partial

Main Category	Category	Terms
		epithelial-to-mesenchymal transition phenotype capable of collective invasion into the uterine myometrium.
Literature	GENERIF_SUMMARY	THE DIAGNOSIS AND RELAPSE PREDICTION OF PATIENTS WITH NON-MUSCLE INVASIVE BLADDER CANCER WITH FGFR3,TP53 AND PIK3CA GENE MUTATION POSSESSED HIGH ACCURACY
Literature	GENERIF_SUMMARY	The study met one of its primary endpoints
Literature	GENERIF_SUMMARY	Upregulation of PIK3CA lead to the clonal amplification of tumor cells during progression of invasive Colorectal Adenocarcinoma.
Literature	GENERIF_SUMMARY	PIK3CA expression may play an important role in tumor immune microenvironment and could alter fraction of tumor associated neutrophils in uterine corpus endometrial carcinoma .
Literature	GENERIF_SUMMARY	PIK3CA mutations in lipomatosis of nerve with or without nerve territory overgrowth.
Literature	GENERIF_SUMMARY	Predisposition to thrombosis is likely multifactorial with risk factors including chronic stasis within vascular malformations, stasis from impaired mobility (e.g., following surgery), decreased anticoagulant proteins, and effects of AKT1 and PIK3CA variants on vascular endothelium
Literature	GENERIF_SUMMARY	Talc-sensitive cells have higher mRNA and protein expression of PI3K catalytic subunit alpha. Both PI3Kalpha and -beta mediate talc-induced cell death and IL-6 secretion in lung carcinoma cells.
Literature	GENERIF_SUMMARY	Genotype correlates with clinical severity in PIK3CA-associated lymphatic malformations.
Literature	GENERIF_SUMMARY	High PIK3CA Expression of PTEN is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	NSAID therapy for PIK3CA-Altered colorectal, breast, and head and neck cancer.
Literature	GENERIF_SUMMARY	Distinct roles of PIK3CA in the enrichment and maintenance of cancer stem cells in head and neck squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Expression of miR-337-3p is down-regulated in epithelial ovarian cancer (EOC) tissues and low expression of miR-337-3p is correlated with advanced pathological grade for patients. PIK3CA and PIK3CB were revealed to be direct targets of miR-337-3p.

Main Category	Category	Terms
		In vivo investigation confirmed that miR-337-3p is a tumor suppressor that control expression of PIK3CA and PIK3CB encoded protein
Literature	GENERIF_SUMMARY	Soft tissue angiomatosis
Literature	GENERIF_SUMMARY	Study investigated the presence of FGFR3 and PIK3CA mutations in neuroblastomas (NBs) and tested NB cell lines for sensitivity to FGFR and PI3K inhibitors. Following treatment with FGFR and PI3K inhibitors, a decrease in viability and proliferation was observed in the majority, but not all, the cell lines. Study findings demonstrate that FGFR3 and PIK3CA mutations are uncommon in patients with NB.
Literature	GENERIF_SUMMARY	Integrative transcriptomic and proteomic analysis reveals CD9/ITGA4/PI3K-Akt axis mediates trabecular meshwork cell apoptosis in human glaucoma.
Literature	GENERIF_SUMMARY	this study identified double mutations in cis as a distinctive and relatively common genomic alteration in PIK3CA, the most frequently mutated oncogene in human cancer
Literature	GENERIF_SUMMARY	PIK3CA mutation is associated with lung cancer risk.
Literature	GENERIF_SUMMARY	Growth hormone deficiency in megalencephaly-capillary malformation syndrome
Literature	GENERIF_SUMMARY	Knockdown of Ski decreases osteosarcoma cell proliferation and migration by suppressing the PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	PIK3CA Mutation in HPV-Associated OPSCC Patients Receiving Deintensified Chemoradiation.
Literature	GENERIF_SUMMARY	Phosphatidylinositol 3-kinase Glu545Lys and His1047Tyr Mutations are not Associated with T2D.
Literature	GENERIF_SUMMARY	The neoantigen-associated mutations PIK3CA (p.H1047R) and TP53 (p.R175H) are common across several cancer types, indicating their potential usage.
Literature	GENERIF_SUMMARY	LINC01518 knockdown inhibited tumorigenicity in ESCC cells by suppression of PIK3CA/Akt pathway through upregulating miR-1-3p
Literature	GENERIF_SUMMARY	Bladder cancer cellsecreted exosomal miR21 activates the PI3K/AKT pathway in macrophages to promote cancer

Main Category	Category	Terms
		progression.
Literature	GENERIF_SUMMARY	Defining How Oncogenic and Developmental Mutations of PIK3R1 Alter the Regulation of Class IA Phosphoinositide 3-Kinases.
Literature	GENERIF_SUMMARY	Mutational status of PIK3ca oncogene in oral cancer-In the new age of PI3K inhibitors.
Literature	GENERIF_SUMMARY	Diffuse capillary malformation with overgrowth contains somatic PIK3CA variants.
Literature	GENERIF_SUMMARY	our study revealed a crucial regulatory network of LINC00152/miR-139/PIK3CA axis in the tumorigenesis of hepatocellular carcinoma
Literature	GENERIF_SUMMARY	PIK3CA mutations and specific treatment
Literature	GENERIF_SUMMARY	PIK3CA gene mutations in the helical domain correlate with high tumor mutation burden and poor prognosis in metastatic breast carcinomas with late-line therapies.
Literature	GENERIF_SUMMARY	BRAF, KRAS and PIK3CA Mutation and Sensitivity to Trastuzumab in Breast Cancer Cell Line Model.
Literature	GENERIF_SUMMARY	Detecting mosaic variants in patients with somatic overgrowth syndromes using cell-free circulating DNA and deep sequencing.
Literature	GENERIF_SUMMARY	Putative functional variants of PI3K/AKT/mTOR pathway are associated with knee osteoarthritis susceptibility.
Literature	GENERIF_SUMMARY	Juvenile papillomatosis of the breast (Swiss cheese disease) has frequent associations with PIK3CA and/or AKT1 mutations.
Literature	GENERIF_SUMMARY	Pik3ca mutations significantly enhance the growth of SHH medulloblastoma and lead to metastatic tumour growth in a novel mouse model.
Literature	GENERIF_SUMMARY	Primary mammary angiosarcomas harbor frequent mutations in KDR and PIK3CA and show evidence of distinct pathogenesis.
Literature	GENERIF_SUMMARY	Co-inhibition of BET proteins and PI3Kalpha triggers mitochondrial apoptosis in rhabdomyosarcoma cells.
Literature	GENERIF_SUMMARY	miR10a increases the cisplatin resistance of lung adenocarcinoma circulating tumor cells via targeting PIK3CA in the PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	A Systematic Literature Review of the Prognostic and Predictive Value of PIK3CA

Main Category	Category	Terms
		Mutations in HR(+)/HER2(-) Metastatic Breast Cancer.
Literature	GENERIF_SUMMARY	MicroRNA3633p inhibits cell proliferation and induces apoptosis in retinoblastoma cells via the Akt/mTOR signaling pathway by targeting PIK3CA.
Literature	GENERIF_SUMMARY	Constitutively active PIK3CA mutations are expressed by lymphatic and vascular endothelial cells in capillary lymphatic venous malformation.
Literature	GENERIF_SUMMARY	Mutation-Associated Phenotypic Heterogeneity in Novel and Canonical PIK3CA Helical and Kinase Domain Mutants.
Literature	GENERIF_SUMMARY	Survival prediction model for non-small cell lung cancer based on somatic mutations.
Literature	GENERIF_SUMMARY	Frequency and spectrum of PIK3CA somatic mutations in breast cancer.
Literature	GENERIF_SUMMARY	Clinicopathological Characteristics of PIK3CA Mutation and Amplification in Korean Patients with Breast Cancers.
Literature	GENERIF_SUMMARY	Clinical overlap between CLAPO syndrome and macrocephaly-capillary malformation syndrome.
Literature	GENERIF_SUMMARY	TRPC1 inhibits the proliferation and migration of estrogen receptor-positive Breast cancer and gives a better prognosis by inhibiting the PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	miR-155 facilitates calcium oxalate crystal-induced HK-2 cell injury via targeting PI3K associated autophagy.
Literature	GENERIF_SUMMARY	Activating AKT1 and PIK3CA Mutations in Metastatic Castration-Resistant Prostate Cancer.
Literature	GENERIF_SUMMARY	The Predictive Role of PIK3CA Mutation Status on PI3K Inhibitors in HR+ Breast Cancer Therapy
Literature	GENERIF_SUMMARY	Blockade of VEGF-C signaling inhibits lymphatic malformations driven by oncogenic PIK3CA mutation.
Literature	GENERIF_SUMMARY	PI3K activation is enhanced by FOXM1D binding to p110 and p85 subunits.
Literature	GENERIF_SUMMARY	Comprehensive genomic profiling reveals ubiquitous KRAS mutations and frequent PIK3CA mutations in ovarian seromucinous borderline tumor.
Literature	GENERIF_SUMMARY	Activating PIK3CA mutation promotes

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		osteogenesis of bone marrow mesenchymal stem cells in macrodactyly.
Literature	GENERIF_SUMMARY	Identifying oncogenic drivers associated with increased risk of late distant recurrence in postmenopausal, estrogen receptor-positive, HER2-negative early breast cancer
Literature	GENERIF_SUMMARY	PIK3CA mutation and CNV status and post-chemoradiotherapy survival in patients with cervical cancer.
Literature	GENERIF_SUMMARY	Frequent activating PIK3CA mutations in sporadic angiolipoma.
Literature	GENERIF_SUMMARY	PIK3CA Mutations and Their Impact on Survival Outcomes of Patients with Cervical Cancer
Literature	GENERIF_SUMMARY	miR1425p inhibits pancreatic cancer cell migration and invasion by targeting PIK3CA.
Literature	GENERIF_SUMMARY	Activating PIK3CA mutation promotes adipogenesis of adipose-derived stem cells in macrodactyly via up-regulation of E2F1.
Literature	GENERIF_SUMMARY	Detailed analysis of phenotypes and genotypes in megalencephaly-capillary malformation-polymicrogyria syndrome caused by somatic mosaicism of PIK3CA mutations.
Literature	GENERIF_SUMMARY	Enforced C-Src Activation Causes Compartmental Dysregulation of PI3K and PTEN Molecules in Lipid Rafts of Tongue Squamous Carcinoma Cells by Attenuating Rac1-Akt-GLUT-1-Mediated Sphingolipid Synthesis.
Literature	GENERIF_SUMMARY	PIK3CA Mutation in the ShortHER Randomized Adjuvant Trial for Patients with Early HER2(+) Breast Cancer
Literature	GENERIF_SUMMARY	A novel method for isolating lymphatic endothelial cells from lymphatic malformations and detecting PIK3CA somatic mutation in these isolated cells.
Literature	GENERIF_SUMMARY	Comprehensive Immuno-Molecular Profiles for Liposarcoma
Literature	GENERIF_SUMMARY	Cerebellar dysplasia related to PIK3CA mutation
Literature	GENERIF_SUMMARY	Clinical Significance of PIK3CA Gene in Non-Small-Cell Lung Cancer
Literature	GENERIF_SUMMARY	PIK3CA C-terminal frameshift mutations are novel oncogenic events that sensitize tumors to PI3K-alpha inhibition.
Literature	GENERIF_SUMMARY	RP11-462C24.1 suppresses proliferation

Main Category	Category	Terms
		and invasion of colorectal carcinoma cells by regulating HSP70 through PI3K/AKT signaling pathway.
Literature	GENERIF_SUMMARY	Metabolic Imaging Detects Resistance to PI3Kalpha Inhibition Mediated by Persistent FOXM1 Expression in ER(+) Breast Cancer.
Literature	GENERIF_SUMMARY	Predictive Role of TP53, PIK3CA and MLL2 in ER+ HER2+ Breast Bancer
Literature	GENERIF_SUMMARY	Clinical-Pathologic Analysis of Breast Cancer With PIK3CA Mutations in Chinese Women.
Literature	GENERIF_SUMMARY	PI3K mutations detected in liquid biopsy are associated to reduced sensitivity to CDK4/6 inhibitors in metastatic breast cancer patients.
Literature	GENERIF_SUMMARY	Phenotypic and genetic spectrum of isolated macrodactyly
Literature	GENERIF_SUMMARY	Phosphatidylinositol-3-OH kinase signalling is spatially organized at endosomal compartments by microtubule-associated protein 4.
Literature	GENERIF_SUMMARY	Prediction of PIK3CA mutations from cancer gene expression data.
Literature	GENERIF_SUMMARY	A novel carcinogenic PI3Kalpha mutation suggesting the role of helical domain in transmitting nSH2 regulatory signals to kinase domain.
Literature	GENERIF_SUMMARY	Phosphorylation and Driver Mutations in PI3Kalpha and PTEN Autoinhibition.
Literature	GENERIF_SUMMARY	The utility of TP53 and PIK3CA mutations as prognostic biomarkers in salivary adenoid cystic carcinoma.
Literature	GENERIF_SUMMARY	N-Phenyl-6-Chloro-4-Hydroxy-2-Quinolone-3-CarboxAmides
Literature	GENERIF_SUMMARY	PIK3CA somatic alterations in invasive breast cancers
Literature	GENERIF_SUMMARY	Relationship between TSHR, BRAF and PIK3CA gene copy number variations and thyroid nodules.
Literature	GENERIF_SUMMARY	Distinct epithelial-to-mesenchymal transitions induced by PIK3CA (H1047R) and PIK3CB.
Literature	GENERIF_SUMMARY	Hypoxia-induced Nur77 activates PI3K/Akt signaling via suppression of Dicer/let-7i-5p to induce epithelial-to-mesenchymal transition.

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Literature	GENERIF_SUMMARY	Biophysical and Structural Characterization of Novel RAS-Binding Domains (RBDs) of PI3Kalpha and PI3Kgamma.
Literature	GENERIF_SUMMARY	Clinical and Biomarker Results from Phase I/II Study of PI3K Inhibitor Alpelisib plus Nab-paclitaxel in HER2-Negative Metastatic Breast Cancer.
Literature	GENERIF_SUMMARY	Regulation of PTEN translation by PI3K signaling maintains pathway homeostasis.
Literature	GENERIF_SUMMARY	Molecular Biomarkers for Contemporary Therapies in Hormone Receptor-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Inhibition of P110alpha and P110delta catalytic subunits of PI3 kinase reverses impaired arterial healing after injury in hypercholesterolemic male mice.
Literature	GENERIF_SUMMARY	PI3K activation promotes resistance to eribulin in HER2-negative breast cancer.
Literature	GENERIF_SUMMARY	Phase 2 study of TAS-117, an allosteric akt inhibitor in advanced solid tumors harboring phosphatidylinositol 3-kinase/v-akt murine thymoma viral oncogene homolog gene mutations.
Literature	GENERIF_SUMMARY	Somatic MAP3K3 and PIK3CA mutations in sporadic cerebral and spinal cord cavernous malformations.
Literature	GENERIF_SUMMARY	The Landscape of PIK3CA Mutations in Colorectal Cancer.
Literature	GENERIF_SUMMARY	PI3Kalpha inhibitor impairs AKT phosphorylation and synergizes with novel angiogenesis inhibitor AL3810 in human hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Alpelisib plus fulvestrant in PIK3CA-mutated, hormone receptor-positive advanced breast cancer after a CDK4/6 inhibitor (BYLieve)
Literature	GENERIF_SUMMARY	MEX3A promotes development and progression of breast cancer through regulation of PIK3CA.
Literature	GENERIF_SUMMARY	Identification of PIK3CA multigene mutation patterns associated with superior prognosis in stomach cancer.
Literature	GENERIF_SUMMARY	Correlation between SNPs of PIK3CA, ERBB2 3'UTR, and their interactions with environmental factors and the risk of epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	Pharmacological and cell-specific genetic PI3Kalpha inhibition worsens cardiac

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		remodeling after myocardial infarction.
Literature	GENERIF_SUMMARY	PIK3CA and CCM mutations fuel cavernomas through a cancer-like mechanism.
Literature	GENERIF_SUMMARY	Secretion of Mutant DNA and mRNA by the Exosomes of Breast Cancer Cells.
Literature	GENERIF_SUMMARY	A Phase I Study of Alpelisib in Combination with Trastuzumab and LJM716 in Patients with PIK3CA-Mutated HER2-Positive Metastatic Breast Cancer.
Literature	GENERIF_SUMMARY	High frequency of PIK3CA and TERT promoter mutations in fibromatosis-like spindle cell carcinomas.
Literature	GENERIF_SUMMARY	Spectrum of PIK3CA/AKT mutations across molecular subtypes of triple-negative breast cancer.
Literature	GENERIF_SUMMARY	Distribution of genetic alterations in high-risk early-stage cervical cancer patients treated with postoperative radiation therapy.
Literature	GENERIF_SUMMARY	The role of different PI3K protein subtypes in the metastasis, angiogenesis and clinical prognosis of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Pancreatic cancer intrinsic PI3Kalpha activity accelerates metastasis and rewires macrophage component.
Literature	GENERIF_SUMMARY	INPP4B promotes PI3Kalpha-dependent late endosome formation and Wnt/beta-catenin signaling in breast cancer.
Literature	GENERIF_SUMMARY	QPCT regulation by CTCF leads to sunitinib resistance in renal cell carcinoma by promoting angiogenesis.
Literature	GENERIF_SUMMARY	Detection of PIK3CA Gene Mutation in Head and Neck Squamous Cell Carcinoma Using Droplet Digital PCR and RT-qPCR.
Literature	GENERIF_SUMMARY	Effect of PIK3CA variants on glioma-related epilepsy and response to treatment.
Literature	GENERIF_SUMMARY	Combined inhibition of DDR1 and CDK4/6 induces synergistic effects in ER-positive, HER2-negative breast cancer with PIK3CA/AKT1 mutations.
Literature	GENERIF_SUMMARY	Frequent PIK3CA mutations in eutopic endometrium of patients with ovarian clear cell carcinoma.
Literature	GENERIF_SUMMARY	Quadruple-editing of the MAPK and PI3K pathways effectively blocks the progression of KRAS-mutated colorectal cancer cells.

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Literature	GENERIF_SUMMARY	Analysis of EGFR, KRAS, and PIK3CA gene mutation rates and clinical distribution in patients with different types of lung cancer.
Literature	GENERIF_SUMMARY	Effects of miR-202-5p silencing PIK3CA gene expression on proliferation, invasion, and epithelial-mesenchymal transition of cervical cancer SiHa cells through inhibiting PI3K/Akt/mTOR signaling pathway activation.
Literature	GENERIF_SUMMARY	Investigation of somatic PIK3CA gene mutations in breast cancer patients.
Literature	GENERIF_SUMMARY	Cx43 deficiency confers EMT-mediated tamoxifen resistance to breast cancer via c-Src/PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Sensitivity to targeted therapy differs between HER2-amplified breast cancer cells harboring kinase and helical domain mutations in PIK3CA.
Literature	GENERIF_SUMMARY	Association of PIK3CA mutation and PTEN loss with expression of CD274 (PD-L1) in colorectal carcinoma.
Literature	GENERIF_SUMMARY	Aberrant PTEN, PIK3CA, pMAPK, and TP53 expression in human scalp and face angiosarcoma.
Literature	GENERIF_SUMMARY	Kaposiform hemangioendothelioma further broadens the phenotype of PIK3CA-related overgrowth spectrum.
Literature	GENERIF_SUMMARY	Predictive Value of the TP53/PIK3CA/ATM Mutation Classifier for Patients With Bladder Cancer Responding to Immune Checkpoint Inhibitor Therapy.
Literature	GENERIF_SUMMARY	PIK3CA mutations in plasma circulating tumor DNA predict survival and treatment outcomes in patients with advanced cancers.
Literature	GENERIF_SUMMARY	Somatic PIK3CA Mutations in Sporadic Cerebral Cavernous Malformations.
Literature	GENERIF_SUMMARY	Cancer-associated mutations in the p85alpha N-terminal SH2 domain activate a spectrum of receptor tyrosine kinases.
Literature	GENERIF_SUMMARY	Double PIK3CA Alterations and Parallel Evolution in Colorectal Cancers.
Literature	GENERIF_SUMMARY	A PLCB1-PI3K-AKT Signaling Axis Activates EMT to Promote Cholangiocarcinoma Progression.
Literature	GENERIF_SUMMARY	A protein network map of head and neck cancer reveals PIK3CA mutant drug sensitivity.

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Literature	GENERIF_SUMMARY	Aberrant muscular hyperplasia of the hand and PIK3CA related overgrowth spectrum disorders. A case series study.", trans "Hiperplasia muscular aberrante de la mano y sindromes de sobrecrecimiento relacionados con el PIK3CA. Serie de casos."
Literature	GENERIF_SUMMARY	[PIK3CA copy number gain in esophageal squamous cell carcinoma and its prognostic significance].
Literature	GENERIF_SUMMARY	Radioresistance and genomic alterations in head and neck squamous cell cancer
Literature	GENERIF_SUMMARY	EZH2 inhibition confers PIK3CA-driven lung tumors enhanced sensitivity to PI3K inhibition.
Literature	GENERIF_SUMMARY	DNA Methylation Profiles of Ovarian Clear Cell Carcinoma.
Literature	GENERIF_SUMMARY	CBX2 Induces Glioma Cell Proliferation and Invasion Through the Akt/PI3K Pathway.
Literature	GENERIF_SUMMARY	PIK3CA mutation affects the proliferation of colorectal cancer cells through the PI3K-MEK/PDK1-GPT2 pathway.
Literature	GENERIF_SUMMARY	Sensitive and robust liquid biopsy-based detection of PIK3CA mutations in hormone-receptor-positive metastatic breast cancer patients.
Literature	GENERIF_SUMMARY	Circulating Tumor DNA Profiling From Breast Cancer Screening Through to Metastatic Disease.
Literature	GENERIF_SUMMARY	Discordance of PIK3CA and TP53 mutations between breast cancer brain metastases and matched primary tumors.
Literature	GENERIF_SUMMARY	Cerebrofacial vascular metameric syndrome is caused by somatic pathogenic variants in PIK3CA.
Literature	GENERIF_SUMMARY	Circulating PIK3CA mutation detection at diagnosis in non-metastatic inflammatory breast cancer patients.
Literature	GENERIF_SUMMARY	PIP3 abundance overcomes PI3K signaling selectivity in invadopodia.
Literature	GENERIF_SUMMARY	A mass spectrometric method for in-depth profiling of phosphoinositide regioisomers and their disease-associated regulation.
Literature	GENERIF_SUMMARY	Association of mutation signature effectuating processes with mutation hotspots in driver genes and non-coding regions.

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Literature	GENERIF_SUMMARY	PIK3CA mutation correlates with mTOR pathway expression but not clinical and pathological features in Fibfibroipose vascular anomaly (FAVA).
Literature	GENERIF_SUMMARY	Activating PIK3CA postzygotic mutations in segmental overgrowth of muscles with bone involvement in the body extremities.
Literature	GENERIF_SUMMARY	Comparison of mutational profiles between triple-negative and hormone receptor-positive/human epidermal growth factor receptor 2-negative breast cancers in T2N0-1M0 stage
Literature	GENERIF_SUMMARY	Introduction and expression of PIK3CA(E545K) in a papillary thyroid cancer BRAF(V600E) cell line leads to a dedifferentiated aggressive phenotype.
Literature	GENERIF_SUMMARY	Intertumor and intratumor heterogeneity of PIK3CA mutations in extramammary Paget's disease.
Literature	GENERIF_SUMMARY	Mutations in the telomerase reverse transcriptase promoter and PIK3CA gene are common events in penile squamous cell carcinoma of Italian and Ugandan patients.
Literature	GENERIF_SUMMARY	PIK3CA-related overgrowth
Literature	GENERIF_SUMMARY	Myoepithelial progenitors as founder cells of hyperplastic human breast lesions upon PIK3CA transformation.
Literature	GENERIF_SUMMARY	Circ-ZEB1 promotes PIK3CA expression by silencing miR-199a-3p and affects the proliferation and apoptosis of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	AKT1 and PIK3CA activating mutations in Moroccan bladder cancer patients biopsies and matched urine.
Literature	GENERIF_SUMMARY	PIK3CA Mutations Drive Therapeutic Resistance in Human Epidermal Growth Factor Receptor 2-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Mutation patterns and prognostic analysis of BRAF/KRAS/PIK3CA in colorectal cancer.
Literature	GENERIF_SUMMARY	CDK6-PI3K signaling axis is an efficient target for attenuating ABCB1/P-gp mediated multi-drug resistance (MDR) in cancer cells.
Literature	GENERIF_SUMMARY	Analysis of PIK3CA Mutation Concordance and Frequency in Primary and Different Distant Metastatic Sites in Breast Cancer.
Literature	GENERIF_SUMMARY	Immunogenicity and therapeutic targeting of a public neoantigen derived from mutated PIK3CA.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Absence of PIK3CA mutation in primary mucinous ovarian carcinoma
Literature	GENERIF_SUMMARY	PIK3CA mutations-mediated downregulation of circLHFPL2 inhibits colorectal cancer progression via upregulating PTEN.
Literature	GENERIF_SUMMARY	Proteomic alterations in extracellular vesicles induced by oncogenic PIK3CA mutations.
Literature	GENERIF_SUMMARY	Reciprocal regulation of Daxx and PIK3CA promotes colorectal cancer cell growth.
Literature	GENERIF_SUMMARY	Prognostic and Predictive Value of PIK3CA Mutations in Metastatic Colorectal Cancer.
Literature	GENERIF_SUMMARY	Mutational landscape of pan-cancer patients with PIK3CA alterations in Chinese population.
Literature	GENERIF_SUMMARY	Genomic landscape of lymphatic malformations
Literature	GENERIF_SUMMARY	Loss of PTEN Expression, PIK3CA Mutations, and Breast Cancer Survival in the Nurses' Health Studies.
Literature	GENERIF_SUMMARY	Hippo-YAP signaling activation and cross-talk with PI3K in oral cancer
Literature	GENERIF_SUMMARY	PIK3CA Mutation is Associated with Poor Response to HER2-Targeted Therapy in Breast Cancer Patients.
Literature	GENERIF_SUMMARY	PIK3CA mutation status, progression and survival in advanced HR + /HER2- breast cancer
Literature	GENERIF_SUMMARY	Energy balance-related factors in childhood and adolescence and risk of colorectal cancer based on KRAS, PIK3CA, and BRAF mutations and MMR status.
Literature	GENERIF_SUMMARY	PIK3CA mutations associated with a poor postoperative prognosis in patients with pulmonary pleomorphic carcinoma
Literature	GENERIF_SUMMARY	PIK3CA-mutations in breast cancer.
Literature	GENERIF_SUMMARY	PIK3CA somatic mutations as potential biomarker for immunotherapy in elder or TP53 mutated gastric cancer patients.
Literature	GENERIF_SUMMARY	Long non-coding RNA ncRuPAR regulates gastric cancer cell proliferation and apoptosis via phosphoinositide 3-kinase/ protein kinase B signaling.
Literature	GENERIF_SUMMARY	Clinical and functional characterization of germline PIK3CA variants in patients with

Main Category	Category	Terms
		PIK3CA-related overgrowth spectrum disorders.
Literature	GENERIF_SUMMARY	JCAD promotes arterial thrombosis through PI3K/Akt modulation
Literature	GENERIF_SUMMARY	[PIK3CA gene mutations in Chinese women with HR[+]/HER2[-] breast cancer].
Literature	GENERIF_SUMMARY	Predictive and prognostic value of PIK3CA mutations in HER2-positive breast cancer treated with tyrosine kinase inhibitors
Literature	GENERIF_SUMMARY	HMGB1 promotes Ox-LDL-induced endothelial cell damage by inhibiting PI3K/Akt signaling pathway.
Literature	GENERIF_SUMMARY	An integrated investigation of structural and pathway alteration caused by PIK3CA and TP53 mutations identified in cfDNA of metastatic breast cancer.
Literature	GENERIF_SUMMARY	Expression and Prognostic Value of PIK3CA, VEGF, IL-8, IL-10, and RIP2 in Diffuse Large B-Cell Lymphoma.
Literature	GENERIF_SUMMARY	Genetic Heterogeneity and Tissue-specific Patterns of Tumors with Multiple PIK3CA Mutations.
Literature	GENERIF_SUMMARY	Oncogenic mutations of PIK3CA lead to increased membrane recruitment driven by reorientation of the ABD, p85 and C-terminus.
Literature	GENERIF_SUMMARY	Solid Organ Transplantation Is Associated with an Increased Rate of Mismatch Repair Deficiency and PIK3CA Mutations in Colorectal Cancer.
Literature	GENERIF_SUMMARY	Early Triple-Negative Breast Cancers in a Singapore Cohort Exhibit High PIK3CA Mutation Rates Associated With Low PD-L1 Expression.
Literature	GENERIF_SUMMARY	Microphysiological model of PIK3CA-driven vascular malformations reveals a role of dysregulated Rac1 and mTORC1/2 in lesion formation.
Literature	GENERIF_SUMMARY	Dysregulation of Mitochondrial Translation Caused by CBFB Deficiency Cooperates with Mutant PIK3CA and Is a Vulnerability in Breast Cancer.
Literature	GENERIF_SUMMARY	Frequent PIK3CA mutation in normal endometrial gland drives spheroid formation and may be involved in stem cell propagation.
Literature	GENERIF_SUMMARY	Driver mutation characteristics of phosphatidylinositol-4,5-bisphosphate 3-

Main Category	Category	Terms
		kinase catalytic subunit alpha (PIK3CA) in advanced non-small cell lung cancer.
Literature	GENERIF_SUMMARY	PI3Kalpha Translocation Mediates Nuclear PtdIns(3,4,5)P3 Effector Signaling in Colorectal Cancer.
Literature	GENERIF_SUMMARY	PIK3CA and PIK3R1 tumor mutational landscape in a pan-cancer patient cohort and its association with pathway activation and treatment efficacy.
Literature	GENERIF_SUMMARY	PIK3CA mutations and their impact on survival outcomes of patients with endometrial cancer
Literature	GENERIF_SUMMARY	CDK12 Promotes the Proliferation, Migration, and Angiogenesis of Gastric Carcinoma via Activating the PI3K/AKT/mTOR Signaling Pathway.
Literature	GENERIF_SUMMARY	PIK3CA Mutations as a Prognostic Factor in Patients With Residual Rectal Cancer After Neoadjuvant Chemoradiotherapy.
Literature	GENERIF_SUMMARY	[Landscape of KRAS, BRAF, and PIK3CA Mutations and Clinical Features of EBV-Associated and Microsatellite Unstable Gastric Cancer].
Literature	GENERIF_SUMMARY	PIK3CA mutation testing as a valuable molecular surrogate for lipomatosis of the median nerve
Literature	GENERIF_SUMMARY	The impact of PIK3CA mutations and PTEN expression on the effect of neoadjuvant therapy for postmenopausal luminal breast cancer patients.
Literature	GENERIF_SUMMARY	PIK3CA mutation in endometriotic epithelial cells promotes viperin-dependent inflammatory response to insulin.
Literature	GENERIF_SUMMARY	Localized Endometrial Proliferations of Pregnancy are Clonal Glandular Outgrowths Characterized by PTEN Loss and PIK3CA Pathogenic Variants.
Literature	GENERIF_SUMMARY	PIK3CA mutations in breast cancer
Literature	GENERIF_SUMMARY	Genetic Alterations Detected by Circulating Tumor DNA in HER2-Low Metastatic Breast Cancer.
Literature	GENERIF_SUMMARY	Expression and Correlation Research of MicroRNA10a-5p and PIK3CA in Middle Ear Cholesteatoma.
Literature	GENERIF_SUMMARY	Breast Cancer Mutations HER2V777L and PIK3CAH1047R Activate the p21-CDK4/6-Cyclin D1 Axis to Drive Tumorigenesis and Drug Resistance.

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Literature	GENERIF_SUMMARY	Loss of PI3Kalpha Mediates Protection From Myocardial Ischemia-Reperfusion Injury Linked to Preserved Mitochondrial Function.
Literature	GENERIF_SUMMARY	Association between PIK3CA activating mutations and outcomes in early-stage invasive lobular breast carcinoma treated with adjuvant systemic therapy.
Literature	GENERIF_SUMMARY	PIK3CA mutation subtype delineates distinct immune profiles in gastric carcinoma.
Literature	GENERIF_SUMMARY	Plasma cell-free tumor DNA, PIK3CA and TP53 mutations predicted inferior endocrine-based treatment outcome in endocrine receptor-positive metastatic breast cancer.
Literature	GENERIF_SUMMARY	Discordance of PIK3CA mutational status between primary and metastatic breast cancer
Literature	GENERIF_SUMMARY	Lipid-correlated alterations in the transcriptome are enriched in several specific pathways in the postmortem prefrontal cortex of Japanese patients with schizophrenia.
Literature	GENERIF_SUMMARY	Knockdown of BUB1 inhibits tumor necrosis factor-alpha-induced proliferation and migration of rheumatoid arthritis synovial fibroblasts by regulating PI3K/Akt pathway.
Literature	GENERIF_SUMMARY	Clinical, Pathological and Molecular Insights on KRAS, NRAS, BRAF, PIK3CA and TP53 Mutations in Metastatic Colorectal Cancer Patients from Northeastern Romania.
Literature	GENERIF_SUMMARY	Somatic mutation spectrum of a Chinese cohort of pediatrics with vascular malformations.
Literature	GENERIF_SUMMARY	Microcystic lymphatic malformations in Turner syndrome are due to somatic mosaicism of PIK3CA.
Literature	GENERIF_SUMMARY	Hemifacial myohyperplasia is due to somatic muscular PIK3CA gain-of-function mutations and responds to pharmacological inhibition.
Literature	GENERIF_SUMMARY	Interplay between ESR1/PIK3CA codon variants, oncogenic pathway alterations and clinical phenotype in patients with metastatic breast cancer (MBC)
Literature	GENERIF_SUMMARY	NOTCH1 and PIK3CA mutation are related to HPV-associated vulvar squamous cell carcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Allosteric PI3Kalpha Inhibition Overcomes On-target Resistance to Orthosteric Inhibitors Mediated by Secondary PIK3CA Mutations.
Literature	GENERIF_SUMMARY	Molecular characterization of 13 patients with PIK3CA-related overgrowth spectrum using a targeted deep sequencing approach.
Literature	GENERIF_SUMMARY	Evaluation of tumor-infiltrating lymphocytes, PD-L1, and PIK3CA mutations and association with prognosis in HER2-positive early stage breast cancer.
Literature	GENERIF_SUMMARY	Oncogenic PIK3CA recruits myeloid-derived suppressor cells to shape the immunosuppressive tumour microenvironment in luminal breast cancer through the 5-lipoxygenase-dependent arachidonic acid pathway.
Literature	GENERIF_SUMMARY	APC/PIK3CA mutations and beta-catenin status predict tankyrase inhibitor sensitivity of patient-derived colorectal cancer cells.
Literature	GENERIF_SUMMARY	The p110alpha/DeltaNp63alpha complex mutations in triple-negative breast cancer
Literature	GENERIF_SUMMARY	Mucinous cystadenocarcinoma of the breast harbours TRPS1 expressions and PIK3CA alterations.
Literature	GENERIF_SUMMARY	Mutational and expressional association of the PIK3CA gene with the risk of breast cancer in the Pakistani population.
Literature	GENERIF_SUMMARY	Coexistent ARID1A-PIK3CA mutations are associated with immune-related pathways in luminal breast cancer.
Literature	GENERIF_SUMMARY	Association of PIK3CA Mutation With Pathologic Complete Response and Outcome by Hormone Receptor Status and Intrinsic Subtype in Early-Stage ERBB2/HER2-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	PIK3CA regulates development of diabetes retinopathy through the PI3K/Akt/mTOR pathway.
Literature	GENERIF_SUMMARY	PIK3CA mutation status in apocrine carcinoma arising in apocrine gland hyperplasia/apocrine nevus
Literature	GENERIF_SUMMARY	PIK3CA mutation as an acquired resistance driver to EGFR-TKIs in non-small cell lung cancer
Literature	GENERIF_SUMMARY	PIK3CA mutation-driven immune signature as a prognostic marker for evaluating the tumor immune microenvironment and therapeutic response in breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Cryo-EM structures reveal two allosteric inhibition modes of PI3Kalpha[H1047R] involving a re-shaping of the activation loop.
Literature	GENERIF_SUMMARY	PI 3-Kinase and the Histone Methyl-Transferase KMT2D Collaborate to Induce Arp2/3-Dependent Migration of Mammary Epithelial Cells.
Literature	GENERIF_SUMMARY	PIK3CA mutations in endocrine-resistant breast cancer.
Literature	GENERIF_SUMMARY	Association of ESR1 Germline Variants with TP53 Somatic Variants in Breast Tumors in a Genome-wide Study.
Literature	GENERIF_SUMMARY	PIK3CA inhibition in models of proliferative glomerulonephritis and lupus nephritis.
Literature	GENERIF_SUMMARY	Analysis of PIK3CA mutations in the primary and recurrent tumors of hormone receptor positive/human epidermal growth factor receptor 2 negative breast cancer.
Literature	GENERIF_SUMMARY	Genetic Insights into Colorectal Cancer
Literature	GENERIF_SUMMARY	Delineation of the phenotypes and genotypes of PIK3CA-related overgrowth spectrum in East asians.
Literature	GENERIF_SUMMARY	Organismal metabolism regulates the expansion of oncogenic PIK3CA mutant clones in normal esophagus.
Literature	GENERIF_SUMMARY	PIK3CA Mutated Colorectal Cancers Without KRAS, NRAS and BRAF Mutations Possess Common and Potentially Targetable Mutations in Epigenetic Modifiers and DNA Damage Response Genes.
Literature	GENERIF_SUMMARY	Constitutional Mutation of PIK3CA
Literature	GENERIF_SUMMARY	Defining the transcriptome of PIK3CA-altered cells in a human capillary malformation using single cell long-read sequencing.
Pathways	KEGG_PATHWAY	Inositol phosphate metabolism
Pathways	KEGG_PATHWAY	Metabolic pathways
Pathways	KEGG_PATHWAY	EGFR tyrosine kinase inhibitor resistance
Pathways	KEGG_PATHWAY	Endocrine resistance
Pathways	KEGG_PATHWAY	Platinum drug resistance
Pathways	KEGG_PATHWAY	ErbB signaling pathway
Pathways	KEGG_PATHWAY	Ras signaling pathway

Main Category	Category	Terms
Pathways	KEGG_PATHWAY	Rap1 signaling pathway
Pathways	KEGG_PATHWAY	cAMP signaling pathway
Pathways	KEGG_PATHWAY	Chemokine signaling pathway
Pathways	KEGG_PATHWAY	HIF-1 signaling pathway
Pathways	KEGG_PATHWAY	FoxO signaling pathway
Pathways	KEGG_PATHWAY	Phosphatidylinositol signaling system
Pathways	KEGG_PATHWAY	Sphingolipid signaling pathway
Pathways	KEGG_PATHWAY	Phospholipase D signaling pathway
Pathways	KEGG_PATHWAY	Hormone signaling
Pathways	KEGG_PATHWAY	Autophagy - animal
Pathways	KEGG_PATHWAY	mTOR signaling pathway
Pathways	KEGG_PATHWAY	PI3K-Akt signaling pathway
Pathways	KEGG_PATHWAY	AMPK signaling pathway
Pathways	KEGG_PATHWAY	Apoptosis
Pathways	KEGG_PATHWAY	Longevity regulating pathway
Pathways	KEGG_PATHWAY	Longevity regulating pathway - multiple species
Pathways	KEGG_PATHWAY	Cellular senescence
Pathways	KEGG_PATHWAY	Axon guidance
Pathways	KEGG_PATHWAY	VEGF signaling pathway
Pathways	KEGG_PATHWAY	Osteoclast differentiation
Pathways	KEGG_PATHWAY	Focal adhesion
Pathways	KEGG_PATHWAY	IgSF CAM signaling
Pathways	KEGG_PATHWAY	Signaling pathways regulating pluripotency of stem cells
Pathways	KEGG_PATHWAY	Platelet activation
Pathways	KEGG_PATHWAY	Neutrophil extracellular trap formation
Pathways	KEGG_PATHWAY	Toll-like receptor signaling pathway
Pathways	KEGG_PATHWAY	C-type lectin receptor signaling pathway
Pathways	KEGG_PATHWAY	JAK-STAT signaling pathway
Pathways	KEGG_PATHWAY	Natural killer cell mediated cytotoxicity
Pathways	KEGG_PATHWAY	T cell receptor signaling pathway
Pathways	KEGG_PATHWAY	B cell receptor signaling pathway

Main Category	Category	Terms
Pathways	KEGG_PATHWAY	Fc epsilon RI signaling pathway
Pathways	KEGG_PATHWAY	Fc gamma R-mediated phagocytosis
Pathways	KEGG_PATHWAY	TNF signaling pathway
Pathways	KEGG_PATHWAY	Leukocyte transendothelial migration
Pathways	KEGG_PATHWAY	Neurotrophin signaling pathway
Pathways	KEGG_PATHWAY	Cholinergic synapse
Pathways	KEGG_PATHWAY	Inflammatory mediator regulation of TRP channels
Pathways	KEGG_PATHWAY	Regulation of actin cytoskeleton
Pathways	KEGG_PATHWAY	Insulin signaling pathway
Pathways	KEGG_PATHWAY	Progesterone-mediated oocyte maturation
Pathways	KEGG_PATHWAY	Estrogen signaling pathway
Pathways	KEGG_PATHWAY	Prolactin signaling pathway
Pathways	KEGG_PATHWAY	Thyroid hormone signaling pathway
Pathways	KEGG_PATHWAY	Regulation of lipolysis in adipocytes
Pathways	KEGG_PATHWAY	Relaxin signaling pathway
Pathways	KEGG_PATHWAY	GnRH secretion
Pathways	KEGG_PATHWAY	Type II diabetes mellitus
Pathways	KEGG_PATHWAY	Insulin resistance
Pathways	KEGG_PATHWAY	Non-alcoholic fatty liver disease
Pathways	KEGG_PATHWAY	AGE-RAGE signaling pathway in diabetic complications
Pathways	KEGG_PATHWAY	Growth hormone synthesis, secretion and action
Pathways	KEGG_PATHWAY	Aldosterone-regulated sodium reabsorption
Pathways	KEGG_PATHWAY	Carbohydrate digestion and absorption
Pathways	KEGG_PATHWAY	Alzheimer disease
Pathways	KEGG_PATHWAY	Spinocerebellar ataxia
Pathways	KEGG_PATHWAY	Prion disease
Pathways	KEGG_PATHWAY	Bacterial invasion of epithelial cells
Pathways	KEGG_PATHWAY	Shigellosis
Pathways	KEGG_PATHWAY	Salmonella infection
Pathways	KEGG_PATHWAY	Yersinia infection
Pathways	KEGG_PATHWAY	Chagas disease

Main Category	Category	Terms
Pathways	KEGG_PATHWAY	Amoebiasis
Pathways	KEGG_PATHWAY	Hepatitis C
Pathways	KEGG_PATHWAY	Hepatitis B
Pathways	KEGG_PATHWAY	Measles
Pathways	KEGG_PATHWAY	Human cytomegalovirus infection
Pathways	KEGG_PATHWAY	Influenza A
Pathways	KEGG_PATHWAY	Human papillomavirus infection
Pathways	KEGG_PATHWAY	Human T-cell leukemia virus 1 infection
Pathways	KEGG_PATHWAY	Kaposi sarcoma-associated herpesvirus infection
Pathways	KEGG_PATHWAY	Herpes simplex virus 1 infection
Pathways	KEGG_PATHWAY	Epstein-Barr virus infection
Pathways	KEGG_PATHWAY	Human immunodeficiency virus 1 infection
Pathways	KEGG_PATHWAY	Coronavirus disease - COVID-19
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	Viral carcinogenesis
Pathways	KEGG_PATHWAY	Proteoglycans in cancer
Pathways	KEGG_PATHWAY	MicroRNAs in cancer
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - receptor activation
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - reactive oxygen species
Pathways	KEGG_PATHWAY	Colorectal cancer
Pathways	KEGG_PATHWAY	Renal cell carcinoma
Pathways	KEGG_PATHWAY	Pancreatic cancer
Pathways	KEGG_PATHWAY	Endometrial cancer
Pathways	KEGG_PATHWAY	Glioma
Pathways	KEGG_PATHWAY	Prostate cancer
Pathways	KEGG_PATHWAY	Melanoma
Pathways	KEGG_PATHWAY	Chronic myeloid leukemia
Pathways	KEGG_PATHWAY	Acute myeloid leukemia
Pathways	KEGG_PATHWAY	Small cell lung cancer
Pathways	KEGG_PATHWAY	Non-small cell lung cancer
Pathways	KEGG_PATHWAY	Breast cancer

Main Category	Category	Terms
Pathways	KEGG_PATHWAY	Hepatocellular carcinoma
Pathways	KEGG_PATHWAY	Gastric cancer
Pathways	KEGG_PATHWAY	Central carbon metabolism in cancer
Pathways	KEGG_PATHWAY	Choline metabolism in cancer
Pathways	KEGG_PATHWAY	PD-L1 expression and PD-1 checkpoint pathway in cancer
Pathways	KEGG_PATHWAY	Diabetic cardiomyopathy
Pathways	KEGG_PATHWAY	Lipid and atherosclerosis
Pathways	KEGG_PATHWAY	Fluid shear stress and atherosclerosis
Pathways	REACTOME_PATHWAY	Hemostasis
Pathways	REACTOME_PATHWAY	PI3K Cascade
Pathways	REACTOME_PATHWAY	IRS-mediated signalling
Pathways	REACTOME_PATHWAY	GPVI-mediated activation cascade
Pathways	REACTOME_PATHWAY	Signaling by FGFR in disease
Pathways	REACTOME_PATHWAY	Signaling by ERBB2
Pathways	REACTOME_PATHWAY	Signaling by ERBB2 in Cancer
Pathways	REACTOME_PATHWAY	Constitutive Signaling by Ligand-Responsive EGFR Cancer Variants
Pathways	REACTOME_PATHWAY	Signaling by ERBB4
Pathways	REACTOME_PATHWAY	PI3K events in ERBB4 signaling
Pathways	REACTOME_PATHWAY	PIP3 activates AKT signaling
Pathways	REACTOME_PATHWAY	Developmental Biology
Pathways	REACTOME_PATHWAY	Cytokine Signaling in Immune system
Pathways	REACTOME_PATHWAY	Adaptive Immune System
Pathways	REACTOME_PATHWAY	Metabolism
Pathways	REACTOME_PATHWAY	Signaling by SCF-KIT
Pathways	REACTOME_PATHWAY	PI Metabolism
Pathways	REACTOME_PATHWAY	Phospholipid metabolism
Pathways	REACTOME_PATHWAY	Cell-Cell communication
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	Signaling by EGFR in Cancer
Pathways	REACTOME_PATHWAY	Synthesis of PIPs at the plasma membrane

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Signaling by NTRKs
Pathways	REACTOME_PATHWAY	Innate Immune System
Pathways	REACTOME_PATHWAY	Immune System
Pathways	REACTOME_PATHWAY	Signaling by EGFR
Pathways	REACTOME_PATHWAY	GAB1 signalosome
Pathways	REACTOME_PATHWAY	Signaling by cytosolic FGFR1 fusion mutants
Pathways	REACTOME_PATHWAY	FGFR1 mutant receptor activation
Pathways	REACTOME_PATHWAY	Downstream signal transduction
Pathways	REACTOME_PATHWAY	Signaling by PDGF
Pathways	REACTOME_PATHWAY	Signaling by NTRK1 (TRKA)
Pathways	REACTOME_PATHWAY	Signaling by FGFR
Pathways	REACTOME_PATHWAY	Signaling by VEGF
Pathways	REACTOME_PATHWAY	Signaling by Rho GTPases
Pathways	REACTOME_PATHWAY	PI3K events in ERBB2 signaling
Pathways	REACTOME_PATHWAY	PI3K/AKT activation
Pathways	REACTOME_PATHWAY	Negative regulation of the PI3K/AKT network
Pathways	REACTOME_PATHWAY	Signaling by ALK
Pathways	REACTOME_PATHWAY	TCR signaling
Pathways	REACTOME_PATHWAY	Downstream TCR signaling
Pathways	REACTOME_PATHWAY	Cell surface interactions at the vascular wall
Pathways	REACTOME_PATHWAY	Fcgamma receptor (FCGR) dependent phagocytosis
Pathways	REACTOME_PATHWAY	Role of phospholipids in phagocytosis
Pathways	REACTOME_PATHWAY	Tie2 Signaling
Pathways	REACTOME_PATHWAY	DAP12 interactions
Pathways	REACTOME_PATHWAY	PI3K/AKT Signaling in Cancer
Pathways	REACTOME_PATHWAY	Constitutive Signaling by Aberrant PI3K in Cancer
Pathways	REACTOME_PATHWAY	Signaling by Type 1 Insulin-like Growth Factor 1 Receptor (IGF1R)
Pathways	REACTOME_PATHWAY	DAP12 signaling
Pathways	REACTOME_PATHWAY	IGF1R signaling cascade
Pathways	REACTOME_PATHWAY	IRS-related events triggered by IGF1R

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Fc epsilon receptor (FCERI) signaling
Pathways	REACTOME_PATHWAY	Role of LAT2/NTAL/LAB on calcium mobilization
Pathways	REACTOME_PATHWAY	Signaling by GPCR
Pathways	REACTOME_PATHWAY	Nephrin family interactions
Pathways	REACTOME_PATHWAY	GPCR downstream signalling
Pathways	REACTOME_PATHWAY	Regulation of T cell activation by CD28 family
Pathways	REACTOME_PATHWAY	Co-stimulation by CD28
Pathways	REACTOME_PATHWAY	CD28 dependent PI3K/Akt signaling
Pathways	REACTOME_PATHWAY	G alpha (q) signalling events
Pathways	REACTOME_PATHWAY	Axon guidance
Pathways	REACTOME_PATHWAY	VEGFA-VEGFR2 Pathway
Pathways	REACTOME_PATHWAY	Signaling by Interleukins
Pathways	REACTOME_PATHWAY	Interleukin-2 family signaling
Pathways	REACTOME_PATHWAY	Interleukin-3, Interleukin-5 and GM-CSF signaling
Pathways	REACTOME_PATHWAY	Metabolism of lipids
Pathways	REACTOME_PATHWAY	Constitutive Signaling by EGFRvIII
Pathways	REACTOME_PATHWAY	Signaling by EGFRvIII in Cancer
Pathways	REACTOME_PATHWAY	Signaling by Ligand-Responsive EGFR Variants in Cancer
Pathways	REACTOME_PATHWAY	Downstream signaling of activated FGFR1
Pathways	REACTOME_PATHWAY	PI-3K cascade:FGFR1
Pathways	REACTOME_PATHWAY	PI-3K cascade:FGFR2
Pathways	REACTOME_PATHWAY	Downstream signaling of activated FGFR2
Pathways	REACTOME_PATHWAY	Downstream signaling of activated FGFR3
Pathways	REACTOME_PATHWAY	PI-3K cascade:FGFR3
Pathways	REACTOME_PATHWAY	Downstream signaling of activated FGFR4
Pathways	REACTOME_PATHWAY	PI-3K cascade:FGFR4
Pathways	REACTOME_PATHWAY	Signaling by FGFR1
Pathways	REACTOME_PATHWAY	Signaling by FGFR2
Pathways	REACTOME_PATHWAY	Signaling by FGFR3
Pathways	REACTOME_PATHWAY	Signaling by FGFR4

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Signaling by FGFR2 in disease
Pathways	REACTOME_PATHWAY	Signaling by FGFR4 in disease
Pathways	REACTOME_PATHWAY	Signaling by FGFR1 in disease
Pathways	REACTOME_PATHWAY	Signaling by FGFR3 in disease
Pathways	REACTOME_PATHWAY	Diseases of signal transduction by growth factor receptors and second messengers
Pathways	REACTOME_PATHWAY	RAF/MAP kinase cascade
Pathways	REACTOME_PATHWAY	MAPK family signaling cascades
Pathways	REACTOME_PATHWAY	MAPK1/MAPK3 signaling
Pathways	REACTOME_PATHWAY	Signaling by MET
Pathways	REACTOME_PATHWAY	PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling
Pathways	REACTOME_PATHWAY	Insulin receptor signalling cascade
Pathways	REACTOME_PATHWAY	Signaling by Insulin receptor
Pathways	REACTOME_PATHWAY	Platelet activation, signaling and aggregation
Pathways	REACTOME_PATHWAY	MET activates PI3K/AKT signaling
Pathways	REACTOME_PATHWAY	RET signaling
Pathways	REACTOME_PATHWAY	ESR-mediated signaling
Pathways	REACTOME_PATHWAY	Cellular responses to stimuli
Pathways	REACTOME_PATHWAY	Signaling by NTRK2 (TRKB)
Pathways	REACTOME_PATHWAY	Signaling by Erythropoietin
Pathways	REACTOME_PATHWAY	Intracellular signaling by second messengers
Pathways	REACTOME_PATHWAY	Signaling by Nuclear Receptors
Pathways	REACTOME_PATHWAY	Signaling by Receptor Tyrosine Kinases
Pathways	REACTOME_PATHWAY	Extra-nuclear estrogen signaling
Pathways	REACTOME_PATHWAY	RHO GTPase cycle
Pathways	REACTOME_PATHWAY	RAC1 GTPase cycle
Pathways	REACTOME_PATHWAY	RAC2 GTPase cycle
Pathways	REACTOME_PATHWAY	Erythropoietin activates Phosphoinositide-3-kinase (PI3K)
Pathways	REACTOME_PATHWAY	Activated NTRK2 signals through PI3K
Pathways	REACTOME_PATHWAY	Signaling by NTRK3 (TRKC)
Pathways	REACTOME_PATHWAY	Interleukin receptor SHC signaling

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Regulation of signaling by CBL
Pathways	REACTOME_PATHWAY	Activated NTRK3 signals through PI3K
Pathways	REACTOME_PATHWAY	FLT3 Signaling
Pathways	REACTOME_PATHWAY	Signaling by ERBB2 KD Mutants
Pathways	REACTOME_PATHWAY	Signaling by ERBB2 ECD mutants
Pathways	REACTOME_PATHWAY	Signaling by KIT in disease
Pathways	REACTOME_PATHWAY	Signaling by phosphorylated juxtamembrane, extracellular and kinase domain KIT mutants
Pathways	REACTOME_PATHWAY	Signaling by PDGFR in disease
Pathways	REACTOME_PATHWAY	Signaling by PDGFRA transmembrane, juxtamembrane and kinase domain mutants
Pathways	REACTOME_PATHWAY	Signaling by PDGFRA extracellular domain mutants
Pathways	REACTOME_PATHWAY	Nervous system development
Pathways	REACTOME_PATHWAY	Signaling by CSF1 (M-CSF) in myeloid cells
Pathways	REACTOME_PATHWAY	FLT3 signaling in disease
Pathways	REACTOME_PATHWAY	Signaling by ALK in cancer
Pathways	REACTOME_PATHWAY	Signaling by FLT3 fusion proteins
Pathways	REACTOME_PATHWAY	Signaling by FLT3 ITD and TKD mutants
Pathways	REACTOME_PATHWAY	Signaling by Rho GTPases, Miro GTPases and RHOBTB3
Pathways	REACTOME_PATHWAY	Signaling by ALK fusions and activated point mutants
Pathways	REACTOME_PATHWAY	Signaling by LTK in cancer
Pathways	REACTOME_PATHWAY	Signaling by LTK
Pathways	REACTOME_PATHWAY	Cellular responses to mechanical stimuli
Pathways	REACTOME_PATHWAY	High laminar flow shear stress activates signaling by PIEZO1 and PECAM1:CDH5:KDR in endothelial cells
Pathways	REACTOME_PATHWAY	Response of endothelial cells to shear stress
Pathways	REACTOME_PATHWAY	Co-stimulation by ICOS
Pathways	WIKIPATHWAYS	AMP activated protein kinase signaling
Pathways	WIKIPATHWAYS	Regulation of toll like receptor signaling
Pathways	WIKIPATHWAYS	Angiogenesis

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	MicroRNAs in cardiomyocyte hypertrophy
Pathways	WIKIPATHWAYS	Sterol regulatory element binding proteins SREBP signaling
Pathways	WIKIPATHWAYS	Thyroid stimulating hormone TSH signaling
Pathways	WIKIPATHWAYS	Prolactin signaling
Pathways	WIKIPATHWAYS	Alzheimer 39 s disease and miRNA effects
Pathways	WIKIPATHWAYS	IL17 signaling
Pathways	WIKIPATHWAYS	Glioblastoma signaling
Pathways	WIKIPATHWAYS	Androgen receptor network in prostate cancer
Pathways	WIKIPATHWAYS	Glycerophospholipid biosynthetic pathway
Pathways	WIKIPATHWAYS	IL1 and megakaryocytes in obesity
Pathways	WIKIPATHWAYS	Transcriptional activation by NRF2 in response to phytochemicals
Pathways	WIKIPATHWAYS	Focal adhesion
Pathways	WIKIPATHWAYS	Hepatocyte growth factor receptor signaling
Pathways	WIKIPATHWAYS	Copper homeostasis
Pathways	WIKIPATHWAYS	RAC1 PAK1 p38 MMP2 pathway
Pathways	WIKIPATHWAYS	Pathways affected in adenoid cystic carcinoma
Pathways	WIKIPATHWAYS	PI3K AKT mTOR signaling and therapeutic opportunities in prostate cancer
Pathways	WIKIPATHWAYS	T cell antigen receptor TCR pathway during Staphylococcus aureus infection
Pathways	WIKIPATHWAYS	RIOK1 and RIOK2 in EGFR and PI3K mediated tumorigenesis
Pathways	WIKIPATHWAYS	4 hydroxytamoxifen dexamethasone and retinoic acids regulation of p27 expression
Pathways	WIKIPATHWAYS	VEGFA VEGFR2 signaling
Pathways	WIKIPATHWAYS	Angiopoietin like protein 8 regulatory pathway
Pathways	WIKIPATHWAYS	Chemokine signaling
Pathways	WIKIPATHWAYS	Focal adhesion PI3K Akt mTOR signaling
Pathways	WIKIPATHWAYS	Microglia pathogen phagocytosis pathway
Pathways	WIKIPATHWAYS	IL4 signaling
Pathways	WIKIPATHWAYS	PDGFR beta pathway

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	miRNA regulation of prostate cancer signaling
Pathways	WIKIPATHWAYS	Fibrin complement receptor 3 signaling
Pathways	WIKIPATHWAYS	PI3K AKT mTOR vitamin D3 signaling
Pathways	WIKIPATHWAYS	Endometrial cancer
Pathways	WIKIPATHWAYS	PI3K Akt signaling
Pathways	WIKIPATHWAYS	MET in type 1 papillary renal cell carcinoma
Pathways	WIKIPATHWAYS	Ebola virus infection in host
Pathways	WIKIPATHWAYS	Ras signaling
Pathways	WIKIPATHWAYS	Epithelial to mesenchymal transition in colorectal cancer
Pathways	WIKIPATHWAYS	Non small cell lung cancer
Pathways	WIKIPATHWAYS	Breast cancer pathway
Pathways	WIKIPATHWAYS	Pancreatic adenocarcinoma pathway
Pathways	WIKIPATHWAYS	NRF2 ARE regulation
Pathways	WIKIPATHWAYS	Nonalcoholic fatty liver disease
Pathways	WIKIPATHWAYS	Resistin as a regulator of inflammation
Pathways	WIKIPATHWAYS	Relationship between inflammation COX 2 and EGFR
Pathways	WIKIPATHWAYS	Synaptic signaling associated with autism spectrum disorder
Pathways	WIKIPATHWAYS	Interferon mediated signaling
Pathways	WIKIPATHWAYS	Neural crest cell migration during development
Pathways	WIKIPATHWAYS	Neural crest cell migration in cancer
Pathways	WIKIPATHWAYS	Translation inhibitors in chronically activated PDGFRA cells
Pathways	WIKIPATHWAYS	Cancer immunotherapy by CTLA4 blockade
Pathways	WIKIPATHWAYS	Measles virus infection
Pathways	WIKIPATHWAYS	Small cell lung cancer
Pathways	WIKIPATHWAYS	Gastrin signaling
Pathways	WIKIPATHWAYS	Hepatitis B infection
Pathways	WIKIPATHWAYS	Head and neck squamous cell carcinoma
Pathways	WIKIPATHWAYS	Melanoma
Pathways	WIKIPATHWAYS	Thyroid hormones production and peripheral

Main Category	Category	Terms
		downstream signaling effects
Pathways	WIKIPATHWAYS	Netrin UNC5B signaling
Pathways	WIKIPATHWAYS	Osteoblast differentiation and related diseases
Pathways	WIKIPATHWAYS	EGFR tyrosine kinase inhibitor resistance
Pathways	WIKIPATHWAYS	Insulin signaling
Pathways	WIKIPATHWAYS	mBDNF and proBDNF regulation of GABA neurotransmission
Pathways	WIKIPATHWAYS	Regucalcin in proximal tubule epithelial kidney cells
Pathways	WIKIPATHWAYS	Kisspeptin kisspeptin receptor system in the ovary
Pathways	WIKIPATHWAYS	Phosphoinositides metabolism
Pathways	WIKIPATHWAYS	Kallmann syndrome
Pathways	WIKIPATHWAYS	Pleural mesothelioma
Pathways	WIKIPATHWAYS	Regulation of actin cytoskeleton
Pathways	WIKIPATHWAYS	Congenital generalized lipodystrophy
Pathways	WIKIPATHWAYS	Familial partial lipodystrophy
Pathways	WIKIPATHWAYS	Alzheimer 39 s disease
Pathways	WIKIPATHWAYS	Resolvin E1 and resolvin D1 signaling decrease inflammation
Pathways	WIKIPATHWAYS	Modulation of PI3K Akt mTOR signaling by bioactive sphingolipids
Pathways	WIKIPATHWAYS	Dravet syndrome
Pathways	WIKIPATHWAYS	Targeted agents in triple negative breast cancer
Pathways	WIKIPATHWAYS	Warburg effect modulated by deubiquitinating enzymes and their substrates
Pathways	WIKIPATHWAYS	G13 signaling
Pathways	WIKIPATHWAYS	Acute myeloid leukemia
Pathways	WIKIPATHWAYS	IL26 signaling
Pathways	WIKIPATHWAYS	Osteoarthritic chondrocyte hypertrophy
Pathways	WIKIPATHWAYS	Phosphatidyl inositol phosphate pathway
Pathways	WIKIPATHWAYS	ADHD and autism ASD pathways
Pathways	WIKIPATHWAYS	Cancer pathways

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	GNAQ pathways in port wine stain
Pathways	WIKIPATHWAYS	Effect of omega 3 PUFA on Huntington 39 s disease pathways
Pathways	WIKIPATHWAYS	Phosphoinositide 3 kinase PI3K family
Pathways	WIKIPATHWAYS	Hallmark of cancer sustaining proliferative signaling
Pathways	WIKIPATHWAYS	Molecular pathway for oxidative stress
Pathways	WIKIPATHWAYS	Inhibition of GNAQ regulated signaling in uveal melanoma
Pathways	WIKIPATHWAYS	Effect of dasatinib on the BCR ABL signaling pathway
Pathways	WIKIPATHWAYS	Targeted therapy in breast cancer
Pathways	WIKIPATHWAYS	Cyclin dependent kinase 4 6 inhibitors in breast cancer
Pathways	WIKIPATHWAYS	AKT signaling and ARTD family members
Pathways	WIKIPATHWAYS	BDNF signaling in neurodevelopment
Pathways	WIKIPATHWAYS	ErbB signaling
Pathways	WIKIPATHWAYS	DNA damage response only ATM dependent
Pathways	WIKIPATHWAYS	Estrogen signaling
Pathways	WIKIPATHWAYS	Serotonin HTR1 group and FOS pathway
Protein Domains	INTERPRO	PI3K_Ras-bd_dom
Protein Domains	INTERPRO	PI3/4_kinase_cat_dom
Protein Domains	INTERPRO	PI3K_accessory_dom
Protein Domains	INTERPRO	PI3K-type_C2_dom
Protein Domains	INTERPRO	PI3K_ABD
Protein Domains	INTERPRO	Kinase-like_dom_sf
Protein Domains	INTERPRO	PI_Kinase
Protein Domains	INTERPRO	ARM-type_fold
Protein Domains	INTERPRO	PI3/4_kinase_CS
Protein	INTERPRO	Ubiquitin-like_domsf

Main Category	Category	Terms
Domains		
Protein Domains	INTERPRO	C2_domain_sf
Protein Domains	INTERPRO	PI3/4_kinase_cat_sf
Protein Domains	INTERPRO	PI3Kalpha_dom
Protein Domains	INTERPRO	PI3K_accessory_sf
Protein Domains	SMART	PI3K_C2
Protein Domains	SMART	PI3K_p85B
Protein Domains	SMART	PI3K_rbd
Protein Domains	SMART	PI3Ka
Protein Domains	SMART	PI3Kc